

FULL PAPER



**PicklePro: A Personal Informatics System for Pickleball Performance Monitoring and
Player Development**

**A Capstone Proposal Presented to Department in Information Technology
College of Computer Studies De La Salle University**

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Prepared and Submitted By:

Orino, Dikie Kirk

Pulido, Kathryn Leigh

Tahilramani, Rohit

Dr. Raphael Gonda

Capstone Adviser

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CHAPTER I Research Background

1.1. Introduction

The increasing adoption of digital technologies has transformed how individuals monitor, analyze, and improve their performance across various domains, including health, fitness, and sports. Advances in computer vision technologies, and artificial intelligence have enabled the automatic collection and analysis of large amounts of performance data, allowing athletes to gain objective insights that support continuous improvement. Rather than relying solely on subjective observations, athletes can now make informed decisions based on measurable evidence collected during training and competition.

Within sports science, these technologies have contributed to the emergence of personal informatics systems, which are designed to help individuals achieve personal goals through continuous data collection, reflection, and behavior change. Personal informatics emphasizes enabling users to understand their own performance by collecting relevant information, presenting meaningful insights, and encouraging actions that support long-term improvement. The process typically involves setting goals, automatically gathering performance data, reflecting on collected information, and adjusting training strategies based on the results.

As pickleball continues to gain popularity among recreational and competitive players, there is an increasing demand for accessible technologies that provide objective performance feedback. While existing solutions offer general fitness metrics or sport-specific analytics for other disciplines, few systems focus on the strategic and technical characteristics that define pickleball. This study proposes the development of a personal informatics system that integrates existing

computer vision technologies to automatically collect and analyze player performance data, provide meaningful performance insights, and support continuous improvement through data-driven decision-making.

1.2. Background of the Study

Pickleball is one of the fastest-growing sports in the world due to its accessibility, social nature, and ability to accommodate players of different ages and skill levels (Sports & Fitness Industry Association [SFIA], n.d.). Unlike many racket sports that rely heavily on physical strength and athleticism, pickleball places significant emphasis on strategy, positioning, consistency, communication, and decision-making (USA Pickleball, n.d.). The sport's smaller court, slower ball speed, and non-volley zone (commonly known as the "kitchen") encourage players to focus on shot placement, anticipation, and tactical play rather than power alone (USA Pickleball, n.d.; EBSCO Information Services, n.d.).

The sport's inclusive nature has contributed to its widespread adoption among recreational players, competitive athletes, students, and older adults. Research has noted that pickleball's lower physical demands compared to sports such as tennis and badminton make it more accessible across age groups and skill levels (MDPI, 2024). This unique characteristic allows beginners, seniors, and competitive players to participate within the same sporting community while pursuing different personal goals.

As participation continues to grow, players increasingly seek ways to improve their performance and better understand their gameplay. However, many pickleball players still rely on self-assessment, informal feedback from playing partners, or occasional coaching sessions to

evaluate their strengths and weaknesses. While coaches can provide valuable guidance, access to regular coaching may be limited by cost, availability, or scheduling constraints.

Recent advancements in computer vision have created new opportunities for sports performance monitoring. These technologies allow athletes to collect and analyze objective data regarding movement, physiological responses, and gameplay behavior. Despite these developments, many existing sports analytics systems focus primarily on professional athletes or are designed for sports such as tennis, basketball, and soccer, leaving an opportunity for solutions specifically tailored to the unique demands of pickleball.

1.3. Statement of the Problem

Despite the increasing availability of digital sports technologies, accessible systems that integrate objective gameplay collection, pickleball-specific performance analysis, community coordination, and collaborative coaching support remain limited. Pickleball players may use match videos, personal observations, informal feedback, and separate community platforms, but these sources are not always connected in a way that supports continuous and evidence-based player development.

Specifically, the study seeks to address the following problems:

P1. Limited objective in-game data collection and reliance on subjective performance evaluation

Pickleball players frequently evaluate their performance through memory, self-observation, or informal comments from coaches and playing partners. These methods may overlook important

gameplay details that occur rapidly during a rally, including player movement, court positioning, kitchen-line presence, court coverage, rally activity, and selected shot events. The limited availability of accessible and systematic in-game data collection consequently makes it difficult for players to establish an objective basis for evaluating their performance.

P2. Fragmented player development tracking and lack of pickleball-specific performance analytics

Player information may be distributed across recorded videos, personal notes, match results, coach feedback, and separate applications. When these records are not integrated into a unified performance history, players may find it difficult to identify recurring strengths, weaknesses, and development patterns across multiple sessions. Furthermore, general sports analytics systems may not sufficiently represent pickleball-specific concepts such as kitchen-line presence, dink activity, court-zone occupancy, rally structure, doubles positioning, and strategic shot selection. This limits the ability of players to interpret their data and monitor improvement over time.

P3. Difficulty finding suitable games and partners and coordinating community-based play

Recreational and developing players often depend on personal networks, social-media groups, or chance encounters when looking for opponents, partners, training groups, or open-play sessions. Information concerning player availability, game format, preferred play type, existing skill information, and community schedules may be communicated through separate or informal channels. This may lead to unsuitable pairings, missed playing opportunities, and inefficient coordination among players, court owners, and community organizers.

P4. Limited timely, contextual, and collaborative coaching support

Access to regular coaching may be constrained by cost, availability, and scheduling. Even when coaches, teammates, or training partners are available, feedback may rely on memory or observations that are not directly connected to the player's recorded performance information. Players may therefore lack a structured mechanism for sharing reports, receiving contextual feedback, reviewing training recommendations, and obtaining supplementary guidance during appropriate pauses in a training session.

Given these problems, there is a need for an integrated pickleball personal informatics system that can capture objective gameplay information, organize performance records, generate pickleball-specific insights, support community coordination, and facilitate evidence-based collaboration among players, coaches, teammates, and court-community organizers.

1.4. Conceptual and Solutions Framework

1.4.1 PPIF Stage Matrix

Stage	Primary Goal	Key Actors	Technologies/ Inputs	Outputs	Example Metrics
1 - Preparation	Define performance targets and focus areas	Players, Coaches	Goal - Setting; Consultation	Performance objectives, list of KPIs to track	Dink consistency, court position targets, shot selection priorities
2 - Collection	Capture objective performance data during play	Players , Sensors	Computer Vision; Motion Tracking	Raw Sensor stream, tagged video clips	Court position, rally structure, shot selection
3 - Integration	Unify all data into a single athlete profile	System , players, coaches	Gameplay data; Context data; Historical records	Athlete Performance Profile, shot maps, heat maps, trend reports	Match history, shot selection, tactical indicators, historical strengths and weaknesses,

4 - Reflection	Analyze processed data to surface insights	Players, Coaches	Dashboards; Visual Analytics; Match Summary	Evidenced-Based feedback, annotated match reviews, training plan input	Dink success-rate, unforced errors, kitchen line time movement type, rally duration,
5 - Action	Convert insights into individualized interventions	Players, Coaches	Reflection outputs; Recommendation engine	Tailored drill plans, tactical adjustments, workload guidance, practice targets for next	Technical session targets, position reminders, workload limits, next session goals

Figure 1.4.1 PPIF Stage Matrix

Note. Based on Li, Dey, and Forlizzi's (2010) Personal Informatics model, adapted for pickleball performance.

The study utilizes the Pickleball Performance Informatics Framework as the conceptual foundation for the four proposed solution frameworks. Computer-vision functions may be delivered through two complementary processing modes: on-demand session processing and web-based post-game processing. The PPIF is a specialized framework designed to support continuous skill development among pickleball players by organizing the system around performance goals, data collection, analytics, reflection, and training action. It is adapted from Li, Dey, and Forlizzi's Personal Informatics model, which outlines five stages of technology-supported self-monitoring: Preparation, Collection, Integration, Reflection, and Action. Within this study, these stages are applied to two possible computer vision implementations: a real-time iPhone-based computer vision framework and a web-based post-game video analytics framework.

In the Preparation Stage, players and coaches identify the specific performance areas that the system must monitor. These may include dink consistency, court positioning, shot selection,

kitchen-line presence, rally duration, movement efficiency, and tactical decision-making. This stage ensures that the system is designed as a performance improvement tool rather than merely a game-recording or scorekeeping application.

In the Collection Stage, the two proposed frameworks differ in how data is captured. The real-time framework collects live gameplay data through an iPhone camera and, when needed, an edge-processing device that can analyze player movement, court zones, rally activity, and pose-related cues during play. Meanwhile, the web-based framework collects recorded match videos uploaded by players or coaches after the game. In both cases, computer vision serves as the primary sensing method, while optional wearable or self-reported data may provide additional information such as fatigue, perceived exertion, recovery condition, or training context.

In the Integration Stage, the system combines gameplay data, visual tracking outputs, match context, and historical records into a unified athlete performance profile. For the real-time framework, this integration supports immediate session-level feedback such as live court position, kitchen time, and rally structure. For the web-based framework, integration supports deeper post-game analysis such as heatmaps, shot maps, rally segmentation, historical trend reports, and player development patterns.

In the Reflection Stage, players and coaches interpret the processed data through dashboards, visual analytics, annotated clips, and performance summaries. The real-time framework supports immediate reflection during or shortly after training, allowing coaches to give timely feedback. The web-based framework supports more detailed post-match review, where players and coaches

can examine performance trends, repeated errors, tactical patterns, and areas for technical improvement.

In the Action Stage, the insights generated by the system are translated into individualized training interventions. These may include technical drills, tactical adjustments, positioning reminders, workload guidance, and next-session performance targets. Through this stage, the framework emphasizes evidence-based coaching decisions rather than relying only on memory, manual observation, or subjective judgment.

Problem	Solution framework	Primary functions	Expected outputs
P1. Limited objective in-game data collection and reliance on subjective evaluation	S1. Objective Gameplay Data Collection Framework	Video capture, player detection and tracking, court calibration, rally segmentation, court-zone analysis, selected shot-event recognition	Structured gameplay events, movement paths, kitchen-zone data, rally records, court-coverage information
P2. Fragmented development tracking and lack of pickleball-specific analytics	S2. Player Personal Informatics and Analytics Framework	Player profiles, goal setting, historical integration, dashboards, post-game reports, reflection, progress monitoring and recommendations	Unified performance profile, visual reports, historical trends, goals, reflections and development recommendations

P3. Difficulty finding suitable games and partners and coordinating community-based play	S3. Court and Community Coordination Framework	Game listings, open-play queues, availability matching, session coordination, partner and opponent suggestions, community announcements, court-owner administration, and access management	Suitable game suggestions, session listings, queue records, participation records, community announcements, and court-community informationcommunity information
P4. Limited timely, contextual and collaborative coaching support	S4. Coach and Team Collaboration Framework	Authorized report access, coach annotations, team review, training feedback, shared development plans and player-initiated audio guidance	Coach feedback, shared reports, training plans, team-progress records and contextual audio cues

Figure 1.4.2 Problem-Solution Matrix

S1. Objective Gameplay Data Collection Framework

The Objective Gameplay Data Collection Framework addresses the fragmented availability of systematic and objective in-game performance data. It utilizes video captured through an accessible camera device, such as a smartphone, together with existing computer-vision technologies to process observable pickleball gameplay.

The framework focuses on feasible performance information such as player detection and tracking, movement paths, court calibration, court-zone occupancy, kitchen-line presence, rally segmentation, court coverage, and selected shot-event classifications. Captured information is converted into structured gameplay records that can be used by the Player Personal Informatics and Analytics Framework.

The framework does not provide automated officiating, official in-or-out decisions, or guaranteed ball-trajectory analysis. Outputs that cannot be detected reliably through the selected camera configuration must be excluded from the core performance metrics or marked according to their level of confidence.

S2. Player Personal Informatics and Analytics Framework

The Player Personal Informatics and Analytics Framework organizes gameplay information into a unified player-centered performance record. It supports the preparation, integration, reflection, and action stages of the Pickleball Performance Informatics Framework by enabling players to establish performance goals, review post-game reports, compare historical and current results, record reflections, and monitor development over time.

The framework integrates structured gameplay events, match context, player goals, historical records, and authorized coach feedback. It presents pickleball-specific information through dashboards, visual summaries, heatmaps, movement reports, rally summaries, and performance trends. Based on the available information, the system may identify priority areas and provide supplementary recommendations for future training sessions.

S3. Court and Community Coordination Framework

The Court and Community Coordination Framework addresses difficulties in finding suitable games, partners, training groups, and community-based playing opportunities. It supports players, court owners, and community organizers by providing centralized game listings, open-play queues, player availability records, session coordination, and partner or opponent suggestions.

The framework recommends suitable playing opportunities using compatibility factors supplied by users, including availability, preferred game format, playing experience, preferred level of competition, training objectives, play style, and location. These factors are intended to improve game suitability and coordination without generating or assigning an official player rating.

Court-owner and community-administration functions are incorporated into this framework because court owners and community organizers are responsible for managing game listings, open-play schedules, player participation, queues, announcements, community records, user roles, and access permissions. The framework centralizes this information to reduce dependence on fragmented social-media posts, private messages, and informal communication channels.

The framework does not include comprehensive facility management, payment processing, financial reporting, tournament-bracket management, or automated competitive ranking.

S4. Coach and Team Collaboration Framework

The Coach and Team Collaboration Framework supports cooperative performance improvement by allowing authorized coaches, teammates, and training partners to review relevant player-performance information. It provides role-based access to reports, annotated gameplay

records, coach comments, training recommendations, shared goals, and team-progress information.

The framework may also include an On-Demand Contextual Audio Coaching Submodule. This feature must be activated by the player after a rally or during an appropriate training break. It may use validated gameplay events, player history, rule-based feedback templates, and approved coaching knowledge to generate a short supplementary audio cue.

The audio feature must not be described as continuous real-time coaching. It does not replace professional coaching and must not provide medical guidance, injury assessment, automated officiating, or advice based on uncertain computer-vision outputs.

1.5 Study Objectives

The study aims to develop an integrated personal informatics system for pickleball players that utilizes existing computer vision and information technologies to support objective gameplay monitoring, player development tracking, community-based play coordination, and collaborative coaching.

Specifically, the study aims to:

1. Develop an Objective Gameplay Data Collection Framework that uses existing computer vision technologies to capture and organize feasible pickleball gameplay indicators, including player movement, court positioning, kitchen-zone presence, rally activity, court coverage, and selected shot events.
2. Develop a Player Personal Informatics and Analytics Framework that integrates current and historical gameplay information into a unified player profile and presents

performance reports, visual analytics, goals, reflections, progress trends, and supplementary development recommendations.

3. Develop a Court and Community Coordination Framework that supports centralized game listings, open-play queues, player availability, session coordination, partner and opponent suggestions, community announcements, participation records, and basic access management for court owners and community organizers.
4. Develop a Coach and Team Collaboration Framework that allows authorized coaches, teammates, and training partners to review relevant player-performance information, provide structured feedback, share development plans, monitor collaborative progress, and deliver player-initiated contextual audio guidance during appropriate training pauses.

1.6. Significance of the Study

This study contributes to the growing application of personal informatics technologies within sports performance monitoring, particularly in the context of pickleball. By utilizing existing computer vision technologies, the proposed system seeks to provide players with objective and meaningful performance information that supports continuous improvement.

The proposed system enables players to better understand their strengths, weaknesses, and gameplay patterns through objective performance analytics. By providing historical performance records and personalized recommendations, players can make more informed decisions regarding their training and monitor their progress over time.

The system also provides coaches with structured performance information that can supplement their observations during training and matches. Access to objective gameplay data allows

coaches to provide more targeted feedback and develop training strategies based on measurable evidence.

Additionally, the study may assist clubs and organizations in promoting player development and balanced matchmaking by introducing technology-assisted performance monitoring that complements existing coaching and training practices.

Lastly, the findings and system design may serve as a foundation for future research and development involving sports performance monitoring, artificial intelligence, and personal informatics applications in other sports or athletic domains.

1.7. Scope and Limitations

This study focuses on the design and development of a personal informatics system for pickleball players that supports continuous performance improvement through automated data collection, performance reflection, and actionable insights. The system is intended primarily for individual players and extends to a cooperative layer where authorized individuals, such as coaches, may access player performance information to provide informed guidance. The study integrates existing computer vision technologies and does not involve the development of new artificial intelligence or machine learning models.

1.7.1 Core Performance Monitoring

The first component is Core Performance Monitoring, which utilizes existing computer vision technologies to analyze pickleball gameplay. The system automatically tracks player movement, classifies common pickleball shots such as dinks, drives, volleys, and

lobs, and evaluates player positioning and court coverage throughout a match. Performance data are presented through an analytics dashboard that enables players to review their strengths and weaknesses, monitor performance trends over time, and compare historical and current gameplay. Consistent with the principles of personal informatics, the system also supports goal setting, personalized performance insights, recommendations, and a continuous feedback loop that encourages players to reflect on collected data and adjust their training strategies accordingly.

1.7.2 Light Skill and Performance Assessment

The second component is Light Skill and Performance Assessment, which provides a general evaluation of player characteristics using collected gameplay data. The system estimates player skill level (e.g., beginner, intermediate, or advanced) and identifies dominant play styles such as aggressive, defensive, or balanced. To encourage participation and interaction within pickleball communities, the system will allow players to share their own performance statistics, analytics, and progress to other players and coaches. In addition, recorded matches are automatically linked to corresponding performance analytics, allowing players to review post-match insights and understand how performance data may be applied to future gameplay and training.

1.7.3 Gameplay Context Support

The third component covers Gameplay Context Support, wherein the system is capable of analyzing and organizing performance data collected from different types of pickleball activities. These include practice or training sessions, casual matches, tournaments, leveling games, singles matches, doubles matches, and individual practice using a wall or

ball machine. Regardless of the gameplay context, the system maintains historical performance records that allow players to monitor long-term progress and evaluate improvements over time.

1.7.4 Post-Game Audio Coaching Assistance

The fourth component is Post-Game Audio Coaching Assistance, which allows players to activate a coaching session through the mobile application during training or practice play. When enabled, the outputs such as player position, kitchen-line presence, court coverage, rally timing, movement patterns, and selected pose-related cues to generate short audio feedback for the player.

The system extends audio feedback to the web-based, non-real-time framework. After a match video is uploaded and processed, the computer vision pipeline produces structured performance data including shot classifications, shot accuracy, and rally outcomes for the full match. This processed data is then converted into a short, spoken audio summary that the player can listen to instead of, or alongside, the visual dashboard.

Unlike the live coaching cues, which are limited to brief, low-latency prompts during play, post-game audio feedback can draw on the complete match dataset, allowing for more evaluative statements. Example outputs include summaries such as shot type distribution and accuracy for the match, and dominant error patterns (for example, a low drive success rate compared to dinks).

This post-game audio feature uses the text-to-speech and feedback-generation service, but is triggered once per processed match rather than continuously during play, and pulls

from the complete post-match performance profile rather than a rolling window of recent frames. Players may choose to enable or disable this feature per upload, and the generated audio is stored alongside the corresponding match record so it can be replayed from the dashboard.

The feature does not replace a professional coach and does not provide officiating, medical, or injury-related decisions. Instead, it serves as an assistive feedback layer that supports player reflection and immediate training adjustment during live or semi-live practice sessions.

1.7.5 Limitations

This study is limited to the collection, analysis, visualization, and presentation of performance-related information. The system is designed to support player self-reflection and decision-making through objective data but does not replace professional coaching, sports training, officiating, or medical expertise. Performance recommendations generated by the system should be considered supplementary guidance rather than definitive prescriptions for player development.

Furthermore, the study is limited to integrating existing computer vision, hardware technologies, and artificial intelligence technologies. The researchers will not develop new computer vision algorithms, machine learning models, or hardware as part of the study.

The following functionalities are outside the scope of the proposed system:

1. Court booking and reservation management.
2. Payment processing and financial transactions related to court usage.
3. Court availability management and administrative functions for court owners or operators.
4. Tournament, league, bracket, or event scheduling and management.
5. Comprehensive team, club, or organization management functions.
6. Medical diagnosis, injury treatment, rehabilitation planning, or clinical health monitoring.
7. Automated officiating, referee decision-making, or official in/out detection.
8. Full replacement of professional coaching, since the live audio coaching feature is limited to supplementary feedback based on detectable computer vision outputs.
9. Development of new computer vision algorithms, machine learning models, or custom hardware. The study is limited to integrating existing technologies, frameworks, and services.
10. The system does not calculate or assign an official skill rating. Instead it uses player-provided preferences, playing experience, availability, and session objectives to identify potentially suitable games, partners, and training groups.

CHAPTER II Review of Related Literature

2.1. Review of Related Concepts

2.1.1 Personal Informatics in Sports Performance Monitoring

Personal informatics refers to the use of technology to help individuals collect, organize, understand, and use personal data for self-improvement. In the context of sports, this concept is important because athletes and recreational players are increasingly using digital tools to track their progress and make better decisions about training and performance. Li et al. (2010) introduced a stage-based model of personal informatics that includes five stages: preparation, collection, integration, reflection, and action. These stages show how users first identify what they want to improve, gather relevant data, combine and interpret the information, reflect on the results, and apply changes based on what they learned.

This concept is closely related to the proposed pickleball system because the study focuses on helping players understand their gameplay through objective performance data. Many pickleball players rely on memory, casual observation, or feedback from playing partners when evaluating their performance. Although these forms of feedback can be helpful, they may not always provide a complete or accurate picture of a player's strengths and weaknesses. Through a personal informatics approach, players can review actual data from their games and use it as a basis for improvement.

Personal informatics has also been explored in sports settings. Rapp and Tirabeni (2018) explained that personal informatics tools are used by both amateur and elite athletes not only to measure performance, but also to understand their bodies, track progress, and

connect their data with their overall sporting experience. In a later study, Rapp and Tirabeni (2020) discussed how self-tracking tools can influence an athlete's motivation, attention, comfort, and lifestyle. These ideas support the direction of the proposed system because pickleball players may benefit from a tool that does more than record statistics. The system should help them reflect on their performance, recognize patterns, and make practical adjustments for future games.

2.1.2 Computer Vision for Sports Performance Analysis

Computer vision is a technology that allows computers to analyze visual information from images or videos. In sports, it has become useful for studying player movements, ball movement, game events, and performance patterns. Naik et al. (2022) explained that computer vision can process sports video data and transform it into measurable information such as player movement, ball trajectories, action recognition, and tactical events. This makes computer vision valuable in sports performance analysis because it can provide objective data that may be difficult to record manually during actual gameplay.

For the proposed pickleball system, computer vision can be used to analyze important gameplay details such as player movement, footwork, court positioning, rally patterns, shot types, reaction time, doubles spacing, and unforced errors. This is especially useful in pickleball because many performance-related actions happen quickly during rallies. A player may not always remember where they were positioned, how often they stayed near the kitchen line, or which shots led to mistakes. By using camera-based analysis, the

system can provide players with a clearer view of how they actually performed during a match.

This concept supports the core performance monitoring component of the proposed system. Instead of depending only on manual scorekeeping or subjective observation, computer vision can help capture gameplay data in a more consistent and detailed manner. The collected information can then be used to generate insights about movement efficiency, positioning, shot selection, and tactical decision-making. In this way, computer vision serves as one of the main technologies that can help make pickleball performance evaluation more data-driven.

2.1.3 Pickleball-Specific Tactical Performance

Pickleball has unique gameplay characteristics that separate it from other racket sports. While sports such as tennis and badminton often emphasize speed, power, and court coverage, pickleball places strong importance on strategy, positioning, consistency, communication, and decision-making. One of the most important features of pickleball is the non-volley zone, commonly known as the kitchen. USA Pickleball (2026) identifies the non-volley zone as a key rule-based area that affects how players move, position themselves, and select shots during rallies.

Because of this, pickleball performance should not be measured only through general fitness or basic match results. Important tactical indicators may include dink consistency, volley success, kitchen-line presence, rally length, shot selection, time spent at the baseline, and unforced errors. These indicators are meaningful because they reflect the actual demands of pickleball gameplay. A player who can position well, control the

kitchen line, and make consistent decisions may perform better than a player who only relies on power.

Recent research also supports the need to study pickleball through technical and tactical indicators. Prieto-Lage et al. (2025) analyzed key match indicators in women's singles pickleball and showed that performance can be examined through measurable gameplay actions. This supports the proposed system's focus on pickleball-specific analytics. By tracking indicators that are directly connected to the sport, the system can provide insights that are more relevant to pickleball players than general sports or fitness tracking tools.

2.1.4 Visual Analytics and Decision Support for Player Improvement

Visual analytics refers to the use of visual tools such as dashboards, charts, graphs, heatmaps, and summaries to help users understand data more easily. In sports performance systems, visual analytics is important because raw numbers alone may not be enough for players and coaches to understand what is happening. A player may see a statistic, but without proper presentation and explanation, it may be difficult to know what action should be taken next. Visual tools can make performance data easier to interpret by showing patterns in movement, shot success, rally trends, fatigue, and progress over time.

Decision support is also important in sports performance monitoring because the purpose of collecting data is not only to store information, but to guide improvement. Wakefield et al. (2014) discussed the informatics needs of amateur athletic coaches and emphasized the importance of considering training context, athlete condition, and performance

information. This supports the idea that sports systems should help users understand what happened, why it happened, and what changes may be needed.

For the proposed pickleball system, visual analytics can help players and coaches review performance in a more meaningful way. Dashboards may show court heatmaps, shot success rates, unforced errors, kitchen-line presence, rally length, and fatigue-related trends. These visual summaries can help players identify which skills need improvement, which shots are effective, and whether their positioning is efficient. As explained in the personal informatics model, collected data becomes more useful when it supports reflection and action (Li et al., 2010). Therefore, visual analytics and decision support are essential parts of the proposed system because they connect performance data to actual player improvement.

2.2. Review of Related Systems

A sports performance monitoring system is a digital tool that helps athletes capture, analyze, and reflect on their gameplay or training data to support improvement. These systems generally fall into three overlapping types: computer-vision systems that analyze video of play or movement, and rating or community systems that quantify skill and connect players. Their core value lies in replacing subjective self-assessment with objective, measurable feedback, often through automatic data collection, performance dashboards, and personalized recommendations. The following systems are assessed on their functionalities and their relevance to a personal informatics system for pickleball players.

2.2.1 Zepp Tennis

Zepp Tennis is a training system for tennis that utilizes a sensor and an app, with a small motion sensor mounted on the handle of the racket. It evaluates key stroke metrics like ball speed, ball spin, and sweet-spot reliability, categorizes strokes by type, and provides Smart Rally video capture that automatically records and tags the top rallies. It offers power, consistency, and intensity metrics, a community featuring leaderboards, and the option to compare a player's accuracy with that of a pro or coach.. Its drawbacks include being tailored for tennis instead of pickleball, relying on proprietary hardware, and its sensor hardware has mostly been phased out and might not connect to contemporary smartphones, restricting current accessibility.

2.2.2 PlaySight SmartCourt

PlaySight signifies the installed facility's multi-camera aspect of computer-vision systems. Its AI platform with connected cameras is used on pickleball, tennis, and padel courts, as well as in professional and college sports, aiding broadcast, instant replay, and player development. Its SmartTracker automatically monitors all on-court activity without an operator, while SmartScore gathers statistics and auto-tags occurrences. Its advantages include extensive automated data collection and analysis that showcase the viability of the proposed system's fundamental monitoring element. Its drawbacks include the need for expensive, permanent infrastructure, a focus on clubs, broadcasting, and officiating rather than the individual player's feedback process, and personalized matchmaking.

2.2.3 Arccos Caddie

Arccos is golf's leading game-tracking system, using lightweight screw-in club sensors that pair with a smartphone or Apple Watch. It automatically tracks every shot without manual input and provides strokes-gained analytics across driving, approach, short game, and putting, along with an AI rangefinder and real-time club and strategy recommendations. Its AI evaluates a player's strengths, weaknesses, playing patterns, course, and past performance to give caddie-style advice. Its strengths are highly relevant to the proposed system: automatic data collection, deep historical analytics, and personalized, data-driven recommendations for improvement. Its limitations are that it is golf-specific, requires proprietary hardware plus an ongoing subscription, and has no computer-vision analysis or player-matchmaking capability.

2.2.4 Padelytics

Padelytics is a platform enhanced by AI for padel, the racket sport closest to pickleball, that transforms regular match footage into performance analytics. It employs machine-learning models and computer-vision algorithms to assess each frame, monitoring the players and the ball, identifying and categorizing player actions, while producing a detailed event stream of all occurrences on the court. A match is captured and uploaded to the cloud, where the AI analyzes each frame and provides insights to the player's phone or website within minutes after the match concludes, instead of in real time. It generates court-location heatmaps, quantifies the distance and speed of each player's runs, monitors fitness improvements over time, analyzes time allocated to defense, offense, and transitions, assesses partner coordination, and determines which shots a player

employs frequently as well as their landing spots, including the most and least effective shots. Its advantages render it one of the nearest analogs to the suggested system, as it executes movement tracking through computer vision, analyzes court positioning, and classifies shots from uploaded match footage on regular cameras, addressing multiple core monitoring objectives of the proposed system simultaneously. It also offers enhancement suggestions, such as advice on improved court positioning. Its shortcomings are that it is designed for padel instead of pickleball, is aimed at clubs equipped with court cameras, and does not provide skill-based matchmaking with opponents.

2.2.5 HomeCourt

HomeCourt, created by the NEX Team and an official NBA partner, is a mobile app for basketball training that transforms a phone or tablet into a computer-vision personal coach. Utilizing the device's camera alongside computer vision and AI, it smartly records a player's shots, dribbling, and movements, analyzing them to provide immediate feedback and comprehensive analytics. Its Shot Science function quantifies advanced metrics including release angle, vertical jump, and player speed, while also executing automatic identification of shot types. It also monitors advancement over time with complete exercise records, data, and skill assessments, and provides a tailored suggested workout daily featuring drills according to the player's proficiency. Its advantages position it as the nearest consumer equivalent to the suggested system, as it showcases a full personal-informatics cycle, seamless data gathering, analysis-driven reflection, and action via tailored recommendations, on user-friendly hardware,

and it incorporates Apple Watch heart rate. Its flaws include being focused solely on basketball, functioning primarily in real-time instead of analyzing recorded game footage, failing to assess racket-court positioning, and providing social competition instead of skill-oriented opponent pairing.

2.2.6 OnForm

OnForm is a sports video analysis and coaching platform that allows athletes and coaches to record, review, and analyze technique using a mobile device. It supports slow-motion playback, frame-by-frame review, side-by-side comparison, voice-over feedback, and remote video sharing between coaches and athletes. Its relevance to the proposed system lies in its use of accessible video-based analysis for performance reflection, allowing athletes to review specific movements and receive structured feedback without requiring expensive facility-based systems. This supports the personal informatics process by helping users collect visual performance data, reflect on their actions, and identify areas for improvement over time. However, OnForm is designed as a general sports coaching tool rather than a pickleball-specific analytics system. It does not automatically classify pickleball shots, track rally patterns, generate court-positioning heatmaps, or provide skill-based matchmaking for players.

2.2.7 Dartfish

Dartfish is a video analysis software platform used in sports performance, coaching, and athlete development. It allows users to capture, analyze, tag, compare, and share sports videos, making it useful for identifying technical errors, tactical patterns, and performance trends. Its tools support video annotation, movement comparison, and performance review, which can help athletes and coaches better understand weaknesses and develop training strategies. Dartfish is relevant to the proposed system because it demonstrates how video can be transformed into measurable and reviewable performance information. It also supports technical and tactical analysis, which is important in racket sports where positioning, timing, and shot selection affect performance. However, Dartfish is not specifically designed for pickleball and may require manual tagging or expert interpretation. It also functions more as a coaching and analysis platform than as an automated personal informatics system with built-in recommendations, matchmaking, and court management features.

2.2.8 CoachNow

CoachNow is an all-in-one sports coaching application that combines video and image analysis, private communication, cloud storage, and feedback tools for coaches and athletes. It includes features such as slow-motion video review, side-by-side comparisons, voice-over feedback, CoachCam, and AI skeleton tracking for analyzing body movement and joint angles. Its relevance to the proposed system is its ability to support continuous improvement through

recorded performance data, coach-athlete communication, and visual feedback. This makes it useful as a reference for the reflection and feedback stage of a personal informatics system. However, CoachNow is not pickleball-specific and is mainly designed around coaching communication rather than automatic match analytics. It does not directly analyze rally structure, shot outcomes, court zones, player coordination, or opponent compatibility, which limits its use as a complete model for pickleball performance monitoring.

2.2.9 DUPR

DUPR, or Dynamic Universal Pickleball Rating, is a pickleball rating system that evaluates player skill through match results and assigns players a numerical rating. Its rating scale begins at 2.000 for beginners and extends to 8.000 for professional-level players. DUPR is widely used for organizing fair matches, leagues, and tournaments by helping players understand their skill level and find more balanced competition. Its relevance to the proposed system lies in its use of player data to support skill-based classification and matchmaking. This is important for pickleball communities because mismatched games can reduce both competitiveness and enjoyment. However, DUPR is mainly result-based rather than performance-based. It does not automatically analyze gameplay footage, identify technical weaknesses, classify shots, or explain the specific movement and tactical behaviors behind a player's rating.

2.2.10 Pickleheads

Pickleheads is a pickleball community platform that helps players find courts, join games, organize sessions, and connect with other players. It provides a court directory, open play schedules, local groups, game sign-ups, and skill-level-based game discovery. Pickleheads is relevant to the proposed system because it addresses the social and logistical side of pickleball participation, particularly court discovery, player connection, and organized play. These features are useful references for the proposed system's court reservation, queueing, and matchmaking components. However, Pickleheads is not primarily a performance monitoring or analytics platform. It does not automatically collect gameplay data, analyze player movement, classify shots, generate performance dashboards, or provide AI-based recommendations for technical improvement. Therefore, it is most useful as a model for community-building and game organization rather than as a direct model for computer-vision-based performance analysis.

2.2.11 Strengths and Weaknesses Table

Software / System	Strengths	Weaknesses
Zepp Tennis	Detailed swing data (speed, spin, sweet spot); stroke classification; tips and drills; community/leaderboards	Tennis-specific; proprietary hardware; hardware largely discontinued; no positioning or matchmaking
PlaySight SmartCourt	Automated multi-camera capture; auto-tracking, stats, and event tagging; supports pickleball; coach access	Requires fixed costly infrastructure; club/broadcast-oriented; no personal matchmaking
Arccos Caddie	Automatic shot tracking;	Golf-specific; proprietary

	strokes-gained analytics; AI-driven personalized recommendations; strong historical insight	hardware + subscription; no computer-vision analysis; no matchmaking
Padelytics	Computer-vision player and ball tracking from uploaded match video; court-positioning heatmaps; shot classification; movement/fitness metrics; improvement tips	Padel-specific; oriented toward clubs with a court-camera setup; no skill-based matchmaking
HomeCourt	Phone-based computer vision; automatic shot, movement, and skill tracking; personalized drills and recommendations; progress history; Apple Watch heart rate; coach sharing	Basketball-specific; mainly real-time; no racket-court positioning; social play rather than skill-based matchmaking
OnForm	Accessible mobile video analysis; slow-motion and frame-by-frame review; side-by-side comparison; voice-over feedback; coach-athlete sharing	General sports coaching tool; not pickleball-specific; no automatic shot classification, heatmaps, or matchmaking
Dartfish	Strong video annotation tools; movement comparison; tagging and sharing; useful for technical and tactical review	Not pickleball-specific; may require manual tagging; relies on coach or expert interpretation; no built-in matchmaking
CoachNow	Video and image analysis; coach feedback tools; cloud storage; side-by-side comparisons; AI skeleton tracking for movement analysis	Coaching-focused rather than match analytics-focused; not pickleball-specific; no rally, shot, court-zone, or matchmaking analysis
DUPR	Pickleball-specific rating system; supports fairer matches, tournaments, and leagues; useful for skill-based classification	Result-based rather than performance-based; no video analysis, shot classification, movement tracking, or technical feedback

Pickleheads	Court finder; game scheduling; open play organization; player connection; supports community and skill-level-based game discovery	Not a performance analytics system; no automatic gameplay data collection, computer-vision analysis, dashboards, or training recommendations
Proposed System	Integrates computer-vision gameplay and movement analysis, automatic shot classification, positioning/court-coverage analysis, skill and play-style assessment, matchmaking, personalized recommendations.	Integrates existing technologies;

2.2.12 Synthesis

The comparison reveals that every current system provides only a portion of what the proposed system intends to integrate. HomeCourt is the most similar in idea, showing a complete personal-informatics cycle with computer-vision capture, analytics, and tailored suggestions on a regular phone, though it is designed for basketball and primarily functions in real-time without analyzing court positioning. Padelytics, focusing on the racket sport most similar to pickleball, demonstrates that computer vision can monitor players, categorize shots, and create court-positioning heatmaps from uploaded match videos, but it is tailored specifically to padel and club-centered. PlaySight demonstrates that automated, multi-angle gameplay capture along with positioning data and coach access is achievable and even accommodates pickleball, though it relies on expensive fixed infrastructure. Zepp Tennis offers advanced sensor-driven swing metrics, although it is tailored for tennis and relies on specific hardware. Arccos illustrates how automatic data gathering can be combined with strokes-gained analysis and AI-based personalized suggestions, but it depends on sensors instead of computer vision and is limited to golf. The proposed solution will be informed by their unique strengths, accessible CV tracking,

positioning analysis, shot classification, swing metrics, historical analytics, and tailored recommendations, while their common limitation, the lack of a single platform that merges all these aspects along with skill-based matchmaking for pickleball within one personal-informatics loop, highlights the research gap that the suggested system aims to fill.

2.2.13 Features Table

Feature	Home Court	Padelytics	PlaySight SmartCourt	Zepp Tennis	Arccos Caddie	On Form	Dart fish	Coach Now	DUPR	Pickle heads	Proposed System
Computer Vision Analysis	✓	✓	✓	✗	✗	△	△	✓	✗	✗	✓
Shot Classification	✓	✓	✓	✓	✓	✗	△	✗	✗	✗	✓
Positioning / Court Coverage	✗	✓	✓	✗	✗	✗	△	✗	✗	✗	✓
Skill & Play-Style Assessment	✓	✗	✗	✗	✓	△	△	△	✓	✗	✓
Matchmaking	✗	✗	✗	✗	✗	✗	✗	✗	△	✓	✓
Personalized Recommendations	✓	✓	✗	✓	✓	△	△	△	✗	✗	✓
Pickleball-Specific	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓	✓

Legend: ✓ = Feature is clearly supported; ✗ = Feature is not supported; △ = Feature is partially supported, manual, coach-driven, or indirectly supported

Figure 2.2.8.1 Features table regarding availability of different functionalities per system

The Features table presents an overview of selected existing systems in relation to the proposed personal informatics system for pickleball players. The systems were evaluated based on key features relevant to sports performance monitoring, including computer vision analysis, shot classification, positioning and court coverage, skill and play-style assessment,

matchmaking, personalized recommendations, and pickleball-specific functionality. These criteria were selected to determine the extent to which existing systems support automated data collection, performance reflection, and data-driven improvement.

The comparison shows that several existing systems provide strong capabilities in specific areas of sports performance analysis. HomeCourt, Padelytics, and PlaySight SmartCourt demonstrate the use of computer vision and automated tracking for performance monitoring. However, these systems are primarily designed for other sports or require specific infrastructure, which may limit their accessibility and direct applicability to pickleball players. Meanwhile, systems such as OnForm, Dartfish, and CoachNow provide video-based coaching and review tools, but their functions are generally broad, coach-driven, or dependent on manual interpretation rather than fully automated pickleball performance analysis.

The table also highlights that DUPR and Pickleheads are more closely related to pickleball in terms of rating, player connection, and community participation. DUPR supports skill-based classification through player ratings, while Pickleheads assists players in finding courts, joining games, and organizing play sessions. However, these systems do not provide detailed gameplay analytics such as shot classification, player movement tracking, court-positioning analysis, or personalized technical recommendations.

While existing systems offer valuable individual features, none of them fully integrate computer vision analysis, shot classification, court-positioning assessment, skill and play-style evaluation, matchmaking, personalized recommendations, and pickleball-specific support within a single

platform. This reveals a functional gap in existing sports performance systems and supports the need for the proposed system, which aims to provide an integrated personal informatics solution specifically designed to help pickleball players monitor, assess, and improve their performance through objective and data-driven feedback.

2.3. Review of Related Methodologies

2.3.1 Scrum

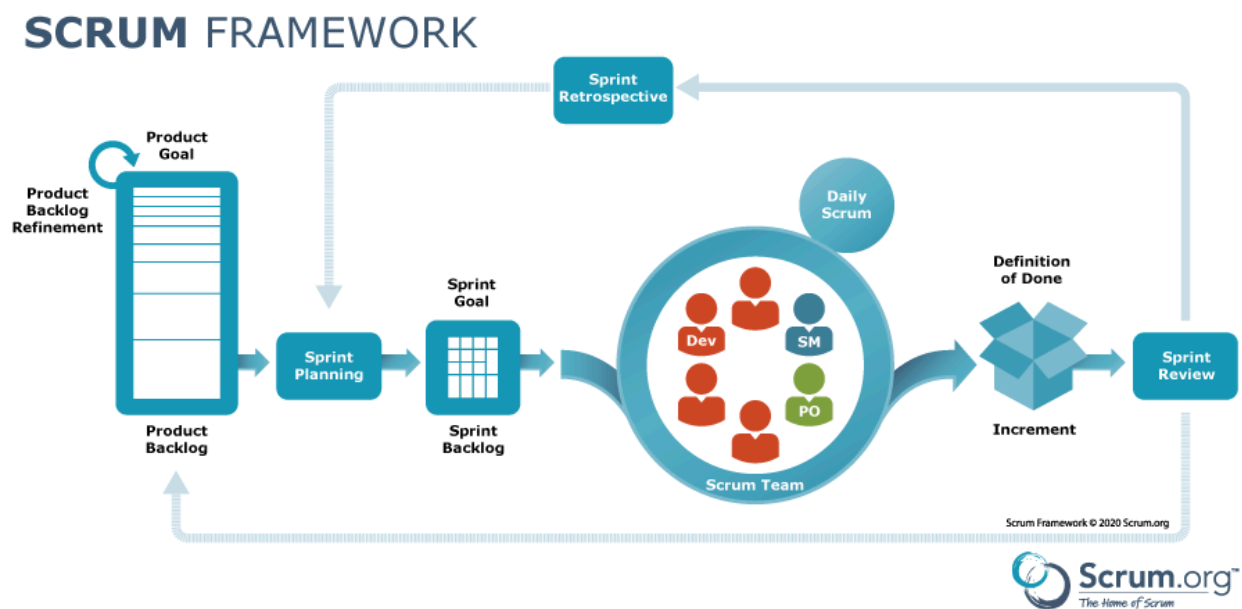


Figure 2.3.1.1 Scrum Framework

Scrum is one of the most popular Agile methodologies for running software projects, and it tends to shine when the work is complicated and the requirements are bound to change along the way (Schwaber & Sutherland, 2020). Instead of trying to map out the entire project before any code is written, Scrum splits the work into short, repeating cycles called sprints, usually two to four weeks each, with the team aiming to finish a working slice of the system by the end of every one. That rhythm fits the proposed pickleball system nicely, since its bigger parts are core

performance monitoring, skill assessment, and gameplay context support can be built and polished one at a time rather than all at once.

Scrum runs on three roles. The Product Owner decides what the team tackles next and keeps the product backlog, the ordered list of features and fixes, up to date. The Scrum Master looks after the process itself, making sure the team sticks to Scrum and clearing out whatever is getting in the way. Everyone else makes up the Development Team, the people who actually build the system and turn backlog items into a working increment each cycle (Schwaber & Sutherland, 2020). At the start of each sprint, the team picks a realistic batch of items from the backlog and commits to getting them done, which keeps everyone focused on a clear goal instead of an open-ended pile of work. And because each sprint produces something that actually works, the people who will use the system later such as the players and coaches can see where things are heading early on and weigh in, rather than waiting until the very end to find out the team built the wrong thing.

A handful of regular meetings hold the whole thing together. Sprint planning sets up what the cycle will cover, the daily stand-up is a quick check-in for sharing progress and raising blockers, and each sprint wraps with a review to show what was built and a retrospective to figure out what to do better next time (Schwaber & Sutherland, 2020). The product backlog, sprint backlog, and increment round it out as shared reference points, so anyone in the team or its stakeholders can tell at a glance what is being worked on and how far along it is.

The real draw, though, is how well Scrum handles change. Requirements almost never stay put, and Agile methods like Scrum were built with exactly that in mind (Beck et al., 2001). Since this project leans on existing computer vision instead of building new models, the team will not

really know what the system can promise until it sees how those tools handle shot classification, movement tracking, positioning, and exertion on actual gameplay. Scrum's iterative, feedback-driven loop makes it easy to work those findings back into the project as they come up, which is a big part of why it is the methodology this study adopts.

2.3.2 DevOps

DevOps is less a step-by-step process and more a culture and a set of practices that bring software development and IT operations together, all aimed at shipping changes quickly, reliably, and without much friction (Kim et al., 2016). It leans hard on automation and on continuous integration and delivery, so that new code can be tested and released in small, frequent batches instead of big, occasional ones. In practice it usually sits on top of an Agile method rather than replacing it, since teams that already work in iterations adopt it to make their releases smoother and their feedback loops faster (Humble & Farley, 2010).

For a system that will eventually be used and updated on a regular basis, that mindset is appealing, because it keeps a product stable while still leaving room for constant improvement. The catch, at least for a capstone, is that DevOps assumes an operational side that a student project usually does not have. It relies on build pipelines, automated testing setups, and ongoing deployment to live environments, which is far more than is needed just to develop and demonstrate a prototype. DevOps tends to pay off most once a system is already running in production and being maintained, rather than during the initial build, which is why it fits this project better as a future direction than as the framework guiding the work now.

2.3.3 Waterfall

Waterfall is the classic, linear approach, where a project moves through a fixed sequence of phases such as requirements, design, implementation, testing, and deployment and each one has to be finished before the next begins (Royce, 1970). It rests on heavy planning right at the start and works best when the requirements are well understood and unlikely to move. Its biggest strengths are clarity and predictability: everyone knows what is being built and in what order, and the thorough documentation it produces can make a project easy to follow.

That same rigidity is exactly where Waterfall runs into trouble for this project. The requirements here are expected to shift as the team learns how the computer vision actually behaves in real gameplay, and Waterfall makes those late changes slow and costly to absorb. Because feedback only shows up at the end of each phase, a problem introduced early can also go unnoticed until much later, when it is far harder to fix. For an exploratory system built around integrating existing technologies, that lack of flexibility is a serious drawback.

2.3.2 Comparison of Methodologies

Feature	Scrum	DevOps	Waterfall
Project structure	Iterative and incremental	Continuous, built on automated delivery	Linear and sequential
Planning	Light upfront, adaptive each sprint	Continuous, focused on delivery pipelines	Extensive upfront, detailed specifications
Requirements	Evolving requirements are accommodated	Evolving, supported through continuous releases	Fixed and cannot be changed
Feedback	Continuous, every sprint	Continuous and automated	Gathered at the end of each phase

Risk management	Addressed throughout the project	Reduced through frequent, small releases	Mainly during the planning phase
Change management	Relatively easy to implement	Easy, with automated testing and deployment	Difficult and costly
Team collaboration	Close collaboration and communication	Tight development-and-operations collaboration	Individual work and handoffs
Project visibility	High, with regular increments	High, with continuous monitoring	Low until the end of each phase
Customer involvement	Encouraged throughout the project	Indirect, through fast release and feedback	Limited until the end of each phase
Tooling / infrastructure	Lightweight, process-focused	Heavy; needs CI/CD and automation setup	Lightweight, document-focused
Best suited for	Iterative builds with evolving requirements	Systems in continuous deployment and maintenance	Projects with fixed, well-known requirements

Figure 2.3.2.1 Comparison of Software Development Methodologies

2.4 Synthesis

2.4.1 Common Themes

The reviewed literature and related systems commonly emphasize the shift from subjective performance evaluation to objective, data-driven feedback. In the related concepts, personal informatics is presented as a process of collecting, integrating, reflecting on, and acting upon personal data for improvement. This theme is reflected in sports performance monitoring, where players are encouraged to move beyond memory-based assessment and instead use measurable information to identify strengths, weaknesses, and progress over time. Computer vision also appears as a recurring theme because it allows gameplay video to be transformed into useful data such as player movement, shot actions, positioning, and tactical patterns. These concepts support the

proposed system's focus on automatic data collection, visual feedback, and player reflection.

Another common theme is the importance of accessibility and usability in sports technologies. Several systems, such as HomeCourt, OnForm, Dartfish, and CoachNow, demonstrate how video-based tools can support performance review and coaching without relying entirely on manual observation. Other systems, such as Padelytics and PlaySight SmartCourt, show the potential of computer vision in tracking court movement, classifying actions, and generating performance statistics. Meanwhile, DUPR and Pickleheads highlight the social and community side of sports participation by supporting skill ratings, fairer play, court discovery, and player connection. Together, these systems show that performance improvement is not only about collecting data, but also about making that data understandable, actionable, and useful within the player's actual sports environment.

The reviewed methodologies also show the importance of flexibility and iterative development. Since the proposed system depends on integrating existing technologies, particularly computer vision, the final system requirements may change as the team tests how these tools perform in actual pickleball gameplay. Scrum is therefore presented as the most suitable methodology because it allows the system to be developed in stages, tested regularly, and refined through continuous feedback.

2.4.2 Problems Currently Addressed

The existing literature and systems currently address several important problems in sports performance monitoring. First, they address the problem of unreliable self-assessment by allowing athletes to review performance through collected data rather than relying only on memory, intuition, or casual feedback. Personal informatics systems help users track their own behavior and performance, while computer vision enables more consistent capture of movement, positioning, and gameplay actions. This is especially useful in sports like pickleball, where rallies happen quickly and players may not clearly remember their positioning, shot choices, or errors after a match.

Second, existing systems address the need for technical and tactical analysis. Systems such as HomeCourt, Padelytics, PlaySight SmartCourt, Zepp Tennis, and Arccos Caddie demonstrate that modern sports tools can classify actions, track performance history, generate statistics, and provide recommendations. Video analysis platforms such as OnForm, Dartfish, and CoachNow also support coaching and reflection by allowing users to review movement, compare performance, and receive feedback. These systems help athletes and coaches identify technical errors, tactical patterns, and areas for improvement.

Third, pickleball-related platforms address problems in player organization and fair competition. DUPR provides a rating system that helps classify players according to skill level, while Pickleheads helps players find courts, join games, and connect with other players. These systems address the social and logistical challenges of pickleball

participation, especially in matching players with suitable games and organizing local play.

Finally, the methodology review addresses the problem of managing uncertainty during development. Waterfall is shown to be too rigid for a project whose requirements may evolve, while DevOps is more appropriate for systems already in continuous deployment and maintenance. Scrum addresses the need for an adaptive development process by allowing the team to build, test, and refine the proposed system through repeated sprint cycles.

2.4.3 Gaps in Literature

Despite the strengths of the reviewed concepts and systems, several gaps remain. The first major gap is the lack of a single pickleball-specific system that integrates computer vision analysis, shot classification, positioning and court coverage, skill and play-style assessment, matchmaking, and personalized recommendations. Existing systems tend to focus on only one or two of these areas. For example, HomeCourt provides strong computer-vision-based basketball training, Padelytics offers padel-focused court and shot analytics, and PlaySight supports automated multi-camera capture but requires fixed infrastructure. These systems demonstrate useful features, but they are not fully designed around pickleball players and their specific gameplay needs.

A second gap is the limited availability of pickleball-specific performance analytics. While DUPR and Pickleheads are directly related to pickleball, they focus mainly on

ratings, community connection, and game organization rather than detailed gameplay analysis. They do not automatically analyze shot types, player movement, rally patterns, court positioning, or technical weaknesses. This shows that pickleball platforms currently support participation and competition more than performance diagnosis and improvement.

A third gap is the limited connection between performance data and actionable improvement. Some systems collect large amounts of data, while others provide coaching tools or recommendations, but few combine automatic data collection, visual analytics, reflection, and personalized action in one continuous personal informatics loop. For the proposed system, this gap is important because the goal is not only to record gameplay data, but also to help players understand what the data means and how they can improve future performance.

Lastly, there is a gap in accessible implementation for recreational and community-level pickleball players. Some systems require proprietary sensors, subscriptions, coach interpretation, or fixed court-camera infrastructure. These limitations may reduce accessibility for ordinary players, schools, clubs, and local communities. The proposed system aims to address this gap by integrating existing technologies into a more focused personal informatics solution for pickleball, combining performance monitoring with player assessment and matchmaking features.

CHAPTER III Methodology

3.1. Description

The Scrum methodology lines up with how this project naturally unfolds. The problem the study is tackling needs a system that can be built and tested in stages instead of delivered in one big drop, and Scrum's sprints let the team stand up the most important pieces first, like the core performance monitoring dashboard, while the rest comes together over time. It takes changing requirements in stride, which matters when so much depends on testing the integrated technologies, and its steady collaboration keeps the computer vision, analytics, and reporting work pulling in the same direction.

3.2.1 Scrum VS Waterfall

When compared to Waterfall, Scrum is the choice that aligns better with the requirements of this project. Waterfall is the simplest to abandon since it requires the requirements to be fixed from the beginning, which they are not in this case; a significant portion of the effort relies on observing the performance of computer vision during actual gameplay before the team can confidently determine what the system will provide. Its inflexibility and tendency to reserve feedback until the conclusion of each phase render it a poor fit for such uncertainty.

In contrast, Scrum allows the system to be built and tested in stages instead of being delivered in one big drop. This makes it more suitable for a project where the team needs to test the core performance monitoring dashboard first, evaluate how the computer vision performs, and adjust the next parts of the system based on the results.

3.2.2 Scrum VS DevOps

DevOps is more appealing, and there are many aspects to appreciate in its automation and ongoing delivery, but fundamentally it is a method for releasing and managing existing software. It also presumes the presence of build pipelines and live deployment configurations that a capstone focused on a prototype does not possess.

Compared to DevOps, Scrum is more practical as the current framework directing the team because it focuses on organizing development through sprints, collaboration, feedback, and changing requirements. DevOps functions most effectively when integrated with an Agile process, indicating that it is better viewed as an evolution for the system in the future rather than as the current framework directing the team.

3.2. Phases

The development of the proposed pickleball system will follow the Scrum framework, organized into seven two-week sprints for CAP-IT1. The sprint sequence is designed to build the system incrementally, prioritizing Core Performance Monitoring first, followed by Light Skill and Performance Assessment and Gameplay Context Support, in line with the MVP scope defined in Chapter 1.

3.2.1 Sprint 1: Project Setup and Video Upload Module

3.2.1.1 Description of the Phase

Sprint 1 will focus on establishing the technical foundation of the system. The team will set up the project repository, define the initial system architecture, and configure the development environment for both the front end and backend services. The primary

deliverable will be a working video upload interface that allows players to submit match footage along with basic match data, laying the groundwork for the computer vision pipeline to be integrated in later sprints.

3.2.1.2 Tools / Techniques / Models to be Used

- **Project management:** Jira for backlog, sprint board, and sprint planning.
- **Development:** Github for version control; Supabase for storage and authentication; front-end framework for the upload interface.
- **Design:** Figma for wireframes of the upload flow.
- **Technique:** User story mapping to break down the “upload match video” epic into backlog items

3.2.1.3 Inputs and Outputs

Input

Product backlog items for Sprint 1, system architecture requirements from Chapter 1 scope, UI wireframes

Output

Working video upload module, configured repository and development environment, updated sprint backlog, sprint review notes

3.2.1.4 Anticipated Risks and Mitigation Strategies

A possible risk is inconsistent handling of large video files during upload, which could cause failed submissions. The team plans to mitigate this by implementing chunked upload handling.

3.2.2 Sprint 2: Computer Vision Pipeline Integration

3.2.2.1 Description of the Phase

Sprint 2 will focus on integrating the core computer vision pipeline for post-match video processing. The team will implement rally segmentation and player tracking using the selected CV models, converting uploaded match footage into structure and analyzable events.

3.2.2.2 Tools / Techniques / Models to be Used

- **Computer vision:** YOLOv8 for object/player detection, ByteTrack for multi-object tracking, 12-point court keypoint detection, and OpenCV homography for bird's-eye court projection.
- **Backend:** Server-side batch processing workers, Supabase for storing processed outputs
- **Technique:** Sprint planning and backlog refinement carried over from Sprint 1

3.2.2.3 Inputs and Outputs

Inputs

Uploaded match videos and metadata from Sprint 1, selected CV model configurations

Outputs

Rally-segmented match data, player tracking data, stored processed video events

3.2.2.4 Anticipated Risks and Mitigation Strategies

Possible risk is degraded tracking accuracy caused by video quality or camera angles from user-submitted footage. The team plans to mitigate this by defining minimum video quality guidelines for uploads and testing the pipeline against a range of sample footage early in the sprint.

3.2.3 Sprint 3: Shot Classification and Court Analytics

3.2.3.1 Description of the Phase

Sprint 3 will build on the tracking data produced in Sprint 2 to classify common pickleball shots (dinks, drives, volleys, lobs) and generate court-based analytics such as heatmaps, kitchen-zone time, and court coverage.

3.2.3.2 Tools / Technique / Models to be used

- **Computer vision:** OpenCV and MediaPipe for movement and shot-pattern analysis, using the existing system.
- **Shot classification:** XGBoost, an existing, pre-built classification algorithm, trained on ball speed, trajectory arc height, and court-zone-at-contact features engineered by the team from Pickleball-Analytics' tracking output.
- **Data processing:** Server side-workers for heatmap and zone-time computation
- **Technique:** Sprint review with sample outputs validated against manually reviewed match clips

3.2.3.3 Inputs and Outputs

Input

Player tracking and rally segmented data from Sprint 2

Output

Shot classification labels, heatmaps, kitchen-zone time metrics, court coverage data

3.2.3.4 Anticipated Risks and Mitigation Strategies

A possible risk is misclassification of visually similar shots, which could reduce the reliability of analytics. The team plans to mitigate this by validating classification outputs against manually labeled sample clips and adjusting detection thresholds accordingly.

3.2.4 Sprint 4: Analytics Dashboard

3.2.4.1 Description of the Phase

Sprint 4 will focus on presenting the processed performance data through a player-facing analytics dashboard, supporting trend monitoring, historical vs current match comparison, and goal setting, consistent with the personal informatics reflection stage.

3.2.4.2 Tools / Technique / Models to be Used

- **Front end:** React/Next.js with charting library for visual analytics
- **Backend:** Supabase queries for aggregating historical performance data
- **Technique:** Usability review of dashboard wireframes with sample users before implementation

3.2.4.3 Inputs and Outputs

Inputs

Shot classification and court analytics data from Sprint 3, dashboard wireframes

Output

Working analytics dashboard, goal-setting feature, historical performance comparison view

3.2.4.4 Anticipated Risks and Mitigation Strategies

A possible risk is information overload from presenting too many metrics at once, reducing usability. The team plans to mitigate this by prioritizing key performance indicators in the default view and placing detailed metrics behind expendable sections.

3.2.5 Sprint 5: Skill Level and Play Style Estimation

3.2.5.1 Description of the Phase

Sprint 5 will implement the Light Skill and Performance Assessment component, estimating the player's skill level (beginner, intermediate, advanced) and identifying their dominant play styles (aggressive, defensive, balanced) based on accumulated performance data.

3.2.5.2 Tools / Techniques / Models to be used

- **Data analysis:** Rule-based or lightweight classification logic applied to aggregated shot and movement data
- **Backend:** Supabase for storing skill and play-style profiles
- **Technique:** Backlog refinement to define clear skill-level thresholds before implementation

3.2.5.3 Inputs and Outputs

Input

Aggregated performance and shot data from Sprints 2-4

Output

Player skill-level estimates, play-style classification, updated player profile

3.2.5.4 Anticipated Risks and Mitigation Strategies

A possible risk is inaccurate skill estimation for players with limited match history. The team plans to mitigate this by requiring a minimum number of analyzed matches before generating a skill estimate and clearly labeling estimates as provisional until more data is collected.

3.2.6 Sprint 6: Gameplay Context Support

3.2.6.1 Description of the Phase

Sprint 6 will implement gameplay context tagging, allowing performance data to be organized by activity type (practice, casual match, tournament, singles, doubles), and ensuring historical records remain consistent across contexts.

3.2.6.2 Tools / Techniques / Models to be used

- **Backend:** Metadata tagging schema in Supabase for match context
- **Front end:** UI Controls for players to label match context during upload
- **Technique:** Sprint planning to prioritize the most common context types first

3.2.6.3 Inputs and Outputs

Input

Video upload module from Sprint 1, dashboard from Sprint 4

Output

Context-tagged match records, filterable historical performance view by activity type

3.2.6.4 Anticipated Risks and Mitigation Strategies

A possible risk is players mislabeling or skipping context tags, reducing data consistency.

The team plans to mitigate this by making context selection a required step in the upload flow with sensible defaults

3.2.7 Sprint 7: Coach Access and Integration Testing

3.2.7.1 Description of the Phase

Sprint 7 will close out CAP-IT1 by implementing the cooperative access layer for coaches, including role-based access control, and conducting integration testing across all modules built during CAP-IT1 to ensure the systems functions cohesively.

3.2.7.2 Tools / Techniques / Models to be Used

- **Backend:** Supabase authentication and RBAC policies
- **Testing:** End-to-end integration testing across upload, CV pipeline, dashboard, and profile modules
- **Technique:** Final sprint review and retrospective for CAP-IT1

3.2.7.3 Inputs and Outputs

Input

All modules developed in Sprints 1-6

Output

Coach access module with RBAC, integrated and tested system covering 80% of proposed functionality, CAP-IT1 sprint review documentation

3.2.7.4 Anticipated Risks and Mitigation Strategies

A possible risk is unforeseen integration conflicts between modules in separate sprints (e.g., data format mismatches between the CV pipeline and dashboard). The researchers plan to mitigate this by conducting incremental integration checks at the end of each sprint rather than deferring all integration to Sprint 7.

3.3. Work Plan / Gantt Chart

[illegible]

Sprint 5: Skill Level, Play Style Estimation, and Player Matching														
Sprint 6: Gameplay Context Support, Find a Game, and Microchallenges														
80% Completion Check and Buffer														
Sprint 7: Coach Access, Integration Testing, and MVP Refinement														
RAG R1: Knowledge Base Preparation														
RAG R2: RAG Feedback / Live Audio Prototype														
System QA and Bug Fixing														
User Testing and Revisions														
Documentation														
Final Polish														

Figure 3.3.1 Project Gantt Chart

Figure 3.3.1 presents the work plan for the development of the proposed Pickleball Performance Analytics System from September to March. The project follows the Scrum framework, with development organized into seven two-week sprints. The schedule is designed to deliver the system incrementally, beginning with the technical foundation before progressing to the core performance monitoring features, player analytics, gameplay support, and final system integration.

Development begins in September with Sprint 1, which focuses on project setup, system architecture, development environment configuration, and the implementation of the video upload module. This is immediately followed by Sprint 2, where the computer vision pipeline is

integrated to support rally segmentation, player detection, and player tracking. Sprint 3 continues the development of the Core Performance Monitoring component by implementing shot classification and generating court analytics such as heatmaps, kitchen-zone metrics, and court coverage.

During October, the project shifts toward the player-facing features. Sprint 4 develops the analytics dashboard together with reflection and recommendation features that allow players to review their historical performance and monitor improvement over time. This is followed by Sprint 5, which implements lightweight skill level estimation, play style classification, and player matching based on accumulated performance data.

In November, Sprint 6 introduces gameplay context support, including match context tagging, the Find-a-Game feature, and microchallenges that encourage continuous player engagement. At the end of the month, an 80% completion check and development buffer is scheduled to verify that the minimum functionality required for Capstone 1 has been achieved while allowing sufficient time to resolve technical issues before the final integration phase.

Beginning in December, Sprint 7 focuses on implementing coach access through role-based access control (RBAC), completing system integration, and refining the minimum viable product (MVP). During the same period, two exploratory RAG development iterations are scheduled. RAG R1 is dedicated to preparing the retrieval knowledge base, while RAG R2 develops a prototype for retrieval-assisted feedback and live audio coaching. These activities are treated as enhancement features and are developed independently of the core MVP to ensure that the primary project objectives remain on schedule.

The remaining project timeline is allocated to system quality assurance and bug fixing, user testing and revisions, documentation, and final polishing. These activities provide sufficient time to validate system functionality, incorporate user feedback, complete the required project documentation, and prepare the system for the Capstone 2 presentation and final evaluation.

3.4 Technology Stack

The proposed system is built by integrating existing, proven computer vision and web technologies rather than developing new models from scratch. The core detection, tracking, and court-mapping layer is built on Pickleball-Analytics, which already combines these into a working pipeline; the table below breaks that pipeline into layers alongside the additional components the team is building on top of it (shot classification, audio feedback, backend, frontend).

Layer	Tools / Technologies	Role in the System
Object Detection	YOLOv8, via Pickleball-Analytics	Detects ball, player, and paddle bounding boxes per frame.
Multi-Object Tracking	ByteTrack / BoT-SORT	Assigns persistent IDs to the detected player and ball across frames so movement can be followed over time, including through brief occlusion. Extended by the team for single-player selection

Pose Estimation	MediaPipe Pose	Extracts player joint landmarks for movement-quality cues in the real-time framework
Court Mapping	OpenCV homography, 12-keypoint model	Projects pixel coordinates for the ball and player onto real-world court coordinates using calibrated reference points, enabling zone-based metrics (kitchen time, court coverage). Extended by the team for half-court, tripod-height framing.
Video Processing	OpenCV, FFmpeg	Frame extraction, resizing, and format handling for uploaded match footage prior to model inference.
Shot Classification	XGBoost (XGBClassifier Api) existing, pre-built classification	Classifies each detected contact as a shot type (dink, drive, lob, serve, volley) using speed, arc, and court-zone features derived from tracking output.
Batch Processing	Server-side workers	Runs the detection, tracking, and classification pipeline on uploaded videos outside the browser for reliability and consistent performance.
Audio Feedback	RAG over player history	Converts structured performance data into short spoken feedback for post-game summaries.

Backend / Storage	Supabase (Postgres, auth, storage)	Stores match videos, tracking output, player profiles, and generated analytics; handles authentication and role-based access.
Frontend	React / Next.js, charting library	Renders the player and coach dashboards, visual analytics, and historical trend views.
Project Management	Jira, Figma	Sprint planning and interface design, per the Scrum workflow in Chapter III.

3.5 Capstone 1 & 2 Targets

The development of the proposed pickleball personal informatics system will be divided into two major completion stages: Capstone 1 and Capstone 2. These stages will ensure that the project is developed, evaluated, and refined progressively. Capstone 1 will focus on completing at least 80% of the proposed system, particularly the major functional components needed for the minimum viable product. Capstone 2 will focus on completing the remaining components, conducting user acceptance testing, refining the system, and preparing the final implementation and documentation.

3.5.1 Capstone 1 Targets

Capstone 1 is expected to cover the main development phase of the system. By the end of Capstone 1, the project should reach approximately 80% completion. This means that the primary system modules must already be functional enough to demonstrate how the

proposed solution addresses the identified problems and opportunities related to pickleball performance tracking, self-reflection, and player improvement. The following components are expected to be completed and evaluated during Capstone 1:

Video Upload and Match Data Module

The system must allow players to upload recorded pickleball match videos and input basic match information such as match date, match type, gameplay context, and player notes. This component serves as the entry point of the system and provides the data needed for performance analysis.

Computer Vision Processing Pipeline

The system must be able to process uploaded match footage using existing computer vision technologies. This includes player detection, movement tracking, rally segmentation, and the extraction of basic gameplay events. The researchers will not develop new computer vision algorithms, but will integrate existing models and frameworks such as YOLOv8, ByteTrack, OpenCV, and MediaPipe.

Shot Classification and Court Analytics

The system must classify common pickleball shots such as dinks, drives, volleys, and lobs based on available video and movement data. It must also generate court-based analytics such as player positioning, heatmaps, kitchen-zone time, and court coverage. These outputs will support objective performance monitoring and allow players to better understand their strengths and weaknesses.

Analytics Dashboard

The system must present processed performance data through a player-facing dashboard. The dashboard should display key performance indicators, match summaries, court analytics, shot distribution, performance trends, and historical comparisons. This component will allow players to review their gameplay in a clear and organized manner.

Goal Setting, Reflection, and Recommendations

The system must support the personal informatics aspect of the study by allowing players to set goals, reflect on post-match performance, and receive basic recommendations based on collected data. These recommendations may include focus areas such as improving court coverage, increasing shot variety, maintaining better positioning, or spending more time near the kitchen line.

Skill Level and Play Style Estimation

The system must provide a light assessment of player characteristics based on accumulated performance data. This includes estimating whether a player is beginner, intermediate, or advanced, and identifying whether the player's dominant play style is aggressive, defensive, or balanced. These classifications will be treated as general estimates and not as official rankings.

Player Matching Support

The system must include basic player matching support based on skill level, play style, availability, and preferred gameplay type. This feature supports the opportunity for players to find suitable opponents or partners within the same pickleball community.

Gameplay Context Support

The system must organize performance records according to different gameplay contexts such as practice, casual matches, tournaments, leveling games, singles, doubles, wall practice, and ball machine practice. This allows players to compare their progress across different playing situations.

3.5.2 Capstone 2 Targets

Capstone 2 will cover the completion of the remaining components, system refinement, user acceptance testing, and final documentation. While Capstone 1 focuses on building the core system, Capstone 2 will focus on ensuring that the system is reliable, usable, and able to address the problems identified in the study. The following components are expected to be completed and evaluated during Capstone 2:

Coach Access and Role-Based Permissions

The system must provide a cooperative access layer where authorized individuals, such as coaches, can view player performance information. Role-based access control must be implemented to ensure that only authorized users can access player data. This feature supports guided improvement while maintaining user privacy and data control.

RAG and Live Audio Coaching Assistance

The RAG component may be used to retrieve coaching knowledge, drill suggestions, player history, and feedback templates to generate more grounded performance recommendations. The Live Audio Coaching Assistance feature may convert selected feedback into short audio cues during practice or training sessions. However, these features will remain supplementary and will not replace professional coaching, medical advice, officiating, or referee decision-making.

Because these features may increase the technical complexity of the project, they will only be pursued if the core system has already met the required level of completion and testing. If not completed within the project timeline, they may be discussed as future enhancements.

Full System Integration

All modules developed during Capstone 1 must be integrated into one cohesive system. This includes the upload module, computer vision pipeline, shot classification, court analytics, dashboard, goal setting, reflection, recommendations, skill assessment, player matching, gameplay context support, Find a Game, microchallenges, and coach access.

System Testing and Bug Fixing

The group must conduct technical testing to identify and resolve errors, data mismatches, broken workflows, and usability issues. Testing must cover the complete system flow, from video upload and processing to dashboard display and coach access.

User Acceptance Testing

User acceptance testing will be conducted with target users such as pickleball players and, when possible, coaches or experienced players. The purpose of this testing is to determine whether the system is understandable, useful, and relevant to the needs of its intended users. Feedback from user testing will be used to refine the dashboard, recommendations, reflection features, player matching, and other user-facing functions.

Final Refinement and Documentation

The remaining period of Capstone 2 will be used for final system improvements, interface polishing, documentation completion, evaluation write-ups, and preparation for the final defense. The group must also ensure that the final manuscript reflects the actual completed system and clearly discusses its scope, limitations, testing results, and recommendations for future development.

3.5.3 Data Requirements for Demonstration and Evaluation

To demonstrate that the developed system addresses the identified problems and opportunities, the group must prepare sufficient sample data for testing and simulation. This includes recorded pickleball videos, sample player profiles, sample match records, gameplay context labels, skill-level examples, play-style examples, availability schedules, sample game listings, and microchallenge records.

The data must be able to simulate different scenarios where the system may be used, such as a beginner reviewing a casual match, an intermediate player tracking improvement over time, a player looking for a suitable match partner, a coach reviewing a player's performance, and a player comparing performance across practice and competitive contexts.

Through these scenarios, the group will be able to demonstrate that the system supports automated data collection, performance reflection, actionable insights, player matching, and continuous improvement within the defined scope of the study.

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Appendix J

Preliminary Related Systems Inventory

This inventory was assembled during early related-systems research. The systems should be discussed in the Review of Related Systems only after their current features and availability have been verified through authoritative sources.

System	Primary Relevance	Original Source	Note
Zepp Tennis	Racket-sport swing and performance tracking	App-store listing	
PlaySight SmartCourt	Video capture, replay, and sports analytics	playsight.com	
Arccos Caddie	Sensor-based golf tracking and recommendations	Google Play listing	
Padelytics	Padel-focused analytics	padelytics.ai	

HomeCourt	Camera-based basketball practice and analytics	homecourt.ai
OnForm	Video analysis and coaching collaboration	onform.com
Dartfish	Video analysis and performance feedback	dartfish.com
CoachNow	Coach-athlete communication and feedback	coachnow.com
DUPR	Pickleball player rating and match-result platform	dupr.com
Pickleheads	Pickleball court, game, and community discovery	pickleheads.com

***Note.** The main reference list should contain the complete, verified citations for all literature, technologies, repositories, and systems cited in the paper. A reference dump should not be placed inside the appendices unless the institution explicitly requires a separate supplementary bibliography.*

CHAPTER I Research Background

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1.1. Introduction

The increasing adoption of digital technologies has transformed how individuals monitor, analyze, and improve their performance across various domains, including health, fitness, and sports. Advances in computer vision technologies, and artificial intelligence have enabled the automatic collection and analysis of large amounts of performance data, allowing athletes to gain objective insights that support continuous improvement. Rather than relying solely on subjective observations, athletes can now make informed decisions based on measurable evidence collected during training and competition.

Within sports science, these technologies have contributed to the emergence of personal informatics systems, which are designed to help individuals achieve personal goals through continuous data collection, reflection, and behavior change. Personal informatics emphasizes enabling users to understand their own performance by collecting relevant information, presenting meaningful insights, and encouraging actions that support long-term improvement. The process typically involves setting goals, automatically gathering performance data, reflecting on collected information, and adjusting training strategies based on the results.

As pickleball continues to gain popularity among recreational and competitive players, there is an increasing demand for accessible technologies that provide objective performance feedback. While existing solutions offer general fitness metrics or sport-specific analytics for other disciplines, few systems focus on the strategic and technical characteristics that define pickleball. This study proposes the development of a personal informatics system that integrates existing

computer vision technologies to automatically collect and analyze player performance data, provide meaningful performance insights, and support continuous improvement through data-driven decision-making.

1.2. Background of the Study

Pickleball is one of the fastest-growing sports in the world due to its accessibility, social nature, and ability to accommodate players of different ages and skill levels (Sports & Fitness Industry Association [SFIA], n.d.). Unlike many racket sports that rely heavily on physical strength and athleticism, pickleball places significant emphasis on strategy, positioning, consistency, communication, and decision-making (USA Pickleball, n.d.). The sport's smaller court, slower ball speed, and non-volley zone (commonly known as the "kitchen") encourage players to focus on shot placement, anticipation, and tactical play rather than power alone (USA Pickleball, n.d.; EBSCO Information Services, n.d.).

The sport's inclusive nature has contributed to its widespread adoption among recreational players, competitive athletes, students, and older adults. Research has noted that pickleball's lower physical demands compared to sports such as tennis and badminton make it more accessible across age groups and skill levels (MDPI, 2024). This unique characteristic allows beginners, seniors, and competitive players to participate within the same sporting community while pursuing different personal goals.

As participation continues to grow, players increasingly seek ways to improve their performance and better understand their gameplay. However, many pickleball players still rely on self-assessment, informal feedback from playing partners, or occasional coaching sessions to

evaluate their strengths and weaknesses. While coaches can provide valuable guidance, access to regular coaching may be limited by cost, availability, or scheduling constraints.

Recent advancements in computer vision have created new opportunities for sports performance monitoring. These technologies allow athletes to collect and analyze objective data regarding movement, physiological responses, and gameplay behavior. Despite these developments, many existing sports analytics systems focus primarily on professional athletes or are designed for sports such as tennis, basketball, and soccer, leaving an opportunity for solutions specifically tailored to the unique demands of pickleball.

1.3. Statement of the Problem

Despite the increasing availability of digital sports technologies, accessible systems that integrate objective gameplay collection, pickleball-specific performance analysis, community coordination, and collaborative coaching support remain limited. Pickleball players may use match videos, personal observations, informal feedback, and separate community platforms, but these sources are not always connected in a way that supports continuous and evidence-based player development.

Specifically, the study seeks to address the following problems:

P1. Limited objective in-game data collection and reliance on subjective performance evaluation

Pickleball players frequently evaluate their performance through memory, self-observation, or informal comments from coaches and playing partners. These methods may overlook important

gameplay details that occur rapidly during a rally, including player movement, court positioning, kitchen-line presence, court coverage, rally activity, and selected shot events. The limited availability of accessible and systematic in-game data collection consequently makes it difficult for players to establish an objective basis for evaluating their performance.

P2. Fragmented player development tracking and lack of pickleball-specific performance analytics

Player information may be distributed across recorded videos, personal notes, match results, coach feedback, and separate applications. When these records are not integrated into a unified performance history, players may find it difficult to identify recurring strengths, weaknesses, and development patterns across multiple sessions. Furthermore, general sports analytics systems may not sufficiently represent pickleball-specific concepts such as kitchen-line presence, dink activity, court-zone occupancy, rally structure, doubles positioning, and strategic shot selection. This limits the ability of players to interpret their data and monitor improvement over time.

P3. Difficulty finding suitable games and partners and coordinating community-based play

Recreational and developing players often depend on personal networks, social-media groups, or chance encounters when looking for opponents, partners, training groups, or open-play sessions. Information concerning player availability, game format, preferred play type, existing skill information, and community schedules may be communicated through separate or informal channels. This may lead to unsuitable pairings, missed playing opportunities, and inefficient coordination among players, court owners, and community organizers.

P4. Limited timely, contextual, and collaborative coaching support

Access to regular coaching may be constrained by cost, availability, and scheduling. Even when coaches, teammates, or training partners are available, feedback may rely on memory or observations that are not directly connected to the player's recorded performance information. Players may therefore lack a structured mechanism for sharing reports, receiving contextual feedback, reviewing training recommendations, and obtaining supplementary guidance during appropriate pauses in a training session.

Given these problems, there is a need for an integrated pickleball personal informatics system that can capture objective gameplay information, organize performance records, generate pickleball-specific insights, support community coordination, and facilitate evidence-based collaboration among players, coaches, teammates, and court-community organizers.

1.4. Conceptual and Solutions Framework

1.4.1 PPIF Stage Matrix

Stage	Primary Goal	Key Actors	Technologies/ Inputs	Outputs	Example Metrics
1 - Preparation	Define performance targets and focus areas	Players, Coaches	Goal - Setting; Consultation	Performance objectives, list of KPIs to track	Dink consistency, court position targets, shot selection priorities
2 - Collection	Capture objective performance data during play	Players , Sensors	Computer Vision; Motion Tracking	Raw Sensor stream, tagged video clips	Court position, rally structure, shot selection
3 - Integration	Unify all data into a single athlete profile	System , players, coaches	Gameplay data; Context data; Historical records	Athlete Performance Profile, shot maps, heat maps, trend reports	Match history, shot selection, tactical indicators, historical strengths and weaknesses,

4 - Reflection	Analyze processed data to surface insights	Players, Coaches	Dashboards; Visual Analytics; Match Summary	Evidenced-Based feedback, annotated match reviews, training plan input	Dink success-rate, unforced errors, kitchen line time movement type, rally duration,
5 - Action	Convert insights into individualized interventions	Players, Coaches	Reflection outputs; Recommendation engine	Tailored drill plans, tactical adjustments, workload guidance, practice targets for next	Technical session targets, position reminders, workload limits, next session goals

Figure 1.4.1 PPIF Stage Matrix

Note. Based on Li, Dey, and Forlizzi's (2010) Personal Informatics model, adapted for pickleball performance.

The study utilizes the Pickleball Performance Informatics Framework as the conceptual foundation for the four proposed solution frameworks. Computer-vision functions may be delivered through two complementary processing modes: on-demand session processing and web-based post-game processing. The PPIF is a specialized framework designed to support continuous skill development among pickleball players by organizing the system around performance goals, data collection, analytics, reflection, and training action. It is adapted from Li, Dey, and Forlizzi's Personal Informatics model, which outlines five stages of technology-supported self-monitoring: Preparation, Collection, Integration, Reflection, and Action. Within this study, these stages are applied to two possible computer vision implementations: a real-time iPhone-based computer vision framework and a web-based post-game video analytics framework.

In the Preparation Stage, players and coaches identify the specific performance areas that the system must monitor. These may include dink consistency, court positioning, shot selection,

kitchen-line presence, rally duration, movement efficiency, and tactical decision-making. This stage ensures that the system is designed as a performance improvement tool rather than merely a game-recording or scorekeeping application.

In the Collection Stage, the two proposed frameworks differ in how data is captured. The real-time framework collects live gameplay data through an iPhone camera and, when needed, an edge-processing device that can analyze player movement, court zones, rally activity, and pose-related cues during play. Meanwhile, the web-based framework collects recorded match videos uploaded by players or coaches after the game. In both cases, computer vision serves as the primary sensing method, while optional wearable or self-reported data may provide additional information such as fatigue, perceived exertion, recovery condition, or training context.

In the Integration Stage, the system combines gameplay data, visual tracking outputs, match context, and historical records into a unified athlete performance profile. For the real-time framework, this integration supports immediate session-level feedback such as live court position, kitchen time, and rally structure. For the web-based framework, integration supports deeper post-game analysis such as heatmaps, shot maps, rally segmentation, historical trend reports, and player development patterns.

In the Reflection Stage, players and coaches interpret the processed data through dashboards, visual analytics, annotated clips, and performance summaries. The real-time framework supports immediate reflection during or shortly after training, allowing coaches to give timely feedback. The web-based framework supports more detailed post-match review, where players and coaches

can examine performance trends, repeated errors, tactical patterns, and areas for technical improvement.

In the Action Stage, the insights generated by the system are translated into individualized training interventions. These may include technical drills, tactical adjustments, positioning reminders, workload guidance, and next-session performance targets. Through this stage, the framework emphasizes evidence-based coaching decisions rather than relying only on memory, manual observation, or subjective judgment.

Problem	Solution framework	Primary functions	Expected outputs
P1. Limited objective in-game data collection and reliance on subjective evaluation	S1. Objective Gameplay Data Collection Framework	Video capture, player detection and tracking, court calibration, rally segmentation, court-zone analysis, selected shot-event recognition	Structured gameplay events, movement paths, kitchen-zone data, rally records, court-coverage information
P2. Fragmented development tracking and lack of pickleball-specific analytics	S2. Player Personal Informatics and Analytics Framework	Player profiles, goal setting, historical integration, dashboards, post-game reports, reflection, progress monitoring and recommendations	Unified performance profile, visual reports, historical trends, goals, reflections and development recommendations

P3. Difficulty finding suitable games and partners and coordinating community-based play	S3. Court and Community Coordination Framework	Game listings, open-play queues, availability matching, session coordination, partner and opponent suggestions, community announcements, court-owner administration, and access management	Suitable game suggestions, session listings, queue records, participation records, community announcements, and court-community information community information
P4. Limited timely, contextual and collaborative coaching support	S4. Coach and Team Collaboration Framework	Authorized report access, coach annotations, team review, training feedback, shared development plans and player-initiated audio guidance	Coach feedback, shared reports, training plans, team-progress records and contextual audio cues

Figure 1.4.2 Problem-Solution Matrix

S1. Objective Gameplay Data Collection Framework

The Objective Gameplay Data Collection Framework addresses the fragmented availability of systematic and objective in-game performance data. It utilizes video captured through an accessible camera device, such as a smartphone, together with existing computer-vision technologies to process observable pickleball gameplay.

The framework focuses on feasible performance information such as player detection and tracking, movement paths, court calibration, court-zone occupancy, kitchen-line presence, rally segmentation, court coverage, and selected shot-event classifications. Captured information is converted into structured gameplay records that can be used by the Player Personal Informatics and Analytics Framework.

The framework does not provide automated officiating, official in-or-out decisions, or guaranteed ball-trajectory analysis. Outputs that cannot be detected reliably through the selected camera configuration must be excluded from the core performance metrics or marked according to their level of confidence.

S2. Player Personal Informatics and Analytics Framework

The Player Personal Informatics and Analytics Framework organizes gameplay information into a unified player-centered performance record. It supports the preparation, integration, reflection, and action stages of the Pickleball Performance Informatics Framework by enabling players to establish performance goals, review post-game reports, compare historical and current results, record reflections, and monitor development over time.

The framework integrates structured gameplay events, match context, player goals, historical records, and authorized coach feedback. It presents pickleball-specific information through dashboards, visual summaries, heatmaps, movement reports, rally summaries, and performance trends. Based on the available information, the system may identify priority areas and provide supplementary recommendations for future training sessions.

S3. Court and Community Coordination Framework

The Court and Community Coordination Framework addresses difficulties in finding suitable games, partners, training groups, and community-based playing opportunities. It supports players, court owners, and community organizers by providing centralized game listings, open-play queues, player availability records, session coordination, and partner or opponent suggestions.

The framework recommends suitable playing opportunities using compatibility factors supplied by users, including availability, preferred game format, playing experience, preferred level of competition, training objectives, play style, and location. These factors are intended to improve game suitability and coordination without generating or assigning an official player rating.

Court-owner and community-administration functions are incorporated into this framework because court owners and community organizers are responsible for managing game listings, open-play schedules, player participation, queues, announcements, community records, user roles, and access permissions. The framework centralizes this information to reduce dependence on fragmented social-media posts, private messages, and informal communication channels.

The framework does not include comprehensive facility management, payment processing, financial reporting, tournament-bracket management, or automated competitive ranking.

S4. Coach and Team Collaboration Framework

The Coach and Team Collaboration Framework supports cooperative performance improvement by allowing authorized coaches, teammates, and training partners to review relevant player-performance information. It provides role-based access to reports, annotated gameplay

records, coach comments, training recommendations, shared goals, and team-progress information.

The framework may also include an On-Demand Contextual Audio Coaching Submodule. This feature must be activated by the player after a rally or during an appropriate training break. It may use validated gameplay events, player history, rule-based feedback templates, and approved coaching knowledge to generate a short supplementary audio cue.

The audio feature must not be described as continuous real-time coaching. It does not replace professional coaching and must not provide medical guidance, injury assessment, automated officiating, or advice based on uncertain computer-vision outputs.

1.5 Study Objectives

The study aims to develop an integrated personal informatics system for pickleball players that utilizes existing computer vision and information technologies to support objective gameplay monitoring, player development tracking, community-based play coordination, and collaborative coaching.

Specifically, the study aims to:

1. Develop an Objective Gameplay Data Collection Framework that uses existing computer vision technologies to capture and organize feasible pickleball gameplay indicators, including player movement, court positioning, kitchen-zone presence, rally activity, court coverage, and selected shot events.
2. Develop a Player Personal Informatics and Analytics Framework that integrates current and historical gameplay information into a unified player profile and presents

performance reports, visual analytics, goals, reflections, progress trends, and supplementary development recommendations.

3. Develop a Court and Community Coordination Framework that supports centralized game listings, open-play queues, player availability, session coordination, partner and opponent suggestions, community announcements, participation records, and basic access management for court owners and community organizers.
4. Develop a Coach and Team Collaboration Framework that allows authorized coaches, teammates, and training partners to review relevant player-performance information, provide structured feedback, share development plans, monitor collaborative progress, and deliver player-initiated contextual audio guidance during appropriate training pauses.

1.6. Significance of the Study

This study contributes to the growing application of personal informatics technologies within sports performance monitoring, particularly in the context of pickleball. By utilizing existing computer vision technologies, the proposed system seeks to provide players with objective and meaningful performance information that supports continuous improvement.

The proposed system enables players to better understand their strengths, weaknesses, and gameplay patterns through objective performance analytics. By providing historical performance records and personalized recommendations, players can make more informed decisions regarding their training and monitor their progress over time.

The system also provides coaches with structured performance information that can supplement their observations during training and matches. Access to objective gameplay data allows

coaches to provide more targeted feedback and develop training strategies based on measurable evidence.

Additionally, the study may assist clubs and organizations in promoting player development and balanced matchmaking by introducing technology-assisted performance monitoring that complements existing coaching and training practices.

Lastly, the findings and system design may serve as a foundation for future research and development involving sports performance monitoring, artificial intelligence, and personal informatics applications in other sports or athletic domains.

1.7. Scope and Limitations

This study focuses on the design and development of a personal informatics system for pickleball players that supports continuous performance improvement through automated data collection, performance reflection, and actionable insights. The system is intended primarily for individual players and extends to a cooperative layer where authorized individuals, such as coaches, may access player performance information to provide informed guidance. The study integrates existing computer vision technologies and does not involve the development of new artificial intelligence or machine learning models.

1.7.1 Core Performance Monitoring

The first component is Core Performance Monitoring, which utilizes existing computer vision technologies to analyze pickleball gameplay. The system automatically tracks player movement, classifies common pickleball shots such as dinks, drives, volleys, and

lobs, and evaluates player positioning and court coverage throughout a match. Performance data are presented through an analytics dashboard that enables players to review their strengths and weaknesses, monitor performance trends over time, and compare historical and current gameplay. Consistent with the principles of personal informatics, the system also supports goal setting, personalized performance insights, recommendations, and a continuous feedback loop that encourages players to reflect on collected data and adjust their training strategies accordingly.

1.7.2 Light Skill and Performance Assessment

The second component is Light Skill and Performance Assessment, which provides a general evaluation of player characteristics using collected gameplay data. The system estimates player skill level (e.g., beginner, intermediate, or advanced) and identifies dominant play styles such as aggressive, defensive, or balanced. To encourage participation and interaction within pickleball communities, the system will allow players to share their own performance statistics, analytics, and progress to other players and coaches. In addition, recorded matches are automatically linked to corresponding performance analytics, allowing players to review post-match insights and understand how performance data may be applied to future gameplay and training.

1.7.3 Gameplay Context Support

The third component covers Gameplay Context Support, wherein the system is capable of analyzing and organizing performance data collected from different types of pickleball activities. These include practice or training sessions, casual matches, tournaments, leveling games, singles matches, doubles matches, and individual practice using a wall or

ball machine. Regardless of the gameplay context, the system maintains historical performance records that allow players to monitor long-term progress and evaluate improvements over time.

1.7.4 Post-Game Audio Coaching Assistance

The fourth component is Post-Game Audio Coaching Assistance, which allows players to activate a coaching session through the mobile application during training or practice play. When enabled, the outputs such as player position, kitchen-line presence, court coverage, rally timing, movement patterns, and selected pose-related cues to generate short audio feedback for the player.

The system extends audio feedback to the web-based, non-real-time framework. After a match video is uploaded and processed, the computer vision pipeline produces structured performance data including shot classifications, shot accuracy, and rally outcomes for the full match. This processed data is then converted into a short, spoken audio summary that the player can listen to instead of, or alongside, the visual dashboard.

Unlike the live coaching cues, which are limited to brief, low-latency prompts during play, post-game audio feedback can draw on the complete match dataset, allowing for more evaluative statements. Example outputs include summaries such as shot type distribution and accuracy for the match, and dominant error patterns (for example, a low drive success rate compared to dinks).

This post-game audio feature uses the text-to-speech and feedback-generation service, but is triggered once per processed match rather than continuously during play, and pulls

from the complete post-match performance profile rather than a rolling window of recent frames. Players may choose to enable or disable this feature per upload, and the generated audio is stored alongside the corresponding match record so it can be replayed from the dashboard.

The feature does not replace a professional coach and does not provide officiating, medical, or injury-related decisions. Instead, it serves as an assistive feedback layer that supports player reflection and immediate training adjustment during live or semi-live practice sessions.

1.7.5 Limitations

This study is limited to the collection, analysis, visualization, and presentation of performance-related information. The system is designed to support player self-reflection and decision-making through objective data but does not replace professional coaching, sports training, officiating, or medical expertise. Performance recommendations generated by the system should be considered supplementary guidance rather than definitive prescriptions for player development.

Furthermore, the study is limited to integrating existing computer vision, hardware technologies, and artificial intelligence technologies. The researchers will not develop new computer vision algorithms, machine learning models, or hardware as part of the study.

The following functionalities are outside the scope of the proposed system:

1. Court booking and reservation management.
2. Payment processing and financial transactions related to court usage.
3. Court availability management and administrative functions for court owners or operators.
4. Tournament, league, bracket, or event scheduling and management.
5. Comprehensive team, club, or organization management functions.
6. Medical diagnosis, injury treatment, rehabilitation planning, or clinical health monitoring.
7. Automated officiating, referee decision-making, or official in/out detection.
8. Full replacement of professional coaching, since the live audio coaching feature is limited to supplementary feedback based on detectable computer vision outputs.
9. Development of new computer vision algorithms, machine learning models, or custom hardware. The study is limited to integrating existing technologies, frameworks, and services.
10. The system does not calculate or assign an official skill rating. Instead it uses player-provided preferences, playing experience, availability, and session objectives to identify potentially suitable games, partners, and training groups.

1.1. Introduction

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The increasing adoption of digital technologies has transformed how individuals monitor, analyze, and improve their performance across various domains, including health, fitness, and sports. Advances in computer vision technologies, and artificial intelligence have enabled the automatic collection and analysis of large amounts of performance data, allowing athletes to gain objective insights that support continuous improvement. Rather than relying solely on subjective observations, athletes can now make informed decisions based on measurable evidence collected during training and competition.

Within sports science, these technologies have contributed to the emergence of personal informatics systems, which are designed to help individuals achieve personal goals through continuous data collection, reflection, and behavior change. Personal informatics emphasizes enabling users to understand their own performance by collecting relevant information, presenting meaningful insights, and encouraging actions that support long-term improvement. The process typically involves setting goals, automatically gathering performance data, reflecting on collected information, and adjusting training strategies based on the results.

As pickleball continues to gain popularity among recreational and competitive players, there is an increasing demand for accessible technologies that provide objective performance feedback. While existing solutions offer general fitness metrics or sport-specific analytics for other disciplines, few systems focus on the strategic and technical characteristics that define pickleball. This study proposes the development of a personal informatics system that integrates existing computer vision technologies to automatically collect and analyze player performance data,

provide meaningful performance insights, and support continuous improvement through data-driven decision-making.

1.2 Background of the Study

1.2. Background of the Study

Pickleball is one of the fastest-growing sports in the world due to its accessibility, social nature, and ability to accommodate players of different ages and skill levels (Sports & Fitness Industry Association [SFIA], n.d.). Unlike many racket sports that rely heavily on physical strength and athleticism, pickleball places significant emphasis on strategy, positioning, consistency, communication, and decision-making (USA Pickleball, n.d.). The sport's smaller court, slower ball speed, and non-volley zone (commonly known as the "kitchen") encourage players to focus on shot placement, anticipation, and tactical play rather than power alone (USA Pickleball, n.d.; EBSCO Information Services, n.d.).

The sport's inclusive nature has contributed to its widespread adoption among recreational players, competitive athletes, students, and older adults. Research has noted that pickleball's lower physical demands compared to sports such as tennis and badminton make it more accessible across age groups and skill levels (MDPI, 2024). This unique characteristic allows beginners, seniors, and competitive players to participate within the same sporting community while pursuing different personal goals.

As participation continues to grow, players increasingly seek ways to improve their performance and better understand their gameplay. However, many pickleball players still rely on self-assessment, informal feedback from playing partners, or occasional coaching sessions to evaluate their strengths and weaknesses. While coaches can provide valuable guidance, access to regular coaching may be limited by cost, availability, or scheduling constraints.

Recent advancements in computer vision have created new opportunities for sports performance monitoring. These technologies allow athletes to collect and analyze objective data regarding movement, physiological responses, and gameplay behavior. Despite these developments, many existing sports analytics systems focus primarily on professional athletes or are designed for sports such as tennis, basketball, and soccer, leaving an opportunity for solutions specifically tailored to the unique demands of pickleball.

1.3 Statement of the Problem

1.3. Statement of the Problem

Despite the increasing availability of digital sports technologies, accessible systems that integrate objective gameplay collection, pickleball-specific performance analysis, community coordination, and collaborative coaching support remain limited. Pickleball players may use match videos, personal observations, informal feedback, and separate community platforms, but these sources are not always connected in a way that supports continuous and evidence-based player development.

Specifically, the study seeks to address the following problems:

P1. Limited objective in-game data collection and reliance on subjective performance evaluation

Pickleball players frequently evaluate their performance through memory, self-observation, or informal comments from coaches and playing partners. These methods may overlook important gameplay details that occur rapidly during a rally, including player movement, court positioning, kitchen-line presence, court coverage, rally activity, and selected shot events. The limited availability of accessible and systematic in-game data collection consequently makes it difficult for players to establish an objective basis for evaluating their performance.

P2. Fragmented player development tracking and lack of pickleball-specific performance analytics

Player information may be distributed across recorded videos, personal notes, match results, coach feedback, and separate applications. When these records are not integrated into a unified performance history, players may find it difficult to identify recurring strengths, weaknesses, and

development patterns across multiple sessions. Furthermore, general sports analytics systems may not sufficiently represent pickleball-specific concepts such as kitchen-line presence, dink activity, court-zone occupancy, rally structure, doubles positioning, and strategic shot selection. This limits the ability of players to interpret their data and monitor improvement over time.

P3. Difficulty finding suitable games and partners and coordinating community-based play

Recreational and developing players often depend on personal networks, social-media groups, or chance encounters when looking for opponents, partners, training groups, or open-play sessions. Information concerning player availability, game format, preferred play type, existing skill information, and community schedules may be communicated through separate or informal channels. This may lead to unsuitable pairings, missed playing opportunities, and inefficient coordination among players, court owners, and community organizers.

P4. Limited timely, contextual, and collaborative coaching support

Access to regular coaching may be constrained by cost, availability, and scheduling. Even when coaches, teammates, or training partners are available, feedback may rely on memory or observations that are not directly connected to the player's recorded performance information. Players may therefore lack a structured mechanism for sharing reports, receiving contextual feedback, reviewing training recommendations, and obtaining supplementary guidance during appropriate pauses in a training session.

Given these problems, there is a need for an integrated pickleball personal informatics system that can capture objective gameplay information, organize performance records, generate

pickleball-specific insights, support community coordination, and facilitate evidence-based collaboration among players, coaches, teammates, and court-community organizers.

1.4. Conceptual and Solutions Framework

1.4. Conceptual and Solutions Framework

1.4.1 PPIF Stage Matrix

Stage	Primary Goal	Key Actors	Technologies/ Inputs	Outputs	Example Metrics
1 - Preparation	Define performance targets and focus areas	Players, Coaches	Goal - Setting, Consultation	Performance objectives, list of KPIs to track	Dink consistency, court position targets, shot selection priorities
2 - Collection	Capture objective gameplay data for subsequent processing	Players , Sensors	Recorded Gameplay Video; Computer Vision; Motion Tracking	Recorded video, tagged video clips, structured gameplay events	Court position, rally structure, shot selection
3 - Integration	Unify all data into a single athlete performance profile	System , players, coaches	Gameplay data; Context data; Historical records	Athlete Performance Profile, shot maps, heat maps, trend reports	Match history, shot selection, tactical indicators, historical strengths and weaknesses,
4 - Reflection	Analyze processed data to surface insights	Players, Coaches	Dashboards; Visual Analytics; Match Summary	Evidenced-Base d feedback, annotated match reviews, training plan input	Dink success-rate, unforced errors, kitchen line time movement type, rally duration,
5 - Action	Convert insights into individualized interventions	Players, Coaches	Reflection outputs; Recommendation engine; Player-initiated audio request	Tailored drill plans, tactical adjustments, workload guidance, next-session practice targets, and contextual audio cues	Technical session targets, position reminders, workload limits, next session goals

Figure 1.4.1 PPIF Stage Matrix

Note. Based on Li, Dey, and Forlizzi's (2010) Personal Informatics model, adapted for pickleball performance.

The study utilizes the Pickleball Performance Informatics Framework as the conceptual foundation for the four proposed solution frameworks. Computer-vision functions are delivered through recorded gameplay capture and post-session video processing. The PPIF is a specialized framework designed to support continuous skill development among pickleball players by organizing the system around performance goals, data collection, analytics, reflection, and training action. It is adapted from Li, Dey, and Forlizzi's Personal Informatics model, which outlines five stages of technology-supported self-monitoring: Preparation, Collection, Integration, Reflection, and Action. Within this study, these stages are applied through a computer-vision implementation that processes recorded pickleball gameplay and generates post-session performance information.

In the Preparation Stage, players and coaches identify the specific performance areas that the system must monitor. These may include dink consistency, court positioning, shot selection, kitchen-line presence, rally duration, movement efficiency, and tactical decision-making. This stage ensures that the system is designed as a performance improvement tool rather than merely a game-recording or scorekeeping application.

In the Collection Stage, gameplay is captured through a smartphone camera positioned to record the playable court area. The recorded gameplay video is subsequently processed by the system to analyze player movement, court zones, rally activity, and pose-related cues. Computer vision serves as the primary sensing method, while self-reported data may provide additional information such as fatigue, perceived exertion, recovery condition, or training context.

In the Integration Stage, the system combines gameplay data, visual tracking outputs, match context, and historical records into a unified athlete performance profile. This integration

supports post-session analysis such as heatmaps, shot maps, rally segmentation, historical trend reports, and player development patterns.

In the Reflection Stage, players and coaches interpret the processed data through dashboards, visual analytics, annotated clips, and performance summaries. The framework supports detailed post-match or post-training review, where players and coaches can examine performance trends, repeated errors, tactical patterns, and areas for technical improvement.

In the Action Stage, the insights generated by the system are translated into individualized training interventions. These may include technical drills, tactical adjustments, positioning reminders, workload guidance, and next-session performance targets. Through this stage, the framework emphasizes evidence-based coaching decisions rather than relying only on memory, manual observation, or subjective judgment. The system also provides player-initiated contextual audio guidance. The audio guidance is requested by the player after a rally, during an appropriate training break, or after the recorded session. It is not delivered continuously while active gameplay is occurring.

Problem	Solution framework	Primary functions	Expected outputs
P1. Limited objective gameplay-data collection and reliance on	S1. Objective Gameplay Data Collection Framework	Video capture, post-session video processing, player detection and tracking, court calibration, rally segmentation, court-zone	Structured gameplay events, movement paths, kitchen-zone data, rally records,

subjective evaluation		analysis, and selected shot-event recognition	court-coverage information
P2. Fragmented development tracking and lack of pickleball-specific analytics	S2. Player Personal Informatics and Analytics Framework	Player profiles, goal setting, historical integration, dashboards, post-game reports, reflection, progress monitoring and recommendations	Unified performance profile, visual reports, historical trends, goals, reflections and development recommendations
P3. Difficulty finding suitable games and partners and coordinating community-based play	S3. Court and Community Coordination Framework	Game listings, open-play queues, availability matching, session coordination, partner and opponent suggestions, community announcements, court-owner administration, and access management	Suitable game suggestions, session listings, queue records, participation records, community announcements, and court-community information
P4. Limited timely, contextual and collaborative coaching support	S4. Coach and Team Collaboration Framework	Authorized report access, coach annotations, team review, training feedback, shared development plans,	Coach feedback, shared reports, training plans, team-progress records and contextual audio cues

		and player-initiated contextual audio guidance	
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Figure 1.4.2 Problem-Solution Matrix

S1. Objective Gameplay Data Collection Framework

The Objective Gameplay Data Collection Framework addresses the fragmented availability of systematic and objective gameplay-performance data. It utilizes video captured through an accessible camera device, such as a smartphone, together with existing computer-vision technologies to process observable pickleball gameplay.

The framework focuses on feasible performance information such as player detection and tracking, movement paths, court calibration, court-zone occupancy, kitchen-line presence, rally segmentation, court coverage, and selected shot-event classifications. Captured information is converted into structured gameplay records that can be used by the Player Personal Informatics and Analytics Framework.

Gameplay is recorded first and then processed by the system to generate the supported performance outputs. The framework does not depend on continuous computer-vision processing while active gameplay is occurring.

The framework does not provide automated officiating, official in-or-out decisions, or guaranteed ball-trajectory analysis. Outputs that cannot be detected reliably through the selected camera configuration must be excluded from the core performance metrics or marked according to their level of confidence.

S2. Player Personal Informatics and Analytics Framework

The Player Personal Informatics and Analytics Framework organizes gameplay information into a unified player-centered performance record. It supports the preparation, integration, reflection, and action stages of the Pickleball Performance Informatics Framework by enabling players to establish performance goals, review post-game reports, compare historical and current results, record reflections, and monitor development over time.

The framework integrates structured gameplay events, match context, player goals, historical records, and authorized coach feedback. It presents pickleball-specific information through dashboards, visual summaries, heatmaps, movement reports, rally summaries, and performance trends. Based on sufficiently reliable available information, the system identifies priority areas and provides supplementary recommendations for future training sessions. The reports, summaries, and recommendations are generated from recorded and processed gameplay information. They are intended for post-session reflection and future training preparation rather than continuous feedback during active gameplay.

S3. Court and Community Coordination Framework

The Court and Community Coordination Framework addresses difficulties in finding suitable games, partners, training groups, and community-based playing opportunities. It supports players, court owners, and community organizers by providing centralized game listings, open-play queues, player availability records, session coordination, and partner or opponent suggestions.

The framework recommends suitable playing opportunities using compatibility factors supplied by users, including availability, preferred game format, playing experience, preferred level of competition, training objectives, play style, and location. These factors are intended to improve game suitability and coordination without generating or assigning an official player rating.

Court-owner and community-administration functions are incorporated into this framework because court owners and community organizers are responsible for managing game listings, open-play schedules, player participation, queues, announcements, community records, user roles, and access permissions. The framework centralizes this information to reduce dependence on fragmented social-media posts, private messages, and informal communication channels.

The framework does not include comprehensive facility management, payment processing, financial reporting, tournament-bracket management, or automated competitive ranking.

S4. Coach and Team Collaboration Framework

The Coach and Team Collaboration Framework supports cooperative performance improvement by allowing authorized coaches, teammates, and training partners to review relevant player-performance information. It provides role-based access to reports, annotated gameplay records, coach comments, training recommendations, shared goals, and team-progress information.

The framework includes a Player-Initiated Contextual Audio Coaching Submodule. This feature is activated by the player after a rally, during an appropriate training break, or after the recorded

session. It uses processed gameplay events, player history, rule-based feedback templates, and approved coaching knowledge to generate a short supplementary audio cue.

The audio guidance is not delivered continuously during active gameplay. It is generated only after the player initiates the feature and when sufficiently reliable processed gameplay information is available.

The audio feature does not replace professional coaching and must not provide medical guidance, injury assessment, automated officiating, or advice based on uncertain computer-vision outputs.

1.5. Study Objectives

1.5 Study Objectives

The study aims to develop an integrated personal informatics system for pickleball players that utilizes existing computer vision and information technologies to support objective gameplay monitoring, player development tracking, community-based play coordination, and collaborative coaching.

Specifically, the study aims to:

1. Develop an Objective Gameplay Data Collection Framework that uses existing computer vision technologies to capture and organize feasible pickleball gameplay indicators, including player movement, court positioning, kitchen-zone presence, rally activity, court coverage, and selected shot events.
2. Develop a Player Personal Informatics and Analytics Framework that integrates current and historical gameplay information into a unified player profile and presents performance reports, visual analytics, goals, reflections, progress trends, and supplementary development recommendations.
3. Develop a Court and Community Coordination Framework that supports centralized game listings, open-play queues, player availability, session coordination, partner and opponent suggestions, community announcements, participation records, and basic access management for court owners and community organizers.
4. Develop a Coach and Team Collaboration Framework that allows authorized coaches, teammates, and training partners to review relevant player-performance information, provide structured feedback, share development plans, monitor collaborative progress, and deliver player-initiated contextual audio guidance during appropriate training pauses.

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CHAPTER II Review of Related Literature

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2.1. Review of Related Concepts

2.1.1 Personal Informatics in Sports Performance Monitoring

Personal informatics refers to the use of technology to help individuals collect, organize, understand, and use personal data for self-improvement. In the context of sports, this concept is important because athletes and recreational players are increasingly using digital tools to track their progress and make better decisions about training and performance. Li et al. (2010) introduced a stage-based model of personal informatics that includes five stages: preparation, collection, integration, reflection, and action. These stages show how users first identify what they want to improve, gather relevant data, combine and interpret the information, reflect on the results, and apply changes based on what they learned.

This concept is closely related to the proposed pickleball system because the study focuses on helping players understand their gameplay through objective performance data. Many pickleball players rely on memory, casual observation, or feedback from playing partners when evaluating their performance. Although these forms of feedback can be helpful, they may not always provide a complete or accurate picture of a player's strengths and weaknesses. Through a personal informatics approach, players can review actual data from their games and use it as a basis for improvement.

Personal informatics has also been explored in sports settings. Rapp and Tirabeni (2018) explained that personal informatics tools are used by both amateur and elite athletes not only to measure performance, but also to understand their bodies, track progress, and

connect their data with their overall sporting experience. In a later study, Rapp and Tirabeni (2020) discussed how self-tracking tools can influence an athlete's motivation, attention, comfort, and lifestyle. These ideas support the direction of the proposed system because pickleball players may benefit from a tool that does more than record statistics. The system should help them reflect on their performance, recognize patterns, and make practical adjustments for future games.

2.1.2 Computer Vision for Sports Performance Analysis

Computer vision is a technology that allows computers to analyze visual information from images or videos. In sports, it has become useful for studying player movements, ball movement, game events, and performance patterns. Naik et al. (2022) explained that computer vision can process sports video data and transform it into measurable information such as player movement, ball trajectories, action recognition, and tactical events. This makes computer vision valuable in sports performance analysis because it can provide objective data that may be difficult to record manually during actual gameplay.

For the proposed pickleball system, computer vision can be used to analyze important gameplay details such as player movement, footwork, court positioning, rally patterns, shot types, reaction time, doubles spacing, and unforced errors. This is especially useful in pickleball because many performance-related actions happen quickly during rallies. A player may not always remember where they were positioned, how often they stayed near the kitchen line, or which shots led to mistakes. By using camera-based analysis, the

system can provide players with a clearer view of how they actually performed during a match.

This concept supports the core performance monitoring component of the proposed system. Instead of depending only on manual scorekeeping or subjective observation, computer vision can help capture gameplay data in a more consistent and detailed manner. The collected information can then be used to generate insights about movement efficiency, positioning, shot selection, and tactical decision-making. In this way, computer vision serves as one of the main technologies that can help make pickleball performance evaluation more data-driven.

2.1.3 Pickleball-Specific Tactical Performance

Pickleball has unique gameplay characteristics that separate it from other racket sports. While sports such as tennis and badminton often emphasize speed, power, and court coverage, pickleball places strong importance on strategy, positioning, consistency, communication, and decision-making. One of the most important features of pickleball is the non-volley zone, commonly known as the kitchen. USA Pickleball (2026) identifies the non-volley zone as a key rule-based area that affects how players move, position themselves, and select shots during rallies.

Because of this, pickleball performance should not be measured only through general fitness or basic match results. Important tactical indicators may include dink consistency, volley success, kitchen-line presence, rally length, shot selection, time spent at the baseline, and unforced errors. These indicators are meaningful because they reflect the actual demands of pickleball gameplay. A player who can position well, control the

kitchen line, and make consistent decisions may perform better than a player who only relies on power.

Recent research also supports the need to study pickleball through technical and tactical indicators. Prieto-Lage et al. (2025) analyzed key match indicators in women's singles pickleball and showed that performance can be examined through measurable gameplay actions. This supports the proposed system's focus on pickleball-specific analytics. By tracking indicators that are directly connected to the sport, the system can provide insights that are more relevant to pickleball players than general sports or fitness tracking tools.

2.1.4 Visual Analytics and Decision Support for Player Improvement

Visual analytics refers to the use of visual tools such as dashboards, charts, graphs, heatmaps, and summaries to help users understand data more easily. In sports performance systems, visual analytics is important because raw numbers alone may not be enough for players and coaches to understand what is happening. A player may see a statistic, but without proper presentation and explanation, it may be difficult to know what action should be taken next. Visual tools can make performance data easier to interpret by showing patterns in movement, shot success, rally trends, fatigue, and progress over time.

Decision support is also important in sports performance monitoring because the purpose of collecting data is not only to store information, but to guide improvement. Wakefield et al. (2014) discussed the informatics needs of amateur athletic coaches and emphasized the importance of considering training context, athlete condition, and performance

information. This supports the idea that sports systems should help users understand what happened, why it happened, and what changes may be needed.

For the proposed pickleball system, visual analytics can help players and coaches review performance in a more meaningful way. Dashboards may show court heatmaps, shot success rates, unforced errors, kitchen-line presence, rally length, and fatigue-related trends. These visual summaries can help players identify which skills need improvement, which shots are effective, and whether their positioning is efficient. As explained in the personal informatics model, collected data becomes more useful when it supports reflection and action (Li et al., 2010). Therefore, visual analytics and decision support are essential parts of the proposed system because they connect performance data to actual player improvement.

2.2. Review of Related Systems

A sports performance monitoring system is a digital tool that helps athletes capture, analyze, and reflect on their gameplay or training data to support improvement. These systems generally fall into three overlapping types: computer-vision systems that analyze video of play or movement, and rating or community systems that quantify skill and connect players. Their core value lies in replacing subjective self-assessment with objective, measurable feedback, often through automatic data collection, performance dashboards, and personalized recommendations. The following systems are assessed on their functionalities and their relevance to a personal informatics system for pickleball players.

2.2.1 Zepp Tennis

Zepp Tennis is a training system for tennis that utilizes a sensor and an app, with a small motion sensor mounted on the handle of the racket. It evaluates key stroke metrics like ball speed, ball spin, and sweet-spot reliability, categorizes strokes by type, and provides Smart Rally video capture that automatically records and tags the top rallies. It offers power, consistency, and intensity metrics, a community featuring leaderboards, and the option to compare a player's accuracy with that of a pro or coach.. Its drawbacks include being tailored for tennis instead of pickleball, relying on proprietary hardware, and its sensor hardware has mostly been phased out and might not connect to contemporary smartphones, restricting current accessibility.

2.2.2 PlaySight SmartCourt

PlaySight signifies the installed facility's multi-camera aspect of computer-vision systems. Its AI platform with connected cameras is used on pickleball, tennis, and padel courts, as well as in professional and college sports, aiding broadcast, instant replay, and player development. Its SmartTracker automatically monitors all on-court activity without an operator, while SmartScore gathers statistics and auto-tags occurrences. Its advantages include extensive automated data collection and analysis that showcase the viability of the proposed system's fundamental monitoring element. Its drawbacks include the need for expensive, permanent infrastructure, a focus on clubs, broadcasting, and officiating rather than the individual player's feedback process, and personalized matchmaking.

2.2.3 Arccos Caddie

Arccos is golf's leading game-tracking system, using lightweight screw-in club sensors that pair with a smartphone or Apple Watch. It automatically tracks every shot without manual input and provides strokes-gained analytics across driving, approach, short game, and putting, along with an AI rangefinder and real-time club and strategy recommendations. Its AI evaluates a player's strengths, weaknesses, playing patterns, course, and past performance to give caddie-style advice. Its strengths are highly relevant to the proposed system: automatic data collection, deep historical analytics, and personalized, data-driven recommendations for improvement. Its limitations are that it is golf-specific, requires proprietary hardware plus an ongoing subscription, and has no computer-vision analysis or player-matchmaking capability.

2.2.4 Padelytics

Padelytics is a platform enhanced by AI for padel, the racket sport closest to pickleball, that transforms regular match footage into performance analytics. It employs machine-learning models and computer-vision algorithms to assess each frame, monitoring the players and the ball, identifying and categorizing player actions, while producing a detailed event stream of all occurrences on the court. A match is captured and uploaded to the cloud, where the AI analyzes each frame and provides insights to the player's phone or website within minutes after the match concludes, instead of in real time. It generates court-location heatmaps, quantifies the distance and speed of each player's runs, monitors fitness improvements over time, analyzes time allocated to defense, offense, and transitions, assesses partner coordination, and determines which shots a player

employs frequently as well as their landing spots, including the most and least effective shots. Its advantages render it one of the nearest analogs to the suggested system, as it executes movement tracking through computer vision, analyzes court positioning, and classifies shots from uploaded match footage on regular cameras, addressing multiple core monitoring objectives of the proposed system simultaneously. It also offers enhancement suggestions, such as advice on improved court positioning. Its shortcomings are that it is designed for padel instead of pickleball, is aimed at clubs equipped with court cameras, and does not provide skill-based matchmaking with opponents.

2.2.5 HomeCourt

HomeCourt, created by the NEX Team and an official NBA partner, is a mobile app for basketball training that transforms a phone or tablet into a computer-vision personal coach. Utilizing the device's camera alongside computer vision and AI, it smartly records a player's shots, dribbling, and movements, analyzing them to provide immediate feedback and comprehensive analytics. Its Shot Science function quantifies advanced metrics including release angle, vertical jump, and player speed, while also executing automatic identification of shot types. It also monitors advancement over time with complete exercise records, data, and skill assessments, and provides a tailored suggested workout daily featuring drills according to the player's proficiency. Its advantages position it as the nearest consumer equivalent to the suggested system, as it showcases a full personal-informatics cycle, seamless data gathering, analysis-driven reflection, and action via tailored recommendations, on user-friendly hardware,

and it incorporates Apple Watch heart rate. Its flaws include being focused solely on basketball, functioning primarily in real-time instead of analyzing recorded game footage, failing to assess racket-court positioning, and providing social competition instead of skill-oriented opponent pairing.

2.2.6 OnForm

OnForm is a sports video analysis and coaching platform that allows athletes and coaches to record, review, and analyze technique using a mobile device. It supports slow-motion playback, frame-by-frame review, side-by-side comparison, voice-over feedback, and remote video sharing between coaches and athletes. Its relevance to the proposed system lies in its use of accessible video-based analysis for performance reflection, allowing athletes to review specific movements and receive structured feedback without requiring expensive facility-based systems. This supports the personal informatics process by helping users collect visual performance data, reflect on their actions, and identify areas for improvement over time. However, OnForm is designed as a general sports coaching tool rather than a pickleball-specific analytics system. It does not automatically classify pickleball shots, track rally patterns, generate court-positioning heatmaps, or provide skill-based matchmaking for players.

2.2.7 Dartfish

Dartfish is a video analysis software platform used in sports performance, coaching, and athlete development. It allows users to capture, analyze, tag, compare, and share sports videos, making it useful for identifying technical errors, tactical patterns, and performance trends. Its tools support video annotation, movement comparison, and performance review, which can help athletes and coaches better understand weaknesses and develop training strategies. Dartfish is relevant to the proposed system because it demonstrates how video can be transformed into measurable and reviewable performance information. It also supports technical and tactical analysis, which is important in racket sports where positioning, timing, and shot selection affect performance. However, Dartfish is not specifically designed for pickleball and may require manual tagging or expert interpretation. It also functions more as a coaching and analysis platform than as an automated personal informatics system with built-in recommendations, matchmaking, and court management features.

2.2.8 CoachNow

CoachNow is an all-in-one sports coaching application that combines video and image analysis, private communication, cloud storage, and feedback tools for coaches and athletes. It includes features such as slow-motion video review, side-by-side comparisons, voice-over feedback, CoachCam, and AI skeleton tracking for analyzing body movement and joint angles. Its relevance to the proposed system is its ability to support continuous improvement through

recorded performance data, coach-athlete communication, and visual feedback. This makes it useful as a reference for the reflection and feedback stage of a personal informatics system. However, CoachNow is not pickleball-specific and is mainly designed around coaching communication rather than automatic match analytics. It does not directly analyze rally structure, shot outcomes, court zones, player coordination, or opponent compatibility, which limits its use as a complete model for pickleball performance monitoring.

2.2.9 DUPR

DUPR, or Dynamic Universal Pickleball Rating, is a pickleball rating system that evaluates player skill through match results and assigns players a numerical rating. Its rating scale begins at 2.000 for beginners and extends to 8.000 for professional-level players. DUPR is widely used for organizing fair matches, leagues, and tournaments by helping players understand their skill level and find more balanced competition. Its relevance to the proposed system lies in its use of player data to support skill-based classification and matchmaking. This is important for pickleball communities because mismatched games can reduce both competitiveness and enjoyment. However, DUPR is mainly result-based rather than performance-based. It does not automatically analyze gameplay footage, identify technical weaknesses, classify shots, or explain the specific movement and tactical behaviors behind a player's rating.

2.2.10 Pickleheads

Pickleheads is a pickleball community platform that helps players find courts, join games, organize sessions, and connect with other players. It provides a court directory, open play schedules, local groups, game sign-ups, and skill-level-based game discovery. Pickleheads is relevant to the proposed system because it addresses the social and logistical side of pickleball participation, particularly court discovery, player connection, and organized play. These features are useful references for the proposed system's court reservation, queueing, and matchmaking components. However, Pickleheads is not primarily a performance monitoring or analytics platform. It does not automatically collect gameplay data, analyze player movement, classify shots, generate performance dashboards, or provide AI-based recommendations for technical improvement. Therefore, it is most useful as a model for community-building and game organization rather than as a direct model for computer-vision-based performance analysis.

2.2.11 Strengths and Weaknesses Table

Software / System	Strengths	Weaknesses
Zepp Tennis	Detailed swing data (speed, spin, sweet spot); stroke classification; tips and drills; community/leaderboards	Tennis-specific; proprietary hardware; hardware largely discontinued; no positioning or matchmaking
PlaySight SmartCourt	Automated multi-camera capture; auto-tracking, stats, and event tagging; supports pickleball; coach access	Requires fixed costly infrastructure; club/broadcast-oriented; no personal matchmaking
Arccos Caddie	Automatic shot tracking;	Golf-specific; proprietary

	strokes-gained analytics; AI-driven personalized recommendations; strong historical insight	hardware + subscription; no computer-vision analysis; no matchmaking
Padelytics	Computer-vision player and ball tracking from uploaded match video; court-positioning heatmaps; shot classification; movement/fitness metrics; improvement tips	Padel-specific; oriented toward clubs with a court-camera setup; no skill-based matchmaking
HomeCourt	Phone-based computer vision; automatic shot, movement, and skill tracking; personalized drills and recommendations; progress history; Apple Watch heart rate; coach sharing	Basketball-specific; mainly real-time; no racket-court positioning; social play rather than skill-based matchmaking
OnForm	Accessible mobile video analysis; slow-motion and frame-by-frame review; side-by-side comparison; voice-over feedback; coach-athlete sharing	General sports coaching tool; not pickleball-specific; no automatic shot classification, heatmaps, or matchmaking
Dartfish	Strong video annotation tools; movement comparison; tagging and sharing; useful for technical and tactical review	Not pickleball-specific; may require manual tagging; relies on coach or expert interpretation; no built-in matchmaking
CoachNow	Video and image analysis; coach feedback tools; cloud storage; side-by-side comparisons; AI skeleton tracking for movement analysis	Coaching-focused rather than match analytics-focused; not pickleball-specific; no rally, shot, court-zone, or matchmaking analysis
DUPR	Pickleball-specific rating system; supports fairer matches, tournaments, and leagues; useful for skill-based classification	Result-based rather than performance-based; no video analysis, shot classification, movement tracking, or technical feedback

Pickleheads	Court finder; game scheduling; open play organization; player connection; supports community and skill-level-based game discovery	Not a performance analytics system; no automatic gameplay data collection, computer-vision analysis, dashboards, or training recommendations
Proposed System	Integrates computer-vision gameplay and movement analysis, automatic shot classification, positioning/court-coverage analysis, skill and play-style assessment, matchmaking, personalized recommendations.	Integrates existing technologies;

2.2.12 Synthesis

The comparison reveals that every current system provides only a portion of what the proposed system intends to integrate. HomeCourt is the most similar in idea, showing a complete personal-informatics cycle with computer-vision capture, analytics, and tailored suggestions on a regular phone, though it is designed for basketball and primarily functions in real-time without analyzing court positioning. Padelytics, focusing on the racket sport most similar to pickleball, demonstrates that computer vision can monitor players, categorize shots, and create court-positioning heatmaps from uploaded match videos, but it is tailored specifically to padel and club-centered. PlaySight demonstrates that automated, multi-angle gameplay capture along with positioning data and coach access is achievable and even accommodates pickleball, though it relies on expensive fixed infrastructure. Zepp Tennis offers advanced sensor-driven swing metrics, although it is tailored for tennis and relies on specific hardware. Arccos illustrates how automatic data gathering can be combined with strokes-gained analysis and AI-based personalized suggestions, but it depends on sensors instead of computer vision and is limited to golf. The proposed solution will be informed by their unique strengths, accessible CV tracking,

positioning analysis, shot classification, swing metrics, historical analytics, and tailored recommendations, while their common limitation, the lack of a single platform that merges all these aspects along with skill-based matchmaking for pickleball within one personal-informatics loop, highlights the research gap that the suggested system aims to fill.

2.2.13 Features Table

Feature	Home Court	Padelytics	PlaySight SmartCourt	Zepp Tennis	Arccos Caddie	On Form	Dart fish	Coach Now	DUPR	Pickle heads	Proposed System
Computer Vision Analysis	✓	✓	✓	✗	✗	△	△	✓	✗	✗	✓
Shot Classification	✓	✓	✓	✓	✓	✗	△	✗	✗	✗	✓
Positioning / Court Coverage	✗	✓	✓	✗	✗	✗	△	✗	✗	✗	✓
Skill & Play-Style Assessment	✓	✗	✗	✗	✓	△	△	△	✓	✗	✓
Matchmaking	✗	✗	✗	✗	✗	✗	✗	✗	△	✓	✓
Personalized Recommendations	✓	✓	✗	✓	✓	△	△	△	✗	✗	✓
Pickleball-Specific	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓	✓

Legend: ✓ = Feature is clearly supported; ✗ = Feature is not supported; △ = Feature is partially supported, manual, coach-driven, or indirectly supported

Figure 2.2.8.1 Features table regarding availability of different functionalities per system

The Features table presents an overview of selected existing systems in relation to the proposed personal informatics system for pickleball players. The systems were evaluated based on key features relevant to sports performance monitoring, including computer vision analysis, shot classification, positioning and court coverage, skill and play-style assessment,

matchmaking, personalized recommendations, and pickleball-specific functionality. These criteria were selected to determine the extent to which existing systems support automated data collection, performance reflection, and data-driven improvement.

The comparison shows that several existing systems provide strong capabilities in specific areas of sports performance analysis. HomeCourt, Padelytics, and PlaySight SmartCourt demonstrate the use of computer vision and automated tracking for performance monitoring. However, these systems are primarily designed for other sports or require specific infrastructure, which may limit their accessibility and direct applicability to pickleball players. Meanwhile, systems such as OnForm, Dartfish, and CoachNow provide video-based coaching and review tools, but their functions are generally broad, coach-driven, or dependent on manual interpretation rather than fully automated pickleball performance analysis.

The table also highlights that DUPR and Pickleheads are more closely related to pickleball in terms of rating, player connection, and community participation. DUPR supports skill-based classification through player ratings, while Pickleheads assists players in finding courts, joining games, and organizing play sessions. However, these systems do not provide detailed gameplay analytics such as shot classification, player movement tracking, court-positioning analysis, or personalized technical recommendations.

While existing systems offer valuable individual features, none of them fully integrate computer vision analysis, shot classification, court-positioning assessment, skill and play-style evaluation, matchmaking, personalized recommendations, and pickleball-specific support within a single

platform. This reveals a functional gap in existing sports performance systems and supports the need for the proposed system, which aims to provide an integrated personal informatics solution specifically designed to help pickleball players monitor, assess, and improve their performance through objective and data-driven feedback.

2.3. Review of Related Methodologies

2.3.1 Scrum

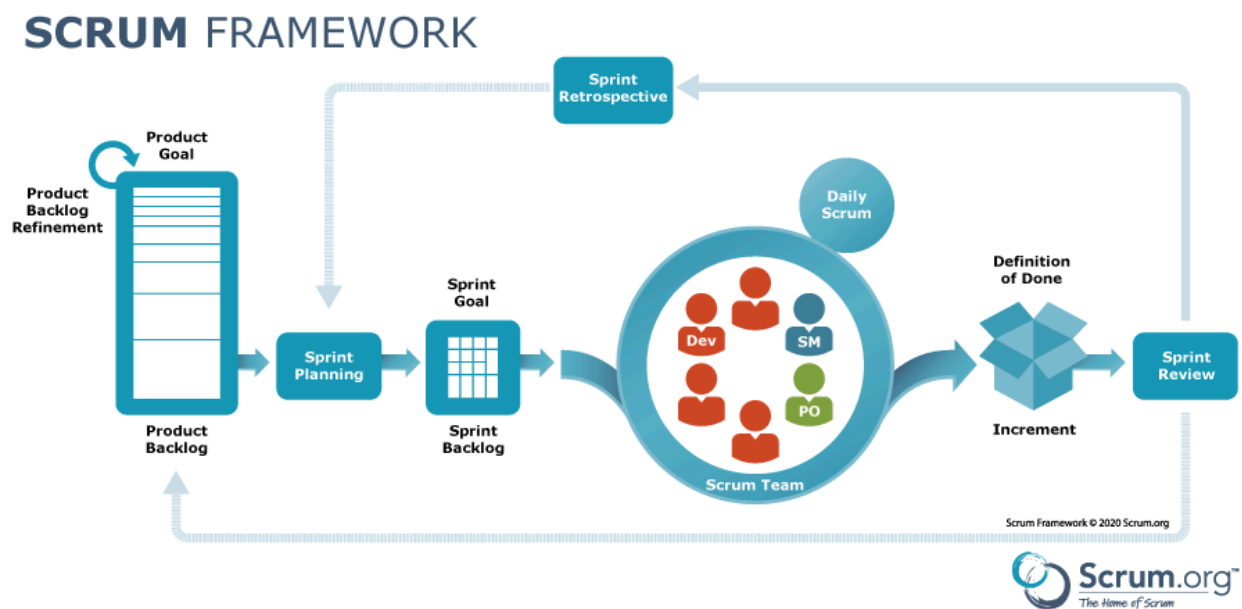


Figure 2.3.1.1 Scrum Framework

Scrum is one of the most popular Agile methodologies for running software projects, and it tends to shine when the work is complicated and the requirements are bound to change along the way (Schwaber & Sutherland, 2020). Instead of trying to map out the entire project before any code is written, Scrum splits the work into short, repeating cycles called sprints, usually two to four weeks each, with the team aiming to finish a working slice of the system by the end of every one. That rhythm fits the proposed pickleball system nicely, since its bigger parts are core

performance monitoring, skill assessment, and gameplay context support can be built and polished one at a time rather than all at once.

Scrum runs on three roles. The Product Owner decides what the team tackles next and keeps the product backlog, the ordered list of features and fixes, up to date. The Scrum Master looks after the process itself, making sure the team sticks to Scrum and clearing out whatever is getting in the way. Everyone else makes up the Development Team, the people who actually build the system and turn backlog items into a working increment each cycle (Schwaber & Sutherland, 2020). At the start of each sprint, the team picks a realistic batch of items from the backlog and commits to getting them done, which keeps everyone focused on a clear goal instead of an open-ended pile of work. And because each sprint produces something that actually works, the people who will use the system later such as the players and coaches can see where things are heading early on and weigh in, rather than waiting until the very end to find out the team built the wrong thing.

A handful of regular meetings hold the whole thing together. Sprint planning sets up what the cycle will cover, the daily stand-up is a quick check-in for sharing progress and raising blockers, and each sprint wraps with a review to show what was built and a retrospective to figure out what to do better next time (Schwaber & Sutherland, 2020). The product backlog, sprint backlog, and increment round it out as shared reference points, so anyone in the team or its stakeholders can tell at a glance what is being worked on and how far along it is.

The real draw, though, is how well Scrum handles change. Requirements almost never stay put, and Agile methods like Scrum were built with exactly that in mind (Beck et al., 2001). Since this project leans on existing computer vision instead of building new models, the team will not

really know what the system can promise until it sees how those tools handle shot classification, movement tracking, positioning, and exertion on actual gameplay. Scrum's iterative, feedback-driven loop makes it easy to work those findings back into the project as they come up, which is a big part of why it is the methodology this study adopts.

2.3.2 DevOps

DevOps is less a step-by-step process and more a culture and a set of practices that bring software development and IT operations together, all aimed at shipping changes quickly, reliably, and without much friction (Kim et al., 2016). It leans hard on automation and on continuous integration and delivery, so that new code can be tested and released in small, frequent batches instead of big, occasional ones. In practice it usually sits on top of an Agile method rather than replacing it, since teams that already work in iterations adopt it to make their releases smoother and their feedback loops faster (Humble & Farley, 2010).

For a system that will eventually be used and updated on a regular basis, that mindset is appealing, because it keeps a product stable while still leaving room for constant improvement. The catch, at least for a capstone, is that DevOps assumes an operational side that a student project usually does not have. It relies on build pipelines, automated testing setups, and ongoing deployment to live environments, which is far more than is needed just to develop and demonstrate a prototype. DevOps tends to pay off most once a system is already running in production and being maintained, rather than during the initial build, which is why it fits this project better as a future direction than as the framework guiding the work now.

2.3.3 Waterfall

Waterfall is the classic, linear approach, where a project moves through a fixed sequence of phases such as requirements, design, implementation, testing, and deployment and each one has to be finished before the next begins (Royce, 1970). It rests on heavy planning right at the start and works best when the requirements are well understood and unlikely to move. Its biggest strengths are clarity and predictability: everyone knows what is being built and in what order, and the thorough documentation it produces can make a project easy to follow.

That same rigidity is exactly where Waterfall runs into trouble for this project. The requirements here are expected to shift as the team learns how the computer vision actually behaves in real gameplay, and Waterfall makes those late changes slow and costly to absorb. Because feedback only shows up at the end of each phase, a problem introduced early can also go unnoticed until much later, when it is far harder to fix. For an exploratory system built around integrating existing technologies, that lack of flexibility is a serious drawback.

2.3.2 Comparison of Methodologies

Feature	Scrum	DevOps	Waterfall
Project structure	Iterative and incremental	Continuous, built on automated delivery	Linear and sequential
Planning	Light upfront, adaptive each sprint	Continuous, focused on delivery pipelines	Extensive upfront, detailed specifications
Requirements	Evolving requirements are accommodated	Evolving, supported through continuous releases	Fixed and cannot be changed
Feedback	Continuous, every sprint	Continuous and automated	Gathered at the end of each phase

Risk management	Addressed throughout the project	Reduced through frequent, small releases	Mainly during the planning phase
Change management	Relatively easy to implement	Easy, with automated testing and deployment	Difficult and costly
Team collaboration	Close collaboration and communication	Tight development-and-operations collaboration	Individual work and handoffs
Project visibility	High, with regular increments	High, with continuous monitoring	Low until the end of each phase
Customer involvement	Encouraged throughout the project	Indirect, through fast release and feedback	Limited until the end of each phase
Tooling / infrastructure	Lightweight, process-focused	Heavy; needs CI/CD and automation setup	Lightweight, document-focused
Best suited for	Iterative builds with evolving requirements	Systems in continuous deployment and maintenance	Projects with fixed, well-known requirements

Figure 2.3.2.1 Comparison of Software Development Methodologies

2.4 Synthesis

2.4.1 Common Themes

The reviewed literature and related systems commonly emphasize the shift from subjective performance evaluation to objective, data-driven feedback. In the related concepts, personal informatics is presented as a process of collecting, integrating, reflecting on, and acting upon personal data for improvement. This theme is reflected in sports performance monitoring, where players are encouraged to move beyond memory-based assessment and instead use measurable information to identify strengths, weaknesses, and progress over time. Computer vision also appears as a recurring theme because it allows gameplay video to be transformed into useful data such as player movement, shot actions, positioning, and tactical patterns. These concepts support the

proposed system's focus on automatic data collection, visual feedback, and player reflection.

Another common theme is the importance of accessibility and usability in sports technologies. Several systems, such as HomeCourt, OnForm, Dartfish, and CoachNow, demonstrate how video-based tools can support performance review and coaching without relying entirely on manual observation. Other systems, such as Padelytics and PlaySight SmartCourt, show the potential of computer vision in tracking court movement, classifying actions, and generating performance statistics. Meanwhile, DUPR and Pickleheads highlight the social and community side of sports participation by supporting skill ratings, fairer play, court discovery, and player connection. Together, these systems show that performance improvement is not only about collecting data, but also about making that data understandable, actionable, and useful within the player's actual sports environment.

The reviewed methodologies also show the importance of flexibility and iterative development. Since the proposed system depends on integrating existing technologies, particularly computer vision, the final system requirements may change as the team tests how these tools perform in actual pickleball gameplay. Scrum is therefore presented as the most suitable methodology because it allows the system to be developed in stages, tested regularly, and refined through continuous feedback.

2.4.2 Problems Currently Addressed

The existing literature and systems currently address several important problems in sports performance monitoring. First, they address the problem of unreliable self-assessment by allowing athletes to review performance through collected data rather than relying only on memory, intuition, or casual feedback. Personal informatics systems help users track their own behavior and performance, while computer vision enables more consistent capture of movement, positioning, and gameplay actions. This is especially useful in sports like pickleball, where rallies happen quickly and players may not clearly remember their positioning, shot choices, or errors after a match.

Second, existing systems address the need for technical and tactical analysis. Systems such as HomeCourt, Padelytics, PlaySight SmartCourt, Zepp Tennis, and Arccos Caddie demonstrate that modern sports tools can classify actions, track performance history, generate statistics, and provide recommendations. Video analysis platforms such as OnForm, Dartfish, and CoachNow also support coaching and reflection by allowing users to review movement, compare performance, and receive feedback. These systems help athletes and coaches identify technical errors, tactical patterns, and areas for improvement.

Third, pickleball-related platforms address problems in player organization and fair competition. DUPR provides a rating system that helps classify players according to skill level, while Pickleheads helps players find courts, join games, and connect with other players. These systems address the social and logistical challenges of pickleball

participation, especially in matching players with suitable games and organizing local play.

Finally, the methodology review addresses the problem of managing uncertainty during development. Waterfall is shown to be too rigid for a project whose requirements may evolve, while DevOps is more appropriate for systems already in continuous deployment and maintenance. Scrum addresses the need for an adaptive development process by allowing the team to build, test, and refine the proposed system through repeated sprint cycles.

2.4.3 Gaps in Literature

Despite the strengths of the reviewed concepts and systems, several gaps remain. The first major gap is the lack of a single pickleball-specific system that integrates computer vision analysis, shot classification, positioning and court coverage, skill and play-style assessment, matchmaking, and personalized recommendations. Existing systems tend to focus on only one or two of these areas. For example, HomeCourt provides strong computer-vision-based basketball training, Padelytics offers padel-focused court and shot analytics, and PlaySight supports automated multi-camera capture but requires fixed infrastructure. These systems demonstrate useful features, but they are not fully designed around pickleball players and their specific gameplay needs.

A second gap is the limited availability of pickleball-specific performance analytics. While DUPR and Pickleheads are directly related to pickleball, they focus mainly on

ratings, community connection, and game organization rather than detailed gameplay analysis. They do not automatically analyze shot types, player movement, rally patterns, court positioning, or technical weaknesses. This shows that pickleball platforms currently support participation and competition more than performance diagnosis and improvement.

A third gap is the limited connection between performance data and actionable improvement. Some systems collect large amounts of data, while others provide coaching tools or recommendations, but few combine automatic data collection, visual analytics, reflection, and personalized action in one continuous personal informatics loop. For the proposed system, this gap is important because the goal is not only to record gameplay data, but also to help players understand what the data means and how they can improve future performance.

Lastly, there is a gap in accessible implementation for recreational and community-level pickleball players. Some systems require proprietary sensors, subscriptions, coach interpretation, or fixed court-camera infrastructure. These limitations may reduce accessibility for ordinary players, schools, clubs, and local communities. The proposed system aims to address this gap by integrating existing technologies into a more focused personal informatics solution for pickleball, combining performance monitoring with player assessment and matchmaking features.

2.1. Review of Related Concepts

2.1. Review of Related Concepts

2.1.1 Personal Informatics in Sports Performance Monitoring

Personal informatics refers to the use of technology to help individuals collect, organize, understand, and use personal data for self-improvement. In the context of sports, this concept is important because athletes and recreational players are increasingly using digital tools to track their progress and make better decisions about training and performance. Li et al. (2010) introduced a stage-based model of personal informatics that includes five stages: preparation, collection, integration, reflection, and action. These stages show how users first identify what they want to improve, gather relevant data, combine and interpret the information, reflect on the results, and apply changes based on what they learned.

This concept is closely related to the proposed pickleball system because the study focuses on helping players understand their gameplay through objective performance data. Many pickleball players rely on memory, casual observation, or feedback from playing partners when evaluating their performance. Although these forms of feedback can be helpful, they may not always provide a complete or accurate picture of a player's strengths and weaknesses. Through a personal informatics approach, players can review actual data from their games and use it as a basis for improvement.

Personal informatics has also been explored in sports settings. Rapp and Tirabeni (2018) explained that personal informatics tools are used by both amateur and elite athletes not only to measure performance, but also to understand their bodies, track progress, and connect their data with their overall sporting experience. In a later study, Rapp and

Tirabeni (2020) discussed how self-tracking tools can influence an athlete's motivation, attention, comfort, and lifestyle. These ideas support the direction of the proposed system because pickleball players may benefit from a tool that does more than record statistics. The system should help them reflect on their performance, recognize patterns, and make practical adjustments for future games.

2.1.2 Computer Vision for Sports Performance Analysis

Computer vision is a technology that allows computers to analyze visual information from images or videos. In sports, it has become useful for studying player movements, ball movement, game events, and performance patterns. Naik et al. (2022) explained that computer vision can process sports video data and transform it into measurable information such as player movement, ball trajectories, action recognition, and tactical events. This makes computer vision valuable in sports performance analysis because it can provide objective data that may be difficult to record manually during actual gameplay.

For the proposed pickleball system, computer vision can be used to analyze important gameplay details such as player movement, footwork, court positioning, rally patterns, shot types, reaction time, doubles spacing, and unforced errors. This is especially useful in pickleball because many performance-related actions happen quickly during rallies. A player may not always remember where they were positioned, how often they stayed near the kitchen line, or which shots led to mistakes. By using camera-based analysis, the system can provide players with a clearer view of how they actually performed during a match.

This concept supports the core performance monitoring component of the proposed system. Instead of depending only on manual scorekeeping or subjective observation, computer vision can help capture gameplay data in a more consistent and detailed manner. The collected information can then be used to generate insights about movement efficiency, positioning, shot selection, and tactical decision-making. In this way, computer vision serves as one of the main technologies that can help make pickleball performance evaluation more data-driven.

2.1.3 Pickleball-Specific Tactical Performance

Pickleball has unique gameplay characteristics that separate it from other racket sports. While sports such as tennis and badminton often emphasize speed, power, and court coverage, pickleball places strong importance on strategy, positioning, consistency, communication, and decision-making. One of the most important features of pickleball is the non-volley zone, commonly known as the kitchen. USA Pickleball (2026) identifies the non-volley zone as a key rule-based area that affects how players move, position themselves, and select shots during rallies.

Because of this, pickleball performance should not be measured only through general fitness or basic match results. Important tactical indicators may include dink consistency, volley success, kitchen-line presence, rally length, shot selection, time spent at the baseline, and unforced errors. These indicators are meaningful because they reflect the actual demands of pickleball gameplay. A player who can position well, control the kitchen line, and make consistent decisions may perform better than a player who only relies on power.

Recent research also supports the need to study pickleball through technical and tactical indicators. Prieto-Lage et al. (2025) analyzed key match indicators in women's singles pickleball and showed that performance can be examined through measurable gameplay actions. This supports the proposed system's focus on pickleball-specific analytics. By tracking indicators that are directly connected to the sport, the system can provide insights that are more relevant to pickleball players than general sports or fitness tracking tools.

2.1.4 Visual Analytics and Decision Support for Player Improvement

Visual analytics refers to the use of visual tools such as dashboards, charts, graphs, heatmaps, and summaries to help users understand data more easily. In sports performance systems, visual analytics is important because raw numbers alone may not be enough for players and coaches to understand what is happening. A player may see a statistic, but without proper presentation and explanation, it may be difficult to know what action should be taken next. Visual tools can make performance data easier to interpret by showing patterns in movement, shot success, rally trends, fatigue, and progress over time.

Decision support is also important in sports performance monitoring because the purpose of collecting data is not only to store information, but to guide improvement. Wakefield et al. (2014) discussed the informatics needs of amateur athletic coaches and emphasized the importance of considering training context, athlete condition, and performance information. This supports the idea that sports systems should help users understand what happened, why it happened, and what changes may be needed.

For the proposed pickleball system, visual analytics can help players and coaches review performance in a more meaningful way. Dashboards may show court heatmaps, shot success rates, unforced errors, kitchen-line presence, rally length, and fatigue-related trends. These visual summaries can help players identify which skills need improvement, which shots are effective, and whether their positioning is efficient. As explained in the personal informatics model, collected data becomes more useful when it supports reflection and action (Li et al., 2010). Therefore, visual analytics and decision support are essential parts of the proposed system because they connect performance data to actual player improvement.

2.2. Review of Related Systems

2.2. Review of Related Systems

A sports performance monitoring system is a digital tool that helps athletes capture, analyze, and reflect on their gameplay or training data to support improvement. These systems generally fall into three overlapping types: computer-vision systems that analyze video of play or movement, and rating or community systems that quantify skill and connect players. Their core value lies in replacing subjective self-assessment with objective, measurable feedback, often through automatic data collection, performance dashboards, and personalized recommendations. The following systems are assessed on their functionalities and their relevance to a personal informatics system for pickleball players.

2.2.1 Zepp Tennis

Zepp Tennis is a training system for tennis that utilizes a sensor and an app, with a small motion sensor mounted on the handle of the racket. It evaluates key stroke metrics like ball speed, ball spin, and sweet-spot reliability, categorizes strokes by type, and provides Smart Rally video capture that automatically records and tags the top rallies. It offers power, consistency, and intensity metrics, a community featuring leaderboards, and the option to compare a player's accuracy with that of a pro or coach.. Its drawbacks include being tailored for tennis instead of pickleball, relying on proprietary hardware, and its sensor hardware has mostly been phased out and might not connect to contemporary smartphones, restricting current accessibility.

2.2.2 PlaySight SmartCourt

PlaySight signifies the installed facility's multi-camera aspect of computer-vision systems. Its AI platform with connected cameras is used on pickleball, tennis, and padel courts, as well as in professional and college sports, aiding broadcast, instant replay, and player development. Its SmartTracker automatically monitors all on-court activity without an operator, while SmartScore gathers statistics and auto-tags occurrences. Its advantages include extensive automated data collection and analysis that showcase the viability of the proposed system's fundamental monitoring element. Its drawbacks include the need for expensive, permanent infrastructure, a focus on clubs, broadcasting, and officiating rather than the individual player's feedback process, and personalized matchmaking.

2.2.3 Arccos Caddie

Arccos is golf's leading game-tracking system, using lightweight screw-in club sensors that pair with a smartphone or Apple Watch. It automatically tracks every shot without manual input and provides strokes-gained analytics across driving, approach, short game, and putting, along with an AI rangefinder and real-time club and strategy recommendations. Its AI evaluates a player's strengths, weaknesses, playing patterns, course, and past performance to give caddie-style advice. Its strengths are highly relevant to the proposed system: automatic data collection, deep historical analytics, and personalized, data-driven recommendations for improvement. Its limitations are that it is golf-specific, requires proprietary hardware plus an ongoing subscription, and has no computer-vision analysis or player-matchmaking capability.

2.2.4 Padelytics

Padelytics is a platform enhanced by AI for padel, the racket sport closest to pickleball, that transforms regular match footage into performance analytics. It employs machine-learning models and computer-vision algorithms to assess each frame, monitoring the players and the ball, identifying and categorizing player actions, while producing a detailed event stream of all occurrences on the court. A match is captured and uploaded to the cloud, where the AI analyzes each frame and provides insights to the player's phone or website within minutes after the match concludes, instead of in real time. It generates court-location heatmaps, quantifies the distance and speed of each player's runs, monitors fitness improvements over time, analyzes time allocated to defense, offense, and transitions, assesses partner coordination, and determines which shots a player employs frequently as well as their landing spots, including the most and least effective shots. Its advantages render it one of the nearest analogs to the suggested system, as it executes movement tracking through computer vision, analyzes court positioning, and classifies shots from uploaded match footage on regular cameras, addressing multiple core monitoring objectives of the proposed system simultaneously. It also offers enhancement suggestions, such as advice on improved court positioning. Its shortcomings are that it is designed for padel instead of pickleball, is aimed at clubs equipped with court cameras, and does not provide skill-based matchmaking with opponents.

2.2.5 HomeCourt

HomeCourt, created by the NEX Team and an official NBA partner, is a mobile app for basketball training that transforms a phone or tablet into a computer-vision personal coach. Utilizing the device's camera alongside computer vision and AI, it smartly records a player's shots, dribbling, and movements, analyzing them to provide immediate feedback and comprehensive analytics. Its Shot Science function quantifies advanced metrics including release angle, vertical jump, and player speed, while also executing automatic identification of shot types. It also monitors advancement over time with complete exercise records, data, and skill assessments, and provides a tailored suggested workout daily featuring drills according to the player's proficiency. Its advantages position it as the nearest consumer equivalent to the suggested system, as it showcases a full personal-informatics cycle, seamless data gathering, analysis-driven reflection, and action via tailored recommendations, on user-friendly hardware, and it incorporates Apple Watch heart rate. Its flaws include being focused solely on basketball, functioning primarily in real-time instead of analyzing recorded game footage, failing to assess racket-court positioning, and providing social competition instead of skill-oriented opponent pairing.

2.2.6 OnForm

OnForm is a sports video analysis and coaching platform that allows athletes and coaches to record, review, and analyze technique using a mobile device. It supports slow-motion playback, frame-by-frame review, side-by-side comparison, voice-over feedback, and remote video sharing between coaches and athletes. Its relevance to the proposed system lies in its use of accessible video-based analysis

for performance reflection, allowing athletes to review specific movements and receive structured feedback without requiring expensive facility-based systems. This supports the personal informatics process by helping users collect visual performance data, reflect on their actions, and identify areas for improvement over time. However, OnForm is designed as a general sports coaching tool rather than a pickleball-specific analytics system. It does not automatically classify pickleball shots, track rally patterns, generate court-positioning heatmaps, or provide skill-based matchmaking for players.

2.2.7 Dartfish

Dartfish is a video analysis software platform used in sports performance, coaching, and athlete development. It allows users to capture, analyze, tag, compare, and share sports videos, making it useful for identifying technical errors, tactical patterns, and performance trends. Its tools support video annotation, movement comparison, and performance review, which can help athletes and coaches better understand weaknesses and develop training strategies. Dartfish is relevant to the proposed system because it demonstrates how video can be transformed into measurable and reviewable performance information. It also supports technical and tactical analysis, which is important in racket sports where positioning, timing, and shot selection affect performance. However, Dartfish is not specifically designed for pickleball and may require manual tagging or expert interpretation. It also functions more as a coaching and analysis platform than as

an automated personal informatics system with built-in recommendations, matchmaking, and court management features.

2.2.8 CoachNow

CoachNow is an all-in-one sports coaching application that combines video and image analysis, private communication, cloud storage, and feedback tools for coaches and athletes. It includes features such as slow-motion video review, side-by-side comparisons, voice-over feedback, CoachCam, and AI skeleton tracking for analyzing body movement and joint angles. Its relevance to the proposed system is its ability to support continuous improvement through recorded performance data, coach-athlete communication, and visual feedback. This makes it useful as a reference for the reflection and feedback stage of a personal informatics system. However, CoachNow is not pickleball-specific and is mainly designed around coaching communication rather than automatic match analytics. It does not directly analyze rally structure, shot outcomes, court zones, player coordination, or opponent compatibility, which limits its use as a complete model for pickleball performance monitoring.

2.2.9 DUPR

DUPR, or Dynamic Universal Pickleball Rating, is a pickleball rating system that evaluates player skill through match results and assigns players a numerical

rating. Its rating scale begins at 2.000 for beginners and extends to 8.000 for professional-level players. DUPR is widely used for organizing fair matches, leagues, and tournaments by helping players understand their skill level and find more balanced competition. Its relevance to the proposed system lies in its use of player data to support skill-based classification and matchmaking. This is important for pickleball communities because mismatched games can reduce both competitiveness and enjoyment. However, DUPR is mainly result-based rather than performance-based. It does not automatically analyze gameplay footage, identify technical weaknesses, classify shots, or explain the specific movement and tactical behaviors behind a player's rating.

2.2.10 Pickleheads

Pickleheads is a pickleball community platform that helps players find courts, join games, organize sessions, and connect with other players. It provides a court directory, open play schedules, local groups, game sign-ups, and skill-level-based game discovery. Pickleheads is relevant to the proposed system because it addresses the social and logistical side of pickleball participation, particularly court discovery, player connection, and organized play. These features are useful references for the proposed system's court reservation, queueing, and matchmaking components. However, Pickleheads is not primarily a performance monitoring or analytics platform. It does not automatically collect gameplay data, analyze player movement, classify shots, generate performance dashboards, or

provide AI-based recommendations for technical improvement. Therefore, it is most useful as a model for community-building and game organization rather than as a direct model for computer-vision-based performance analysis.

2.2.11 Strengths and Weaknesses Table

Software / System	Strengths	Weaknesses
Zepp Tennis	Detailed swing data (speed, spin, sweet spot); stroke classification; tips and drills; community/leaderboards	Tennis-specific; proprietary hardware; hardware largely discontinued; no positioning or matchmaking
PlaySight SmartCourt	Automated multi-camera capture; auto-tracking, stats, and event tagging; supports pickleball; coach access	Requires fixed costly infrastructure; club/broadcast-oriented; no personal matchmaking
Arccos Caddie	Automatic shot tracking; strokes-gained analytics; AI-driven personalized recommendations; strong historical insight	Golf-specific; proprietary hardware + subscription; no computer-vision analysis; no matchmaking
Padelytics	Computer-vision player and ball tracking from uploaded match video; court-positioning heatmaps; shot classification; movement/fitness metrics; improvement tips	Padel-specific; oriented toward clubs with a court-camera setup; no skill-based matchmaking
HomeCourt	Phone-based computer vision; automatic shot, movement, and skill tracking; personalized drills and recommendations; progress history; Apple Watch heart rate; coach sharing	Basketball-specific; mainly real-time; no racket-court positioning; social play rather than skill-based matchmaking
OnForm	Accessible mobile video analysis; slow-motion and frame-by-frame review; side-by-side comparison;	General sports coaching tool; not pickleball-specific; no automatic shot classification, heatmaps, or matchmaking

	voice-over feedback; coach-athlete sharing	
Dartfish	Strong video annotation tools; movement comparison; tagging and sharing; useful for technical and tactical review	Not pickleball-specific; may require manual tagging; relies on coach or expert interpretation; no built-in matchmaking
CoachNow	Video and image analysis; coach feedback tools; cloud storage; side-by-side comparisons; AI skeleton tracking for movement analysis	Coaching-focused rather than match analytics-focused; not pickleball-specific; no rally, shot, court-zone, or matchmaking analysis
DUPR	Pickleball-specific rating system; supports fairer matches, tournaments, and leagues; useful for skill-based classification	Result-based rather than performance-based; no video analysis, shot classification, movement tracking, or technical feedback
Pickleheads	Court finder; game scheduling; open play organization; player connection; supports community and skill-level-based game discovery	Not a performance analytics system; no automatic gameplay data collection, computer-vision analysis, dashboards, or training recommendations
Proposed System	Integrates computer-vision gameplay and movement analysis, automatic shot classification, positioning/court-coverage analysis, skill and play-style assessment, matchmaking, personalized recommendations.	Integrates existing technologies;

2.2.12 Synthesis

The comparison reveals that every current system provides only a portion of what the proposed system intends to integrate. HomeCourt is the most similar in idea, showing a complete personal-informatics cycle with computer-vision capture, analytics, and tailored suggestions on a

regular phone, though it is designed for basketball and primarily functions in real-time without analyzing court positioning. Padelytics, focusing on the racket sport most similar to pickleball, demonstrates that computer vision can monitor players, categorize shots, and create court-positioning heatmaps from uploaded match videos, but it is tailored specifically to padel and club-centered. PlaySight demonstrates that automated, multi-angle gameplay capture along with positioning data and coach access is achievable and even accommodates pickleball, though it relies on expensive fixed infrastructure. Zepp Tennis offers advanced sensor-driven swing metrics, although it is tailored for tennis and relies on specific hardware. Arccos illustrates how automatic data gathering can be combined with strokes-gained analysis and AI-based personalized suggestions, but it depends on sensors instead of computer vision and is limited to golf. The proposed solution will be informed by their unique strengths, accessible CV tracking, positioning analysis, shot classification, swing metrics, historical analytics, and tailored recommendations, while their common limitation, the lack of a single platform that merges all these aspects along with skill-based matchmaking for pickleball within one personal-informatics loop, highlights the research gap that the suggested system aims to fill.

2.2.13 Features Table

Feature	Home Court	Padelytics	PlaySight SmartCourt	Zepp Tennis	Arccos Caddie	On Form	Dart fish	Coach Now	DUPR	Pickle heads	Proposed System
Computer Vision Analysis	✓	✓	✓	✗	✗	△	△	✓	✗	✗	✓
Shot Classification	✓	✓	✓	✓	✓	✗	△	✗	✗	✗	✓
Positioning / Court Coverage	✗	✓	✓	✗	✗	✗	△	✗	✗	✗	✓

Skill & Play-Style Assessment	✓	✗	✗	✗	✓	△	△	△	✓	✗	✓
Matchmaking	✗	✗	✗	✗	✗	✗	✗	✗	△	✓	✓
Personalized Recommendations	✓	✓	✗	✓	✓	△	△	△	✗	✗	✓
Pickleball-Specific	✗	✗	✗	✗	✗	✗	✗	✗	✓	✓	✓

Legend: ✓ = Feature is clearly supported; ✗ = Feature is not supported; △ = Feature is partially supported, manual, coach-driven, or indirectly supported

Figure 2.2.8.1 Features table regarding availability of different functionalities per system

The Features table presents an overview of selected existing systems in relation to the proposed personal informatics system for pickleball players. The systems were evaluated based on key features relevant to sports performance monitoring, including computer vision analysis, shot classification, positioning and court coverage, skill and play-style assessment, matchmaking, personalized recommendations, and pickleball-specific functionality. These criteria were selected to determine the extent to which existing systems support automated data collection, performance reflection, and data-driven improvement.

The comparison shows that several existing systems provide strong capabilities in specific areas of sports performance analysis. HomeCourt, Padelytics, and PlaySight SmartCourt demonstrate the use of computer vision and automated tracking for performance monitoring. However, these systems are primarily designed for other sports or require specific infrastructure, which may limit their accessibility and direct applicability to pickleball players. Meanwhile, systems such as OnForm, Dartfish, and CoachNow provide video-based coaching and review tools, but their

functions are generally broad, coach-driven, or dependent on manual interpretation rather than fully automated pickleball performance analysis.

The table also highlights that DUPR and Pickleheads are more closely related to pickleball in terms of rating, player connection, and community participation. DUPR supports skill-based classification through player ratings, while Pickleheads assists players in finding courts, joining games, and organizing play sessions. However, these systems do not provide detailed gameplay analytics such as shot classification, player movement tracking, court-positioning analysis, or personalized technical recommendations.

While existing systems offer valuable individual features, none of them fully integrate computer vision analysis, shot classification, court-positioning assessment, skill and play-style evaluation, matchmaking, personalized recommendations, and pickleball-specific support within a single platform. This reveals a functional gap in existing sports performance systems and supports the need for the proposed system, which aims to provide an integrated personal informatics solution specifically designed to help pickleball players monitor, assess, and improve their performance through objective and data-driven feedback.

2.3. Review of Related Methodologies

2.3. Review of Related Methodologies

2.3.1 Scrum

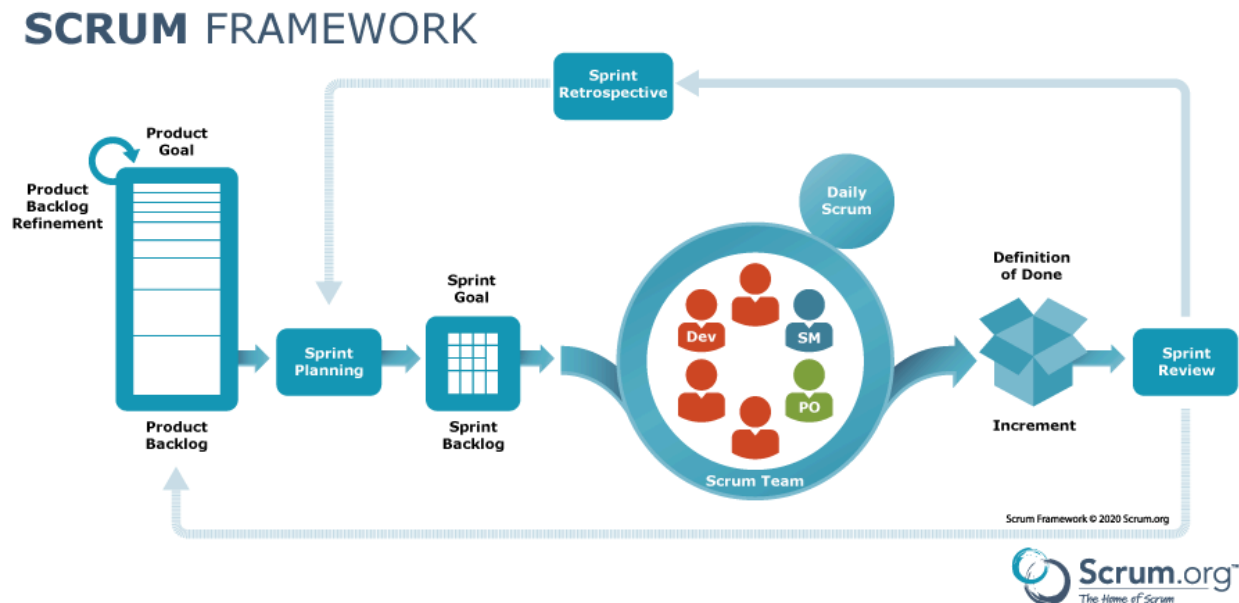


Figure 2.3.1.1 Scrum Framework

Scrum is one of the most popular Agile methodologies for running software projects, and it tends to shine when the work is complicated and the requirements are bound to change along the way (Schwaber & Sutherland, 2020). Instead of trying to map out the entire project before any code is written, Scrum splits the work into short, repeating cycles called sprints, usually two to four weeks each, with the team aiming to finish a working slice of the system by the end of every one. That rhythm fits the proposed pickleball system nicely, since its bigger parts are core performance monitoring, skill assessment, and gameplay context support can be built and polished one at a time rather than all at once.

Scrum runs on three roles. The Product Owner decides what the team tackles next and keeps the product backlog, the ordered list of features and fixes, up to date. The Scrum Master looks after

the process itself, making sure the team sticks to Scrum and clearing out whatever is getting in the way. Everyone else makes up the Development Team, the people who actually build the system and turn backlog items into a working increment each cycle (Schwaber & Sutherland, 2020). At the start of each sprint, the team picks a realistic batch of items from the backlog and commits to getting them done, which keeps everyone focused on a clear goal instead of an open-ended pile of work. And because each sprint produces something that actually works, the people who will use the system later such as the players and coaches can see where things are heading early on and weigh in, rather than waiting until the very end to find out the team built the wrong thing.

A handful of regular meetings hold the whole thing together. Sprint planning sets up what the cycle will cover, the daily stand-up is a quick check-in for sharing progress and raising blockers, and each sprint wraps with a review to show what was built and a retrospective to figure out what to do better next time (Schwaber & Sutherland, 2020). The product backlog, sprint backlog, and increment round it out as shared reference points, so anyone in the team or its stakeholders can tell at a glance what is being worked on and how far along it is.

The real draw, though, is how well Scrum handles change. Requirements almost never stay put, and Agile methods like Scrum were built with exactly that in mind (Beck et al., 2001). Since this project leans on existing computer vision instead of building new models, the team will not really know what the system can promise until it sees how those tools handle shot classification, movement tracking, positioning, and exertion on actual gameplay. Scrum's iterative, feedback-driven loop makes it easy to work those findings back into the project as they come up, which is a big part of why it is the methodology this study adopts.

2.3.2 DevOps

DevOps is less a step-by-step process and more a culture and a set of practices that bring software development and IT operations together, all aimed at shipping changes quickly, reliably, and without much friction (Kim et al., 2016). It leans hard on automation and on continuous integration and delivery, so that new code can be tested and released in small, frequent batches instead of big, occasional ones. In practice it usually sits on top of an Agile method rather than replacing it, since teams that already work in iterations adopt it to make their releases smoother and their feedback loops faster (Humble & Farley, 2010).

For a system that will eventually be used and updated on a regular basis, that mindset is appealing, because it keeps a product stable while still leaving room for constant improvement. The catch, at least for a capstone, is that DevOps assumes an operational side that a student project usually does not have. It relies on build pipelines, automated testing setups, and ongoing deployment to live environments, which is far more than is needed just to develop and demonstrate a prototype. DevOps tends to pay off most once a system is already running in production and being maintained, rather than during the initial build, which is why it fits this project better as a future direction than as the framework guiding the work now.

2.3.3 Waterfall

Waterfall is the classic, linear approach, where a project moves through a fixed sequence of phases such as requirements, design, implementation, testing, and deployment and each one has to be finished before the next begins (Royce, 1970). It rests on heavy planning right at the start and works best when the requirements are well understood and unlikely to move. Its biggest

strengths are clarity and predictability: everyone knows what is being built and in what order, and the thorough documentation it produces can make a project easy to follow.

That same rigidity is exactly where Waterfall runs into trouble for this project. The requirements here are expected to shift as the team learns how the computer vision actually behaves in real gameplay, and Waterfall makes those late changes slow and costly to absorb. Because feedback only shows up at the end of each phase, a problem introduced early can also go unnoticed until much later, when it is far harder to fix. For an exploratory system built around integrating existing technologies, that lack of flexibility is a serious drawback.

2.3.2 Comparison of Methodologies

Feature	Scrum	DevOps	Waterfall
Project structure	Iterative and incremental	Continuous, built on automated delivery	Linear and sequential
Planning	Light upfront, adaptive each sprint	Continuous, focused on delivery pipelines	Extensive upfront, detailed specifications
Requirements	Evolving requirements are accommodated	Evolving, supported through continuous releases	Fixed and cannot be changed
Feedback	Continuous, every sprint	Continuous and automated	Gathered at the end of each phase
Risk management	Addressed throughout the project	Reduced through frequent, small releases	Mainly during the planning phase
Change management	Relatively easy to implement	Easy, with automated testing and deployment	Difficult and costly
Team collaboration	Close collaboration and communication	Tight development-and-operations collaboration	Individual work and handoffs

Project visibility	High, with regular increments	High, with continuous monitoring	Low until the end of each phase
Customer involvement	Encouraged throughout the project	Indirect, through fast release and feedback	Limited until the end of each phase
Tooling infrastructure /	Lightweight, process-focused	Heavy; needs CI/CD and automation setup	Lightweight, document-focused
Best suited for	Iterative builds with evolving requirements	Systems in continuous deployment and maintenance	Projects with fixed, well-known requirements

Figure 2.3.2.1 Comparison of Software Development Methodologies

2.4 Synthesis

2.4 Synthesis

2.4.1 Common Themes

The reviewed literature and related systems commonly emphasize the shift from subjective performance evaluation to objective, data-driven feedback. In the related concepts, personal informatics is presented as a process of collecting, integrating, reflecting on, and acting upon personal data for improvement. This theme is reflected in sports performance monitoring, where players are encouraged to move beyond memory-based assessment and instead use measurable information to identify strengths, weaknesses, and progress over time. Computer vision also appears as a recurring theme because it allows gameplay video to be transformed into useful data such as player movement, shot actions, positioning, and tactical patterns. These concepts support the proposed system's focus on automatic data collection, visual feedback, and player reflection.

Another common theme is the importance of accessibility and usability in sports technologies. Several systems, such as HomeCourt, OnForm, Dartfish, and CoachNow, demonstrate how video-based tools can support performance review and coaching without relying entirely on manual observation. Other systems, such as Padelytics and PlaySight SmartCourt, show the potential of computer vision in tracking court movement, classifying actions, and generating performance statistics. Meanwhile, DUPR and Pickleheads highlight the social and community side of sports participation by supporting skill ratings, fairer play, court discovery, and player connection. Together, these systems show that performance improvement is not only about collecting data, but

also about making that data understandable, actionable, and useful within the player's actual sports environment.

The reviewed methodologies also show the importance of flexibility and iterative development. Since the proposed system depends on integrating existing technologies, particularly computer vision, the final system requirements may change as the team tests how these tools perform in actual pickleball gameplay. Scrum is therefore presented as the most suitable methodology because it allows the system to be developed in stages, tested regularly, and refined through continuous feedback.

2.4.2 Problems Currently Addressed

The existing literature and systems currently address several important problems in sports performance monitoring. First, they address the problem of unreliable self-assessment by allowing athletes to review performance through collected data rather than relying only on memory, intuition, or casual feedback. Personal informatics systems help users track their own behavior and performance, while computer vision enables more consistent capture of movement, positioning, and gameplay actions. This is especially useful in sports like pickleball, where rallies happen quickly and players may not clearly remember their positioning, shot choices, or errors after a match.

Second, existing systems address the need for technical and tactical analysis. Systems such as HomeCourt, Padelytics, PlaySight SmartCourt, Zepp Tennis, and Arccos Caddie demonstrate that modern sports tools can classify actions, track performance history,

generate statistics, and provide recommendations. Video analysis platforms such as OnForm, Dartfish, and CoachNow also support coaching and reflection by allowing users to review movement, compare performance, and receive feedback. These systems help athletes and coaches identify technical errors, tactical patterns, and areas for improvement.

Third, pickleball-related platforms address problems in player organization and fair competition. DUPR provides a rating system that helps classify players according to skill level, while Pickleheads helps players find courts, join games, and connect with other players. These systems address the social and logistical challenges of pickleball participation, especially in matching players with suitable games and organizing local play.

Finally, the methodology review addresses the problem of managing uncertainty during development. Waterfall is shown to be too rigid for a project whose requirements may evolve, while DevOps is more appropriate for systems already in continuous deployment and maintenance. Scrum addresses the need for an adaptive development process by allowing the team to build, test, and refine the proposed system through repeated sprint cycles.

2.4.3 Gaps in Literature

Despite the strengths of the reviewed concepts and systems, several gaps remain. The first major gap is the lack of a single pickleball-specific system that integrates computer

vision analysis, shot classification, positioning and court coverage, skill and play-style assessment, matchmaking, and personalized recommendations. Existing systems tend to focus on only one or two of these areas. For example, HomeCourt provides strong computer-vision-based basketball training, Padelytics offers padel-focused court and shot analytics, and PlaySight supports automated multi-camera capture but requires fixed infrastructure. These systems demonstrate useful features, but they are not fully designed around pickleball players and their specific gameplay needs.

A second gap is the limited availability of pickleball-specific performance analytics. While DUPR and Pickleheads are directly related to pickleball, they focus mainly on ratings, community connection, and game organization rather than detailed gameplay analysis. They do not automatically analyze shot types, player movement, rally patterns, court positioning, or technical weaknesses. This shows that pickleball platforms currently support participation and competition more than performance diagnosis and improvement.

A third gap is the limited connection between performance data and actionable improvement. Some systems collect large amounts of data, while others provide coaching tools or recommendations, but few combine automatic data collection, visual analytics, reflection, and personalized action in one continuous personal informatics loop. For the proposed system, this gap is important because the goal is not only to record gameplay data, but also to help players understand what the data means and how they can improve future performance.

Lastly, there is a gap in accessible implementation for recreational and community-level pickleball players. Some systems require proprietary sensors, subscriptions, coach interpretation, or fixed court-camera infrastructure. These limitations may reduce accessibility for ordinary players, schools, clubs, and local communities. The proposed system aims to address this gap by integrating existing technologies into a more focused personal informatics solution for pickleball, combining performance monitoring with player assessment and matchmaking features.

CHAPTER III Methodology

CHAPTER III Methodology

3.1. Description

The Scrum methodology lines up with how this project naturally unfolds. The problem the study is tackling needs a system that can be built and tested in stages instead of delivered in one big drop, and Scrum's sprints let the team stand up the most important pieces first, like the core performance monitoring dashboard, while the rest comes together over time. It takes changing requirements in stride, which matters when so much depends on testing the integrated technologies, and its steady collaboration keeps the computer vision, analytics, and reporting work pulling in the same direction.

3.2.1 Scrum VS Waterfall

When compared to Waterfall, Scrum is the choice that aligns better with the requirements of this project. Waterfall is the simplest to abandon since it requires the requirements to be fixed from the beginning, which they are not in this case; a significant portion of the effort relies on observing the performance of computer vision during actual gameplay before the team can confidently determine what the system will provide. Its inflexibility and tendency to reserve feedback until the conclusion of each phase render it a poor fit for such uncertainty.

In contrast, Scrum allows the system to be built and tested in stages instead of being delivered in one big drop. This makes it more suitable for a project where the team needs to test the core performance monitoring dashboard first, evaluate how the computer vision performs, and adjust the next parts of the system based on the results.

3.2.2 Scrum VS DevOps

DevOps is more appealing, and there are many aspects to appreciate in its automation and ongoing delivery, but fundamentally it is a method for releasing and managing existing software. It also presumes the presence of build pipelines and live deployment configurations that a capstone focused on a prototype does not possess.

Compared to DevOps, Scrum is more practical as the current framework directing the team because it focuses on organizing development through sprints, collaboration, feedback, and changing requirements. DevOps functions most effectively when integrated with an Agile process, indicating that it is better viewed as an evolution for the system in the future rather than as the current framework directing the team.

3.2. Phases

The development of the proposed pickleball system will follow the Scrum framework, organized into seven two-week sprints for CAP-IT1. The sprint sequence is designed to build the system incrementally, prioritizing Core Performance Monitoring first, followed by Light Skill and Performance Assessment and Gameplay Context Support, in line with the MVP scope defined in Chapter 1.

3.2.1 Sprint 1: Project Setup and Video Upload Module

3.2.1.1 Description of the Phase

Sprint 1 will focus on establishing the technical foundation of the system. The team will set up the project repository, define the initial system architecture, and configure the

development environment for both the front end and backend services. The primary deliverable will be a working video upload interface that allows players to submit match footage along with basic match data, laying the groundwork for the computer vision pipeline to be integrated in later sprints.

3.2.1.2 Tools / Techniques / Models to be Used

- **Project management:** Jira for backlog, sprint board, and sprint planning.
- **Development:** Github for version control; Supabase for storage and authentication; front-end framework for the upload interface.
- **Design:** Figma for wireframes of the upload flow.
- **Technique:** User story mapping to break down the “upload match video” epic into backlog items

3.2.1.3 Inputs and Outputs

Input

Product backlog items for Sprint 1, system architecture requirements from Chapter 1 scope, UI wireframes

Output

Working video upload module, configured repository and development environment, updated sprint backlog, sprint review notes

3.2.1.4 Anticipated Risks and Mitigation Strategies

A possible risk is inconsistent handling of large video files during upload, which could cause failed submissions. The team plans to mitigate this by implementing chunked upload handling.

3.2.2 Sprint 2: Computer Vision Pipeline Integration

3.2.2.1 Description of the Phase

Sprint 2 will focus on integrating the core computer vision pipeline for post-match video processing. The team will implement rally segmentation and player tracking using the selected CV models, converting uploaded match footage into structure and analyzable events.

3.2.2.2 Tools / Techniques / Models to be Used

- **Computer vision:** YOLOv8 for object/player detection, ByteTrack for multi-object tracking, 12-point court keypoint detection, and OpenCV homography for bird's-eye court projection.
- **Backend:** Server-side batch processing workers, Supabase for storing processed outputs
- **Technique:** Sprint planning and backlog refinement carried over from Sprint 1

3.2.2.3 Inputs and Outputs

Inputs

Uploaded match videos and metadata from Sprint 1, selected CV model configurations

Outputs

Rally-segmented match data, player tracking data, stored processed video events

3.2.2.4 Anticipated Risks and Mitigation Strategies

Possible risk is degraded tracking accuracy caused by video quality or camera angles from user-submitted footage. The team plans to mitigate this by defining minimum video quality guidelines for uploads and testing the pipeline against a range of sample footage early in the sprint.

3.2.3 Sprint 3: Shot Classification and Court Analytics

3.2.3.1 Description of the Phase

Sprint 3 will build on the tracking data produced in Sprint 2 to classify common pickleball shots (dinks, drives, volleys, lobs) and generate court-based analytics such as heatmaps, kitchen-zone time, and court coverage.

3.2.3.2 Tools / Technique / Models to be used

- **Computer vision:** OpenCV and MediaPipe for movement and shot-pattern analysis, using the existing system.
- **Shot classification:** XGBoost, an existing, pre-built classification algorithm, trained on ball speed, trajectory arc height, and court-zone-at-contact features engineered by the team from Pickleball-Analytics' tracking output.
- **Data processing:** Server side-workers for heatmap and zone-time computation
- **Technique:** Sprint review with sample outputs validated against manually reviewed match clips

3.2.3.3 Inputs and Outputs

Input

Player tracking and rally segmented data from Sprint 2

Output

Shot classification labels, heatmaps, kitchen-zone time metrics, court coverage data

3.2.3.4 Anticipated Risks and Mitigation Strategies

A possible risk is misclassification of visually similar shots, which could reduce the reliability of analytics. The team plans to mitigate this by validating classification outputs against manually labeled sample clips and adjusting detection thresholds accordingly.

3.2.4 Sprint 4: Analytics Dashboard

3.2.4.1 Description of the Phase

Sprint 4 will focus on presenting the processed performance data through a player-facing analytics dashboard, supporting trend monitoring, historical vs current match comparison, and goal setting, consistent with the personal informatics reflection stage.

3.2.4.2 Tools / Technique / Models to be Used

- **Front end:** React/Next.js with charting library for visual analytics
- **Backend:** Supabase queries for aggregating historical performance data
- **Technique:** Usability review of dashboard wireframes with sample users before implementation

3.2.4.3 Inputs and Outputs

Inputs

Shot classification and court analytics data from Sprint 3, dashboard wireframes

Output

Working analytics dashboard, goal-setting feature, historical performance comparison view

3.2.4.4 Anticipated Risks and Mitigation Strategies

A possible risk is information overload from presenting too many metrics at once, reducing usability. The team plans to mitigate this by prioritizing key performance indicators in the default view and placing detailed metrics behind expendable sections.

3.2.5 Sprint 5: Skill Level and Play Style Estimation

3.2.5.1 Description of the Phase

Sprint 5 will implement the Light Skill and Performance Assessment component, estimating the player's skill level (beginner, intermediate, advanced) and identifying their dominant play styles (aggressive, defensive, balanced) based on accumulated performance data.

3.2.5.2 Tools / Techniques / Models to be used

- **Data analysis:** Rule-based or lightweight classification logic applied to aggregated shot and movement data

- **Backend:** Supabase for storing skill and play-style profiles
- **Technique:** Backlog refinement to define clear skill-level thresholds before implementation

3.2.5.3 Inputs and Outputs

Input

Aggregated performance and shot data from Sprints 2-4

Output

Player skill-level estimates, play-style classification, updated player profile

3.2.5.4 Anticipated Risks and Mitigation Strategies

A possible risk is inaccurate skill estimation for players with limited match history. The team plans to mitigate this by requiring a minimum number of analyzed matches before generating a skill estimate and clearly labeling estimates as provisional until more data is collected.

3.2.6 Sprint 6: Gameplay Context Support

3.2.6.1 Description of the Phase

Sprint 6 will implement gameplay context tagging, allowing performance data to be organized by activity type (practice, casual match, tournament, singles, doubles), and ensuring historical records remain consistent across contexts.

3.2.6.2 Tools / Techniques / Models to be used

- **Backend:** Metadata tagging schema in Supabase for match context
- **Front end:** UI Controls for players to label match context during upload
- **Technique:** Sprint planning to prioritize the most common context types first

3.2.6.3 Inputs and Outputs

Input

Video upload module from Sprint 1, dashboard from Sprint 4

Output

Context-tagged match records, filterable historical performance view by activity type

3.2.6.4 Anticipated Risks and Mitigation Strategies

A possible risk is players mislabeling or skipping context tags, reducing data consistency.

The team plans to mitigate this by making context selection a required step in the upload flow with sensible defaults

3.2.7 Sprint 7: Post-Game Audio Feedback

3.2.7.1 Description of the Phase

Sprint 7 will implement the audio feedback layer for web-based frameworks, using retrieval-augmented generation (RAG) to produce the underlying feedback text. This includes converting live computer vision outputs into short spoken coaching cues during play, and converting completed post-game performance data — shot accuracy, shot type distribution, kitchen-zone activity — into a spoken match summary players can replay from the dashboard.

3.2.7.2 Tools / Techniques / Models to be Used

- **Tools:** Google Cloud TTS (speech) + RAG stack (pgvector → LLM API) for the feedback text\
- **Technique:** Sprint review - compare generated feedback/audio against manually written samples for accuracy and tone.
- **Backend:** Supabase - stores audio files, player history embeddings, match records

3.2.7.3 Inputs and Outputs

Input

Live CV data (during play) + processed match stats (after) + player history

Output

replayable post-game audio summary

3.2.7.4 Anticipated Risk and Mitigation

Audio Feedback is not accurate. The team plans to not overcomplicate the match statistics to help the Feedback be more accurate.

3.2.8 Sprint 8: Coach Access and Integration Testing

3.2.8.1 Description of the Phase

Sprint 7 will close out CAP-IT1 by implementing the cooperative access layer for coaches, including role-based access control, and conducting integration testing across all modules built during CAP-IT1 to ensure the systems functions cohesively.

3.2.8.2 Tools / Techniques / Models to be Used

- **Backend:** Supabase authentication and RBAC policies

- **Testing:** End-to-end integration testing across upload, CV pipeline, dashboard, and profile modules
- **Technique:** Final sprint review and retrospective for CAP-IT1

3.2.8.3 Inputs and Outputs

Input

All modules developed in Sprints 1-6

Output

Coach access module with RBAC, integrated and tested system covering 80% of proposed functionality, CAP-IT1 sprint review documentation

3.2.8.4 Anticipated Risks and Mitigation Strategies

A possible risk is unforeseen integration conflicts between modules in separate sprints (e.g., data format mismatches between the CV pipeline and dashboard). The researchers plan to mitigate this by conducting incremental integration checks at the end of each sprint rather than deferring all integration to Sprint 7.

3.3. Work Plan / Gantt Chart

[illegible]

Classification and Court Analytics														
Sprint 4: Analytics Dashboard, Reflection, and Recommendations														
Sprint 5: Skill Level, Play Style Estimation, and Player Matching														
Sprint 6: Gameplay Context Support, Find a Game, and Microchallenges														
80% Completion Check and Buffer														
Sprint 7: Coach Access, Integration Testing, and MVP Refinement														
RAG R1: Knowledge Base Preparation														
RAG R2: RAG Feedback / Live Audio Prototype														
System QA and Bug Fixing														
User Testing and Revisions														
Documentation														
Final Polish														

Figure 3.3.1 Project Gantt Chart

Figure 3.3.1 presents the work plan for the development of the proposed Pickleball Performance Analytics System from September to March. The project follows the Scrum framework, with development organized into seven two-week sprints. The schedule is designed to deliver the system incrementally, beginning with the technical foundation before progressing to the core performance monitoring features, player analytics, gameplay support, and final system integration.

Development begins in September with Sprint 1, which focuses on project setup, system architecture, development environment configuration, and the implementation of the video upload module. This is immediately followed by Sprint 2, where the computer vision pipeline is integrated to support rally segmentation, player detection, and player tracking. Sprint 3 continues the development of the Core Performance Monitoring component by implementing shot classification and generating court analytics such as heatmaps, kitchen-zone metrics, and court coverage.

During October, the project shifts toward the player-facing features. Sprint 4 develops the analytics dashboard together with reflection and recommendation features that allow players to review their historical performance and monitor improvement over time. This is followed by Sprint 5, which implements lightweight skill level estimation, play style classification, and player matching based on accumulated performance data.

In November, Sprint 6 introduces gameplay context support, including match context tagging, the Find-a-Game feature, and microchallenges that encourage continuous player engagement. At the end of the month, an 80% completion check and development buffer is scheduled to verify that the minimum functionality required for Capstone 1 has been achieved while allowing sufficient time to resolve technical issues before the final integration phase.

Beginning in December, Sprint 7 focuses on implementing coach access through role-based access control (RBAC), completing system integration, and refining the minimum viable product (MVP). During the same period, two exploratory RAG development iterations are scheduled. RAG R1 is dedicated to preparing the retrieval knowledge base, while RAG R2 develops a prototype for retrieval-assisted feedback and live audio coaching. These activities are treated as

enhancement features and are developed independently of the core MVP to ensure that the primary project objectives remain on schedule.

The remaining project timeline is allocated to system quality assurance and bug fixing, user testing and revisions, documentation, and final polishing. These activities provide sufficient time to validate system functionality, incorporate user feedback, complete the required project documentation, and prepare the system for the Capstone 2 presentation and final evaluation.

3.4 Technology Stack

The proposed system is built by integrating existing, proven computer vision and web technologies rather than developing new models from scratch. The core detection, tracking, and court-mapping layer is built on Pickleball-Analytics, which already combines these into a working pipeline; the table below breaks that pipeline into layers alongside the additional components the team is building on top of it (shot classification, audio feedback, backend, frontend).

Layer	Tools / Technologies	Role in the System
Object Detection	YOLOv8, via Pickleball-Analytics	Detects ball, player, and paddle bounding boxes per frame.
Multi-Object Tracking	ByteTrack / BoT-SORT	Assigns persistent IDs to the detected player and ball across frames so movement can be followed over time, including

		through brief occlusion. Extended by the team for single-player selection
Pose Estimation	MediaPipe Pose	Extracts player joint landmarks for movement-quality cues in the real-time framework
Court Mapping	OpenCV homography, 12-keypoint model	Projects pixel coordinates for the ball and player onto real-world court coordinates using calibrated reference points, enabling zone-based metrics (kitchen time, court coverage). Extended by the team for half-court, tripod-height framing.
Video Processing	OpenCV, FFmpeg	Frame extraction, resizing, and format handling for uploaded match footage prior to model inference.
Shot Classification	XGBoost (XGBClassifier Api) existing, pre-built classification	Classifies each detected contact as a shot type (dink, drive, lob, serve, volley) using speed, arc, and court-zone features derived from tracking output.
Batch Processing	Server-side workers	Runs the detection, tracking, and classification pipeline on uploaded videos outside the browser for reliability and consistent performance.

Audio Feedback	RAG over player history	Converts structured performance data into short spoken feedback for post-game summaries.
Backend / Storage	Supabase (Postgres, auth, storage)	Stores match videos, tracking output, player profiles, and generated analytics; handles authentication and role-based access.
Frontend	React / Next.js, charting library	Renders the player and coach dashboards, visual analytics, and historical trend views.
Project Management	Jira, Figma	Sprint planning and interface design, per the Scrum workflow in Chapter III.

3.4.1 Data Specification

To make the system's data collection concrete, the table below specifies the exact fields captured or derived at each stage of the computer vision pipeline, distinct from the general metric categories already listed in Figure 1.4.1 (PPIF Stage Matrix).

Data Category	Specific Fields Captured / Derived	Source
Raw detection (per frame)	Ball bounding box (x, y, width, height, confidence); player bounding box; paddle bounding box; frame timestamp	YOLOv8 inference
Tracking (per frame)	Persistent track ID for player and ball; frame-to-frame displacement	ByteTrack / BoT-SORT

Court-mapped position (per frame)	Ball position in real-world court coordinates (x, y in feet/meters from baseline and sideline); player position in court coordinates; court zone flag (kitchen / mid-court / baseline)	OpenCV homography using calibrated court reference points
Derived ball kinematics (per shot)	Ball speed at contact (court-units/sec, computed from consecutive frame positions); trajectory arc height (max vertical pixel/position deviation between contact and next bounce or contact); time-to-bounce	Computed from tracked ball positions
Contact / shot event	Contact frame index and timestamp; contact court-zone; ball speed and direction change at that frame; paddle-ball bounding-box proximity at contact	Derived from ball + paddle tracking
Shot classification output	Shot type label (dink, drive, lob, serve, volley) per detected contact; classifier confidence score	Trajectory-feature classifier
Match-level aggregates	Shot type distribution and accuracy per match; kitchen-zone dwell time; court coverage/heatmap data; session date and context tag (practice, casual, tournament)	Aggregated from rally-level data, stored in player profile

3.5 Capstone 1 & 2 Targets

The development of the proposed pickleball personal informatics system will be divided into two major completion stages: Capstone 1 and Capstone 2. These stages will ensure that the project is developed, evaluated, and refined progressively. Capstone 1 will focus on completing at least 80% of the proposed system, particularly the major functional components needed for the minimum viable product. Capstone 2 will focus on completing the remaining components, conducting user acceptance testing, refining the system, and preparing the final implementation and documentation.

3.5.1 Capstone 1 Targets

Capstone 1 is expected to cover the main development phase of the system. By the end of Capstone 1, the project should reach approximately 80% completion. This means that the primary system modules must already be functional enough to demonstrate how the proposed solution addresses the identified problems and opportunities related to pickleball performance tracking, self-reflection, and player improvement. The following components are expected to be completed and evaluated during Capstone 1:

Video Upload and Match Data Module

The system must allow players to upload recorded pickleball match videos and input basic match information such as match date, match type, gameplay context, and player notes. This component serves as the entry point of the system and provides the data needed for performance analysis.

Computer Vision Processing Pipeline

The system must be able to process uploaded match footage using existing computer vision technologies. This includes player detection, movement tracking, rally segmentation, and the extraction of basic gameplay events. The researchers will not develop new computer vision algorithms, but will integrate existing models and frameworks such as YOLOv8, ByteTrack, OpenCV, and MediaPipe.

Shot Classification and Court Analytics

The system must classify common pickleball shots such as dinks, drives, volleys, and lobs based on available video and movement data. It must also generate court-based analytics such as player positioning, heatmaps, kitchen-zone time, and court coverage. These outputs will support objective performance monitoring and allow players to better understand their strengths and weaknesses.

Analytics Dashboard

The system must present processed performance data through a player-facing dashboard. The dashboard should display key performance indicators, match summaries, court analytics, shot distribution, performance trends, and historical comparisons. This component will allow players to review their gameplay in a clear and organized manner.

Goal Setting, Reflection, and Recommendations

The system must support the personal informatics aspect of the study by allowing players to set goals, reflect on post-match performance, and receive basic recommendations based on collected data. These recommendations may include focus areas such as improving court coverage, increasing shot variety, maintaining better positioning, or spending more time near the kitchen line.

Skill Level and Play Style Estimation

The system must provide a light assessment of player characteristics based on accumulated performance data. This includes estimating whether a player is beginner, intermediate, or advanced, and identifying whether the player's dominant play style is

aggressive, defensive, or balanced. These classifications will be treated as general estimates and not as official rankings.

Player Matching Support

The system must include basic player matching support based on skill level, play style, availability, and preferred gameplay type. This feature supports the opportunity for players to find suitable opponents or partners within the same pickleball community.

Gameplay Context Support

The system must organize performance records according to different gameplay contexts such as practice, casual matches, tournaments, leveling games, singles, doubles, wall practice, and ball machine practice. This allows players to compare their progress across different playing situations.

3.5.2 Capstone 2 Targets

Capstone 2 will cover the completion of the remaining components, system refinement, user acceptance testing, and final documentation. While Capstone 1 focuses on building the core system, Capstone 2 will focus on ensuring that the system is reliable, usable, and able to address the problems identified in the study. The following components are expected to be completed and evaluated during Capstone 2:

Coach Access and Role-Based Permissions

The system must provide a cooperative access layer where authorized individuals, such as coaches, can view player performance information. Role-based access control must be implemented to ensure that only authorized users can access player data. This feature supports guided improvement while maintaining user privacy and data control.

RAG and Audio Coaching Assistance

The RAG component may be used to retrieve coaching knowledge, drill suggestions, player history, and feedback templates to generate more grounded performance recommendations. The Audio Coaching Assistance feature may convert selected feedback into short audio cues during practice or training sessions. However, these features will remain supplementary and will not replace professional coaching, medical advice, officiating, or referee decision-making.

Because these features may increase the technical complexity of the project, they will only be pursued if the core system has already met the required level of completion and testing. If not completed within the project timeline, they may be discussed as future enhancements.

Full System Integration

All modules developed during Capstone 1 must be integrated into one cohesive system. This includes the upload module, computer vision pipeline, shot classification, court analytics, dashboard, goal setting, reflection, recommendations, skill assessment, player matching, gameplay context support, Find a Game, microchallenges, and coach access.

System Testing and Bug Fixing

The group must conduct technical testing to identify and resolve errors, data mismatches, broken workflows, and usability issues. Testing must cover the complete system flow, from video upload and processing to dashboard display and coach access.

User Acceptance Testing

User acceptance testing will be conducted with target users such as pickleball players and, when possible, coaches or experienced players. The purpose of this testing is to determine whether the system is understandable, useful, and relevant to the needs of its intended users. Feedback from user testing will be used to refine the dashboard, recommendations, reflection features, player matching, and other user-facing functions.

Final Refinement and Documentation

The remaining period of Capstone 2 will be used for final system improvements, interface polishing, documentation completion, evaluation write-ups, and preparation for the final defense. The group must also ensure that the final manuscript reflects the actual completed system and clearly discusses its scope, limitations, testing results, and recommendations for future development.

3.5.3 Data Requirements for Demonstration and Evaluation

To demonstrate that the developed system addresses the identified problems and opportunities, the group must prepare sufficient sample data for testing and simulation.

This includes recorded pickleball videos, sample player profiles, sample match records, gameplay context labels, skill-level examples, play-style examples, availability schedules, sample game listings, and microchallenge records.

The data must be able to simulate different scenarios where the system may be used, such as a beginner reviewing a casual match, an intermediate player tracking improvement over time, a player looking for a suitable match partner, a coach reviewing a player's performance, and a player comparing performance across practice and competitive contexts.

Through these scenarios, the group will be able to demonstrate that the system supports automated data collection, performance reflection, actionable insights, player matching, and continuous improvement within the defined scope of the study.

3.1. Description

3.1. Description

The Scrum methodology lines up with how this project naturally unfolds. The problem the study is tackling needs a system that can be built and tested in stages instead of delivered in one big drop, and Scrum's sprints let the team stand up the most important pieces first, like the core performance monitoring dashboard, while the rest comes together over time. It takes changing requirements in stride, which matters when so much depends on testing the integrated technologies, and its steady collaboration keeps the computer vision, analytics, and reporting work pulling in the same direction.

3.2.1 Scrum VS Waterfall

When compared to Waterfall, Scrum is the choice that aligns better with the requirements of this project. Waterfall is the simplest to abandon since it requires the requirements to be fixed from the beginning, which they are not in this case; a significant portion of the effort relies on observing the performance of computer vision during actual gameplay before the team can confidently determine what the system will provide. Its inflexibility and tendency to reserve feedback until the conclusion of each phase render it a poor fit for such uncertainty.

In contrast, Scrum allows the system to be built and tested in stages instead of being delivered in one big drop. This makes it more suitable for a project where the team needs to test the core performance monitoring dashboard first, evaluate how the computer vision performs, and adjust the next parts of the system based on the results.

3.2.2 Scrum VS DevOps

DevOps is more appealing, and there are many aspects to appreciate in its automation and ongoing delivery, but fundamentally it is a method for releasing and managing existing software. It also presumes the presence of build pipelines and live deployment configurations that a capstone focused on a prototype does not possess.

Compared to DevOps, Scrum is more practical as the current framework directing the team because it focuses on organizing development through sprints, collaboration, feedback, and changing requirements. DevOps functions most effectively when integrated with an Agile process, indicating that it is better viewed as an evolution for the system in the future rather than as the current framework directing the team.

3.2. Phases

3.2. Phases

The development of the proposed pickleball system will follow the Scrum framework, organized into seven two-week sprints for CAP-IT1. The sprint sequence is designed to build the system incrementally, prioritizing Core Performance Monitoring first, followed by Light Skill and Performance Assessment and Gameplay Context Support, in line with the MVP scope defined in Chapter 1.

3.2.1 Sprint 1: Project Setup and Video Upload Module

3.2.1.1 Description of the Phase

Sprint 1 will focus on establishing the technical foundation of the system. The team will set up the project repository, define the initial system architecture, and configure the development environment for both the front end and backend services. The primary deliverable will be a working video upload interface that allows players to submit match footage along with basic match data, laying the groundwork for the computer vision pipeline to be integrated in later sprints.

3.2.1.2 Tools / Techniques / Models to be Used

- **Project management:** Jira for backlog, sprint board, and sprint planning.
- **Development:** Github for version control; Supabase for storage and authentication; front-end framework for the upload interface.
- **Design:** Figma for wireframes of the upload flow.
- **Technique:** User story mapping to break down the “upload match video” epic into backlog items

3.2.1.3 Inputs and Outputs

Input

Product backlog items for Sprint 1, system architecture requirements from Chapter 1 scope, UI wireframes

Output

Working video upload module, configured repository and development environment, updated sprint backlog, sprint review notes

3.2.1.4 Anticipated Risks and Mitigation Strategies

A possible risk is inconsistent handling of large video files during upload, which could cause failed submissions. The team plans to mitigate this by implementing chunked upload handling.

3.2.2 Sprint 2: Computer Vision Pipeline Integration

3.2.2.1 Description of the Phase

Sprint 2 will focus on integrating the core computer vision pipeline for post-match video processing. The team will implement rally segmentation and player tracking using the selected CV models, converting uploaded match footage into structure and analyzable events.

3.2.2.2 Tools / Techniques / Models to be Used

- **Computer vision:** YOLOv8 for object/player detection, ByteTrack for multi-object tracking, 12-point court keypoint detection, and OpenCV homography for bird's-eye court projection.
- **Backend:** Server-side batch processing workers, Supabase for storing processed outputs
- **Technique:** Sprint planning and backlog refinement carried over from Sprint 1

3.2.2.3 Inputs and Outputs

Inputs

Uploaded match videos and metadata from Sprint 1, selected CV model configurations

Outputs

Rally-segmented match data, player tracking data, stored processed video events

3.2.2.4 Anticipated Risks and Mitigation Strategies

Possible risk is degraded tracking accuracy caused by video quality or camera angles from user-submitted footage. The team plans to mitigate this by defining minimum video quality guidelines for uploads and testing the pipeline against a range of sample footage early in the sprint.

3.2.3 Sprint 3: Shot Classification and Court Analytics

3.2.3.1 Description of the Phase

Sprint 3 will build on the tracking data produced in Sprint 2 to classify common pickleball shots (dinks, drives, volleys, lobs) and generate court-based analytics such as heatmaps, kitchen-zone time, and court coverage.

3.2.3.2 Tools / Technique / Models to be used

- **Computer vision:** OpenCV and MediaPipe for movement and shot-pattern analysis, using the existing system.
- **Shot classification:** XGBoost, an existing, pre-built classification algorithm, trained on ball speed, trajectory arc height, and court-zone-at-contact features engineered by the team from Pickleball-Analytics' tracking output.
- **Data processing:** Server side-workers for heatmap and zone-time computation
- **Technique:** Sprint review with sample outputs validated against manually reviewed match clips

3.2.3.3 Inputs and Outputs

Input

Player tracking and rally segmented data from Sprint 2

Output

Shot classification labels, heatmaps, kitchen-zone time metrics, court coverage data

3.2.3.4 Anticipated Risks and Mitigation Strategies

A possible risk is misclassification of visually similar shots, which could reduce the reliability of analytics. The team plans to mitigate this by validating classification outputs against manually labeled sample clips and adjusting detection thresholds accordingly.

3.2.4 Sprint 4: Analytics Dashboard

3.2.4.1 Description of the Phase

Sprint 4 will focus on presenting the processed performance data through a player-facing analytics dashboard, supporting trend monitoring, historical vs current match comparison, and goal setting, consistent with the personal informatics reflection stage.

3.2.4.2 Tools / Technique / Models to be Used

- **Front end:** React/Next.js with charting library for visual analytics
- **Backend:** Supabase queries for aggregating historical performance data
- **Technique:** Usability review of dashboard wireframes with sample users before implementation

3.2.4.3 Inputs and Outputs

Inputs

Shot classification and court analytics data from Sprint 3, dashboard wireframes

Output

Working analytics dashboard, goal-setting feature, historical performance comparison view

3.2.4.4 Anticipated Risks and Mitigation Strategies

A possible risk is information overload from presenting too many metrics at once, reducing usability. The team plans to mitigate this by prioritizing key performance indicators in the default view and placing detailed metrics behind expendable sections.

3.2.5 Sprint 5: Skill Level and Play Style Estimation

3.2.5.1 Description of the Phase

Sprint 5 will implement the Light Skill and Performance Assessment component, estimating the player's skill level (beginner, intermediate, advanced) and identifying their dominant play styles (aggressive, defensive, balanced) based on accumulated performance data.

3.2.5.2 Tools / Techniques / Models to be used

- **Data analysis:** Rule-based or lightweight classification logic applied to aggregated shot and movement data
- **Backend:** Supabase for storing skill and play-style profiles
- **Technique:** Backlog refinement to define clear skill-level thresholds before implementation

3.2.5.3 Inputs and Outputs

Input

Aggregated performance and shot data from Sprints 2-4

Output

Player skill-level estimates, play-style classification, updated player profile

3.2.5.4 Anticipated Risks and Mitigation Strategies

A possible risk is inaccurate skill estimation for players with limited match history. The team plans to mitigate this by requiring a minimum number of analyzed matches before generating a skill estimate and clearly labeling estimates as provisional until more data is collected.

3.2.6 Sprint 6: Gameplay Context Support

3.2.6.1 Description of the Phase

Sprint 6 will implement gameplay context tagging, allowing performance data to be organized by activity type (practice, casual match, tournament, singles, doubles), and ensuring historical records remain consistent across contexts.

3.2.6.2 Tools / Techniques / Models to be used

- **Backend:** Metadata tagging schema in Supabase for match context
- **Front end:** UI Controls for players to label match context during upload
- **Technique:** Sprint planning to prioritize the most common context types first

3.2.6.3 Inputs and Outputs

Input

Video upload module from Sprint 1, dashboard from Sprint 4

Output

Context-tagged match records, filterable historical performance view by activity type

3.2.6.4 Anticipated Risks and Mitigation Strategies

A possible risk is players mislabeling or skipping context tags, reducing data consistency.

The team plans to mitigate this by making context selection a required step in the upload flow with sensible defaults

3.2.7 Sprint 7: Post-Game Audio Feedback

3.2.7.1 Description of the Phase

Sprint 7 will implement the audio feedback layer for web-based frameworks, using retrieval-augmented generation (RAG) to produce the underlying feedback text. This includes converting live computer vision outputs into short spoken coaching cues during play, and converting completed post-game performance data — shot accuracy, shot type distribution, kitchen-zone activity — into a spoken match summary players can replay from the dashboard.

3.2.7.2 Tools / Techniques / Models to be Used

- **Tools:** Google Cloud TTS (speech) + RAG stack (pgvector → LLM API) for the feedback text\
- **Technique:** Sprint review - compare generated feedback/audio against manually written samples for accuracy and tone.
- **Backend:** Supabase - stores audio files, player history embeddings, match records

3.2.7.3 Inputs and Outputs

Input

Live CV data (during play) + processed match stats (after) + player history

Output

replayable post-game audio summary

3.2.7.4 Anticipated Risk and Mitigation

Audio Feedback is not accurate. The team plans to not overcomplicate the match statistics to help the Feedback be more accurate.

3.2.8 Sprint 8: Coach Access and Integration Testing

3.2.8.1 Description of the Phase

Sprint 7 will close out CAP-IT1 by implementing the cooperative access layer for coaches, including role-based access control, and conducting integration testing across all modules built during CAP-IT1 to ensure the systems functions cohesively.

3.2.8.2 Tools / Techniques / Models to be Used

- **Backend:** Supabase authentication and RBAC policies
- **Testing:** End-to-end integration testing across upload, CV pipeline, dashboard, and profile modules
- **Technique:** Final sprint review and retrospective for CAP-IT1

3.2.8.3 Inputs and Outputs

Input

All modules developed in Sprints 1-6

Output

Coach access module with RBAC, integrated and tested system covering 80% of proposed functionality, CAP-IT1 sprint review documentation

3.2.8.4 Anticipated Risks and Mitigation Strategies

A possible risk is unforeseen integration conflicts between modules in separate sprints (e.g., data format mismatches between the CV pipeline and dashboard). The researchers plan to mitigate this by conducting incremental integration checks at the end of each sprint rather than deferring all integration to Sprint 7.

3.3. Work Plan / Gantt Chart

3.3. Work Plan / Gantt Chart

Task	Sept		Oct		Nov		Dec		Jan		Feb		Mar	
	Wk 1-2	Wk 3-4	Wk 1-2	Wk 3-4	Wk 1-2	Wk 3-4	Wk 1-2	Wk 3-4	Wk 1-2	Wk 3-4	Wk 1-2	Wk 3-4	Wk 1-2	Wk 3-4
Sprint 1: Project Setup and Video Upload Module														
Sprint 2: Computer Vision Pipeline Integration														
Sprint 3: Shot Classification and Court Analytics														
Sprint 4: Analytics Dashboard, Reflection, and Recommendations														
Sprint 5: Skill Level, Play Style Estimation, and Player Matching														
Sprint 6: Gameplay Context Support, Find a Game, and Microchallenges														
80% Completion Check and Buffer														
RAG R1/ Sprint 7: Knowledge Base Preparation														
RAG R2: RAG Feedback / Audio Prototype														
Sprint 8: Coach Access, Integration Testing, and MVP Refinement														
System QA and Bug Fixing														
User Testing and Revisions														
Documentation														
Final Polish														

Figure 3.3.1 Project Gantt Chart

Figure 3.3.1 presents the work plan for the development of the proposed Pickleball Performance Analytics System from September to March. The project follows the Scrum framework, with development organized into seven two-week sprints. The schedule is designed to deliver the system incrementally, beginning with the technical foundation before progressing to the core performance monitoring features, player analytics, gameplay support, and final system integration.

Development begins in September with Sprint 1, which focuses on project setup, system architecture, development environment configuration, and the implementation of the video upload module. This is immediately followed by Sprint 2, where the computer vision pipeline is integrated to support rally segmentation, player detection, and player tracking. Sprint 3 continues the development of the Core Performance Monitoring component by implementing shot classification and generating court analytics such as heatmaps, kitchen-zone metrics, and court coverage.

During October, the project shifts toward the player-facing features. Sprint 4 develops the analytics dashboard together with reflection and recommendation features that allow players to review their historical performance and monitor improvement over time. This is followed by Sprint 5, which implements lightweight skill level estimation, play style classification, and player matching based on accumulated performance data.

In November, Sprint 6 introduces gameplay context support, including match context tagging, the Find-a-Game feature, and microchallenges that encourage continuous player engagement. At the end of the month, an 80% completion check and development buffer is scheduled to verify

that the minimum functionality required for Capstone 1 has been achieved while allowing sufficient time to resolve technical issues before the final integration phase.

Beginning in December, Sprint 7 focuses on implementing coach access through role-based access control (RBAC), completing system integration, and refining the minimum viable product (MVP). During the same period, two exploratory RAG development iterations are scheduled. RAG R1 is dedicated to preparing the retrieval knowledge base, while RAG R2 develops a prototype for retrieval-assisted feedback and live audio coaching. These activities are treated as enhancement features and are developed independently of the core MVP to ensure that the primary project objectives remain on schedule.

The remaining project timeline is allocated to system quality assurance and bug fixing, user testing and revisions, documentation, and final polishing. These activities provide sufficient time to validate system functionality, incorporate user feedback, complete the required project documentation, and prepare the system for the Capstone 2 presentation and final evaluation.

3.4 Technology Stack

3.4 Technology Stack

The proposed system is built by integrating existing, proven computer vision and web technologies rather than developing new models from scratch. The core detection, tracking, and court-mapping layer is built on Pickleball-Analytics, which already combines these into a working pipeline; the table below breaks that pipeline into layers alongside the additional components the team is building on top of it (shot classification, audio feedback, backend, frontend).

Layer	Tools / Technologies	Role in the System
Object Detection	YOLOv8, via Pickleball-Analytics	Detects ball, player, and paddle bounding boxes per frame.
Multi-Object Tracking	ByteTrack / BoT-SORT	Assigns persistent IDs to the detected player and ball across frames so movement can be followed over time, including through brief occlusion. Extended by the team for single-player selection
Pose Estimation	MediaPipe Pose	Extracts player joint landmarks for movement-quality cues in the real-time framework

Court Mapping	OpenCV homography, 12-keypoint model	Projects pixel coordinates for the ball and player onto real-world court coordinates using calibrated reference points, enabling zone-based metrics (kitchen time, court coverage). Extended by the team for half-court, tripod-height framing.
Video Processing	OpenCV, FFmpeg	Frame extraction, resizing, and format handling for uploaded match footage prior to model inference.
Shot Classification	XGBoost (XGBClassifier Api) existing, pre-built classification	Classifies each detected contact as a shot type (dink, drive, lob, serve, volley) using speed, arc, and court-zone features derived from tracking output.
Batch Processing	Server-side workers	Runs the detection, tracking, and classification pipeline on uploaded videos outside the browser for reliability and consistent performance.
Audio Feedback	RAG over player history	Converts structured performance data into short spoken feedback for post-game summaries.
Backend / Storage	Supabase (Postgres, auth, storage)	Stores match videos, tracking output, player profiles, and generated analytics; handles authentication and role-based access.

Frontend	React / Next.js, charting library	Renders the player and coach dashboards, visual analytics, and historical trend views.
Project Management	Jira, Figma	Sprint planning and interface design, per the Scrum workflow in Chapter III.

3.4.1 Data Specification

To make the system's data collection concrete, the table below specifies the exact fields captured or derived at each stage of the computer vision pipeline, distinct from the general metric categories already listed in Figure 1.4.1 (PPIF Stage Matrix).

Data Category	Specific Fields Captured / Derived	Source
Raw detection (per frame)	Ball bounding box (x, y, width, height, confidence); player bounding box; paddle bounding box; frame timestamp	YOLOv8 inference
Tracking (per frame)	Persistent track ID for player and ball; frame-to-frame displacement	ByteTrack / BoT-SORT
Court-mapped position (per frame)	Ball position in real-world court coordinates (x, y in feet/meters from baseline and sideline); player position in court coordinates; court zone flag (kitchen / mid-court / baseline)	OpenCV homography using calibrated court reference points
Derived ball kinematics (per shot)	Ball speed at contact (court-units/sec, computed from consecutive frame positions); trajectory arc height (max vertical pixel/position deviation between contact and next bounce or contact); time-to-bounce	Computed from tracked ball positions
Contact / shot event	Contact frame index and timestamp; contact court-zone; ball speed and direction change at that frame; paddle–ball bounding-box proximity at contact	Derived from ball + paddle tracking

Shot classification output	Shot type label (dink, drive, lob, serve, volley) per detected contact; classifier confidence score	Trajectory-feature classifier
Match-level aggregates	Shot type distribution and accuracy per match; kitchen-zone dwell time; court coverage/heatmap data; session date and context tag (practice, casual, tournament)	Aggregated from rally-level data, stored in player profile

3.5 Capstone 1 & 2 Targets

3.5 Capstone 1 & 2 Targets

The development of the proposed pickleball personal informatics system will be divided into two major completion stages: Capstone 1 and Capstone 2. These stages will ensure that the project is developed, evaluated, and refined progressively. Capstone 1 will focus on completing at least 80% of the proposed system, particularly the major functional components needed for the minimum viable product. Capstone 2 will focus on completing the remaining components, conducting user acceptance testing, refining the system, and preparing the final implementation and documentation.

3.5.1 Capstone 1 Targets

Capstone 1 is expected to cover the main development phase of the system. By the end of Capstone 1, the project should reach approximately 80% completion. This means that the primary system modules must already be functional enough to demonstrate how the proposed solution addresses the identified problems and opportunities related to pickleball performance tracking, self-reflection, and player improvement. The following components are expected to be completed and evaluated during Capstone 1:

Video Upload and Match Data Module

The system must allow players to upload recorded pickleball match videos and input basic match information such as match date, match type, gameplay context, and player notes. This component serves as the entry point of the system and provides the data needed for performance analysis.

Computer Vision Processing Pipeline

The system must be able to process uploaded match footage using existing computer vision technologies. This includes player detection, movement tracking, rally segmentation, and the extraction of basic gameplay events. The researchers will not develop new computer vision algorithms, but will integrate existing models and frameworks such as YOLOv8, ByteTrack, OpenCV, and MediaPipe.

Shot Classification and Court Analytics

The system must classify common pickleball shots such as dinks, drives, volleys, and lobs based on available video and movement data. It must also generate court-based analytics such as player positioning, heatmaps, kitchen-zone time, and court coverage. These outputs will support objective performance monitoring and allow players to better understand their strengths and weaknesses.

Analytics Dashboard

The system must present processed performance data through a player-facing dashboard. The dashboard should display key performance indicators, match summaries, court analytics, shot distribution, performance trends, and historical comparisons. This component will allow players to review their gameplay in a clear and organized manner.

Goal Setting, Reflection, and Recommendations

The system must support the personal informatics aspect of the study by allowing players to set goals, reflect on post-match performance, and receive basic recommendations

based on collected data. These recommendations may include focus areas such as improving court coverage, increasing shot variety, maintaining better positioning, or spending more time near the kitchen line.

Skill Level and Play Style Estimation

The system must provide a light assessment of player characteristics based on accumulated performance data. This includes estimating whether a player is beginner, intermediate, or advanced, and identifying whether the player's dominant play style is aggressive, defensive, or balanced. These classifications will be treated as general estimates and not as official rankings.

Player Matching Support

The system must include basic player matching support based on skill level, play style, availability, and preferred gameplay type. This feature supports the opportunity for players to find suitable opponents or partners within the same pickleball community.

Gameplay Context Support

The system must organize performance records according to different gameplay contexts such as practice, casual matches, tournaments, leveling games, singles, doubles, wall practice, and ball machine practice. This allows players to compare their progress across different playing situations.

3.5.2 Capstone 2 Targets

Capstone 2 will cover the completion of the remaining components, system refinement, user acceptance testing, and final documentation. While Capstone 1 focuses on building the core system, Capstone 2 will focus on ensuring that the system is reliable, usable, and able to address the problems identified in the study. The following components are expected to be completed and evaluated during Capstone 2:

Coach Access and Role-Based Permissions

The system must provide a cooperative access layer where authorized individuals, such as coaches, can view player performance information. Role-based access control must be implemented to ensure that only authorized users can access player data. This feature supports guided improvement while maintaining user privacy and data control.

RAG and Live Audio Coaching Assistance

The RAG component may be used to retrieve coaching knowledge, drill suggestions, player history, and feedback templates to generate more grounded performance recommendations. The Live Audio Coaching Assistance feature may convert selected feedback into short audio cues during practice or training sessions. However, these features will remain supplementary and will not replace professional coaching, medical advice, officiating, or referee decision-making.

Because these features may increase the technical complexity of the project, they will only be pursued if the core system has already met the required level of completion and

testing. If not completed within the project timeline, they may be discussed as future enhancements.

Full System Integration

All modules developed during Capstone 1 must be integrated into one cohesive system. This includes the upload module, computer vision pipeline, shot classification, court analytics, dashboard, goal setting, reflection, recommendations, skill assessment, player matching, gameplay context support, Find a Game, microchallenges, and coach access.

System Testing and Bug Fixing

The group must conduct technical testing to identify and resolve errors, data mismatches, broken workflows, and usability issues. Testing must cover the complete system flow, from video upload and processing to dashboard display and coach access.

User Acceptance Testing

User acceptance testing will be conducted with target users such as pickleball players and, when possible, coaches or experienced players. The purpose of this testing is to determine whether the system is understandable, useful, and relevant to the needs of its intended users. Feedback from user testing will be used to refine the dashboard, recommendations, reflection features, player matching, and other user-facing functions.

Final Refinement and Documentation

The remaining period of Capstone 2 will be used for final system improvements, interface polishing, documentation completion, evaluation write-ups, and preparation for the final defense. The group must also ensure that the final manuscript reflects the actual completed system and clearly discusses its scope, limitations, testing results, and recommendations for future development.

3.5.3 Data Requirements for Demonstration and Evaluation

To demonstrate that the developed system addresses the identified problems and opportunities, the group must prepare sufficient sample data for testing and simulation. This includes recorded pickleball videos, sample player profiles, sample match records, gameplay context labels, skill-level examples, play-style examples, availability schedules, sample game listings, and microchallenge records.

The data must be able to simulate different scenarios where the system may be used, such as a beginner reviewing a casual match, an intermediate player tracking improvement over time, a player looking for a suitable match partner, a coach reviewing a player's performance, and a player comparing performance across practice and competitive contexts.

Through these scenarios, the group will be able to demonstrate that the system supports automated data collection, performance reflection, actionable insights, player matching, and continuous improvement within the defined scope of the study.

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APPENDICES

Appendix A

Player Interview Guide

This interview guide was prepared to gather information about pickleball players' current performance-improvement practices, information needs, comfort with video-based analysis, and interest in community features. The interviewer should explain the study, request informed consent, and remind the participant that participation is voluntary before beginning.

A. Participant Background

1. How long have you been playing pickleball?
2. How often do you play pickleball? (Daily; 3–5 times per week; 1–2 times per week; occasionally)
3. What types of games do you usually play? (Practice or training; casual games; leveling games; tournaments; singles; doubles; other)
4. Are you part of a pickleball club, community, or regular playing group? If yes, please describe it briefly.
5. How did you learn to play pickleball? (Self-taught; coach; friends or family; online videos; clinics or workshops; other)
6. When you teach yourself, what resources do you use to improve your skills?

B. Current Performance-Improvement Practices

7. Do you currently track your pickleball performance or progress? Why or why not?
8. How do you evaluate whether you are improving as a pickleball player?

9. How confident are you in your ability to assess your own performance accurately? Why?
10. When you track your performance, what tools, devices, or methods do you use?
11. What information or metrics do you monitor?
12. Which metrics do you find most useful?
13. How satisfied are you with your current tracking method? Why?
14. What limitations or frustrations have you experienced?
15. When you do not track your performance, what prevents you from doing so?
16. What information would you want a performance-tracking system to provide?

C. Challenges and Information Needs

17. Which aspects of your game are the most difficult to improve?
18. What information do you wish you had after finishing a match?
19. Have you ever been unsure about what to improve next? Please describe the experience.
20. What do current sports or pickleball performance tools do well?
21. What do you think these tools are missing?

D. Validation of the Proposed System

22. Imagine a system that records a pickleball match and analyzes gameplay using computer vision. How useful would this system be for improving your performance? Why?
23. Which performance insights would be most valuable to you? (Shot classification; court positioning; player movement; court coverage; match statistics; strengths and weaknesses; progress over time; personalized recommendations; goal tracking; other)
24. How would you use these insights during future training sessions or matches?

25. Would reviewing your historical performance motivate you to continue improving?
Why or why not?

E. Computer Vision and Data Privacy

26. Would you be comfortable having your matches recorded for performance analysis?
Why or why not?
27. What concerns would you have regarding video-based performance analysis?
28. What privacy, access-control, or data-security measures would make you comfortable using the system?

F. Preliminary Wearable-Technology Questions

29. Do you currently use a smartwatch, fitness tracker, or sports wearable? If yes, what device do you use?
30. Would you be interested in combining wearable information with gameplay analysis?
Why or why not?
31. Which wearable information would you consider valuable? (Heart rate; calories burned; activity load; recovery status; fatigue; swing speed; swing angle; impact; injury-risk indicators; other)
32. What concerns would you have about using a wearable device or paddle-mounted sensor during play?

***Note.** These questions formed part of the preliminary requirements interview. Wearable integration was later removed from the final project scope, but the questions are retained here to document the requirements-development process.*

G. Matchmaking and Community Features

- 33. How do you usually find opponents or playing partners?
- 34. Have you experienced difficulty finding players with a similar skill level? Please describe.
- 35. Would a player-matching feature be useful to you? Why or why not?
- 36. Would you use a feature that recommends nearby games or events based on skill level and availability?
- 37. Would community challenges or achievement-based goals encourage you to play more consistently? Why or why not?

H. Closing Questions

- 38. If you could design the ideal pickleball performance-tracking system, which features would you include?
- 39. Is there anything else you would like to share that could improve the proposed system?

Appendix B

Player Interview Transcript: Participant P1

***Note.** The participant is identified by code to protect privacy. Responses were lightly edited for grammar and readability without changing their substantive meaning. “No response provided” indicates that the participant left an item unanswered.*

A. Participant Background

How long have you been playing pickleball?

Response: Approximately three to four months.

How often do you play pickleball?

Response: One to two times per week.

What types of games do you usually play?

Response: Casual games.

Are you part of a pickleball club, community, or regular playing group?

Response: No.

How did you learn to play pickleball?

Response: Through friends or family.

If you are self-taught, what resources do you use?

Response: Not applicable.

B. Current Performance-Improvement Practices

How do you currently evaluate whether you are improving?

Response: I record myself while playing and compare my current games with previous ones. I also notice whether I make fewer mistakes, move better on the court, and become more consistent during rallies.

How confident are you in your ability to assess your own performance accurately?

Response: Somewhat confident. I can identify obvious mistakes, but I might miss technical details or bad habits because I do not have a coach or a performance-analysis tool.

Do you currently use a structured method or tool to track your performance?

Response: No. I do not really know how to track my performance or which tools and resources I should use.

What information would you want a performance-tracking system to provide?

Response: I would want it to show my progress over time, identify my strengths and weaknesses, point out the skills that need improvement, and recommend what I should work on next.

C. Challenges and Information Needs

Which aspect of your game is the most difficult to improve?

Response: Learning and consistently executing my backhand.

What information do you wish you had after finishing a match?

Response: I want to know what I did well, what mistakes I made, which areas need improvement, and what I should focus on in my next match.

Have you ever been unsure about what to improve next?

Response: Yes. Since I am still new to pickleball, I sometimes know that I did not play well but cannot tell whether the problem is my footwork, positioning, shot selection, or technique. Without feedback, it is difficult to know what to prioritize.

Are you familiar with existing pickleball or sports performance-tracking systems?

Response: No.

What do you think current performance-tracking tools do well?

Response: They can help players monitor performance and provide statistics that are useful for identifying strengths and weaknesses.

What do you think current tools are missing?

Response: They could provide more personalized feedback, easier-to-understand insights for beginners, and specific recommendations based on the player's performance.

D. Validation of the Proposed System

How useful would a system that records and analyzes your matches be for improving your performance?

Response: Very useful. It would provide objective feedback instead of relying only on my own observations. It would help me understand what I am doing correctly, what needs improvement, and how I can train more effectively.

Which types of performance insights would be most valuable?

Response: Player movement tracking; strengths and weaknesses; progress over time; personalized recommendations; and goal tracking.

How would you use these insights?

Response: I would focus my training on weak areas, monitor improvement over time, and apply the recommendations during practice and actual matches.

Would reviewing historical performance motivate you to continue improving?

Response: Yes. Seeing my progress would motivate me to keep practicing because I could clearly see how much I had improved and identify new goals.

E. Computer Vision and Data Privacy

Would you be comfortable having your matches recorded for performance analysis?

Response: Yes, provided that the recordings are secure, used only for performance analysis, and that I control who can access my data.

What concerns would you have regarding video-based performance analysis?

Response: No additional concern was stated.

What would make you comfortable using this type of system?

Response: No response provided.

F. Preliminary Wearable-Technology Questions

Do you currently use a smartwatch, fitness tracker, or sports wearable?

Response: No.

Would you be interested in combining wearable data with gameplay analysis?

Response: Yes. It could provide a more complete picture of performance by connecting physical activity with gameplay.

Which wearable information would you consider valuable?

Response: Calories burned, activity load, swing speed, swing angle, and swing impact.

How useful would workload- or exertion-based injury-risk warnings be?

Response: Very useful because they could indicate when I am overexerting myself or should rest before my condition becomes worse.

What concerns would you have about using wearable devices during pickleball?

Response: Comfort, battery life, accuracy, and whether the device might interfere with movement.

Would you use an attachable paddle sensor that measures swing speed, swing angle, and impact?

Response: Yes. It could help me understand my swing mechanics and identify ways to improve technique and consistency.

***Note.** Wearable integration was later removed from the final system scope. These responses remain part of the historical requirements evidence.*

G. Matchmaking and Community Features

How do you usually find opponents or playing partners?

Response: Through the website used by open-play organizers.

Have you experienced difficulty finding players with a similar skill level?

Response: Yes. There are several skill categories, such as beginner, semi-beginner, intermediate, and advanced. It is not always easy to find a close match, so games can become too easy or too difficult, affecting both learning and enjoyment.

Would a player-matching feature be useful?

Response: Yes. It would make it easier to find players with similar skill levels and create more balanced, enjoyable, and competitive games.

Would you use recommendations for nearby games or events based on skill level and availability?

Response: Yes.

Would community challenges or achievement-based goals encourage you to play more consistently?

Response: Yes. Goals, achievements, and challenges would make training more enjoyable, support regular play, and help me track progress.

H. Closing Questions

Which features would you include in an ideal pickleball performance-tracking system?

Response: No response provided.

Is there anything else you would like to share that could improve the proposed system?

Response: No response provided.

Appendix C

Coach Interview Guide

This semi-structured guide is intended for pickleball coaches or trainers. It examines current coaching practices, objective information needs, expectations for automated analytics, and the appropriate boundary between system recommendations and professional judgment.

A. Coach Background

1. How long have you been coaching pickleball?
2. How many years have you been playing pickleball?
3. What types or skill levels of players do you usually coach?
4. Do you coach individuals, groups, or both?
5. How often do you conduct coaching sessions?

B. Current Coaching Practices

6. How do you currently evaluate a player's performance?
7. Which aspects of performance do you observe during training sessions or matches?
8. How do you identify a player's strengths and weaknesses?
9. How do you monitor whether a player is improving over time?
10. Do you keep records or notes on player progress? If yes, what do you record?
11. What challenges do you encounter when evaluating player performance?
12. When do you wish you had more objective performance data? Please describe.

C. Performance Information Needs

13. Which performance metrics would help you provide better coaching?
14. How would you use those insights during coaching sessions?
15. How important is it to compare a player's current performance with previous matches?

D. Validation of the Proposed System

16. How useful would an automated video-analysis system be in supporting your coaching?
Why?
17. Which system features would provide the greatest value?
18. How could automated performance analytics improve your coaching process?
19. Which coaching decisions should remain based on human judgment rather than system-generated recommendations?
20. What types of recommendations would you expect the system to generate?

E. Computer Vision and Trust

21. Would you be comfortable using video-based match analysis as part of your coaching process? Why or why not?
22. What level of accuracy, validation, or supporting evidence would you require before trusting the system's analytics?

F. Player Development

23. Which factors do you consider when determining a player's skill level?
24. How do you identify a player's playing style, such as aggressive, defensive, or balanced?
25. How useful would historical performance trends be when planning future coaching sessions?
26. Would personal goal setting and progress monitoring between coaching sessions help your players? Why or why not?

G. Matchmaking and Community

27. How important is it for players to compete against others with similar skill levels?
28. Would player matching based on skill level and availability benefit your players? Why or why not?

H. Closing Questions

29. What concerns or recommendations would you have before adopting a system like this?
30. Is there anything else you would like to share that could improve the proposed system?

Appendix D

Proposed Pickleball Performance Metrics and Sample Reports

The following metrics were identified in the preliminary system-design notes. Final implementation should define the detection event, denominator, success condition, and minimum confidence threshold for every percentage-based metric.

Table D1

Proposed Core Pickleball Statistics

Stat	Meaning
PTS	Points scored
ACE	Service aces
WIN	Winning shots that directly end a rally

DRV%	Drive accuracy or success rate
DNK%	Dink accuracy or success rate
VOL%	Volley accuracy or success rate
UER	Unforced errors
KF	Kitchen or non-volley-zone faults
CC%	Court coverage efficiency
KLC%	Kitchen-line control

Example Player Stat Line

Juan Dela Cruz

11 PTS | 3 ACE | 8 WIN | 82% DRV | 76% DNK | 88% VOL | 4 UER | 1 KF

- Scored 11 points.
- Recorded three service aces.
- Hit eight winning shots.
- Achieved an 82% drive success rate.
- Achieved a 76% dink success rate.
- Achieved an 88% volley success rate.
- Committed four unforced errors.
- Committed one kitchen fault.

***Note.** The preliminary example does not contain values for court coverage efficiency (CC%) or kitchen-line control (KLC%). These fields should be populated only after their formulas and court-zone thresholds have been operationally defined.*

Example Doubles Match Report

Team A vs. Team B. Final score: 11–8.

Player	Stat	Value
Player A	PTS	6
Player A	ACE	2
Player A	WIN	5
Player A	DRV%	84%
Player A	DNK%	79%
Player A	VOL%	91%
Player A	UER	2
Player A	KF	0
Player B	PTS	5
Player B	ACE	1
Player B	WIN	3

Player B	DRV%	80%
Player B	DNK%	75%
Player B	VOL%	85%
Player B	UER	3
Player B	KF	1

Appendix E

RAG-Supported Audio Coaching Workflow

***Note.** The initial concept described continuous live coaching. The refined capstone scope uses RAG-supported coaching after video processing or during natural pauses, rather than constant audio during active rallies. This reduces distraction and makes the feature easier to test and defend as an MVP.*

The workflow converts computer-vision outputs into short, contextual coaching cues. Retrieval-augmented generation (RAG) is used to retrieve approved pickleball knowledge, player history, and drill templates before the language model produces a recommendation.

Table E1

RAG-Supported Coaching Process

Stage	Process	Output
1. Start session	The player records or uploads a practice or match video and selects a session type.	Video and session metadata
2. Detect court and players	Computer vision identifies the court, player, and ball and tracks their positions.	Coordinates, tracking IDs, court zones
3. Extract performance metrics	Tracking data are converted into metrics such as kitchen-line distance, baseline time, movement path, rally duration, and court coverage.	Structured performance metrics
4. Identify coaching events	Rules and thresholds identify moments such as staying too deep, failing to recover, or leaving a zone uncovered.	Detected gameplay event
5. Filter and prioritize	Confidence, repetition, severity, and cooldown rules prevent excessive or low-quality feedback.	Approved event
6. Build retrieval query	The event is translated into a query containing the player's skill level, session context, and detected behavior.	RAG query
7. Retrieve evidence	The system searches coach-approved tactics, shot guidance, drill templates, cue templates, and the player's history.	Relevant context

8. Generate recommendation	A language model produces one concise and actionable cue grounded in the retrieved material.	Text recommendation
9. Apply safety and quality checks	The system rejects medical advice, officiating decisions, unsupported claims, harsh language, or distracting instructions.	Approved cue
10. Convert to audio	Text-to-speech converts the approved recommendation into audio.	Audio coaching cue
11. Deliver feedback	The player receives the cue after processing, between drills, or during a natural pause.	Player-facing feedback
12. Log and reflect	The event, recommendation, timestamp, and later outcome are saved for review on the dashboard.	Session history and reflection data

Suggested Knowledge-Base Contents

Knowledge-Base Section	Example Contents
Pickleball tactics	Kitchen control, third-shot strategy, recovery position, doubles spacing
Shot-specific guidance	Dinks, drives, volleys, lobs, serves, and returns

Skill-level guidance	Beginner, intermediate, and advanced cue variants
Drill library	Dink drills, reset drills, return-and-advance drills, and footwork-recovery drills
Player history	Repeated weaknesses, recent progress, goals, and prior recommendations
Coach-approved cue templates	Short, single-action, constructive feedback patterns
Safety rules	No medical diagnosis, no officiating decisions, and no unsafe or distracting commands

Illustrative Example

Detected event:

- The player remains near the baseline for 4.2 seconds after returning serve.
- The player is approximately 5.8 meters from the kitchen line.
- Session type: doubles training.
- Skill level: beginner to intermediate.
- Detection confidence: 0.86.

Retrieved coaching principle:

After returning serve in doubles, the returning team generally benefits from advancing toward the non-volley zone to establish stronger court control.

Retrieved drill:

Return-and-advance drill: return deep, move forward, and split-step before the opponent contacts the ball.

Generated cue:

“Move forward after your return. Take the kitchen earlier.”

Appendix F

Candidate Computer Vision Resources and Proposed Technology Stack

***Note.** The project should integrate existing models, repositories, and services rather than develop a new artificial-intelligence model from scratch. Every external resource remains subject to testing, licensing review, device requirements, and compatibility with the approved capstone scope.*

A. Pickleball-Specific Candidate Resources

5urabhi/Pickle_ball_tracking

Tracks a pickleball in video, including bounce status and whether the ball is inside or outside court boundaries. Uses OpenCV, Ultralytics, NumPy, Pandas, Matplotlib, and FilterPy.

[Source](#)

kpp91302/Pickleball-Analytics

Computer-vision pipeline that detects players, ball position, and court layout and produces heatmaps, rally tempo, and kitchen-zone usage. [Source](#)

AndrewDettor/TrackNet-Pickleball

Deep-learning model for pickleball location tracking and candidate outputs such as ball velocity, shot type, and location. [Source](#)

Roboflow Pickleball Detection Dataset

Candidate annotated pickleball-detection dataset that may support evaluation or integration of an existing detector. [Source](#)

Pickleball Vision: CV-Driven Match Analytics

Pipeline described as using YOLOv8, a court-keypoint regressor, and homography refinement for player tracking, ball tracking, court geometry, a minimap, and speed or shot analytics. *Candidate public project/article; verify source before implementation*

bradautomates/claude-video

General video-processing utility that extracts frames and transcripts for model-assisted analysis. This is supplementary rather than a pickleball-specific tracker. [Source](#)

B. Proposed Integration Stack

Layer	Candidate Technology	Primary Output
Video input	Fixed court-side camera or smartphone recording	Match or training video
Object detection	Existing Ultralytics YOLO model or approved pickleball-specific detector	Player, ball, and object detections

Tracking	ByteTrack, TrackNet-Pickleball, FilterPy-based tracking, or an existing repository	Frame-to-frame identities and ball trajectory
Court mapping	OpenCV perspective transformation and homography	Court-space coordinates and zones
Dataset management	Roboflow or an equivalent annotation and dataset-management platform	Dataset versions and evaluation samples
Metric extraction	Python and OpenCV rules using tracked coordinates and timestamps	Movement paths, zone occupancy, rally timing, and candidate shot features
Storage	Supabase/PostgreSQL and object storage	Player profiles, match records, video references, analytics, and recommendations
RAG retrieval	OpenAI vector stores, Supabase pgvector, or Pinecone	Retrieved coaching knowledge and player context
Recommendation generation	OpenAI API or another approved language-model API	Grounded coaching text

Audio output	Approved text-to-speech service or device TTS	Player-facing audio cue
User interface	Web application with player and coach views	Upload, reflection, trends, goals, and coach access

C. Minimum Computer-Vision Outputs

- Player location and movement trajectory.
- Ball location and trajectory when detector confidence is sufficient.
- Court boundaries, non-volley zone, baseline, and other defined court zones.
- Time spent near the baseline and non-volley zone.
- Court coverage and position-recovery patterns.
- Rally duration and event timestamps.
- Candidate shot classification only when technically validated using existing resources and an explicit operational definition.

Appendix G

Personal Informatics Framework Mapping

The proposed system is structured around the five stages of personal informatics: preparation, collection, integration, reflection, and action. The coach operates as a cooperative stakeholder

who can interpret player-authorized information without replacing the player as the primary user.

Table G1

System Activities Across the Personal Informatics Stages

Stage	System Activity	Output
Preparation	The player selects a session type, reviews prior goals, and chooses a performance focus. A beginner may receive starter goals or explanations of pickleball terms.	Session objective, selected metrics, and goal
Collection	The player records or uploads gameplay. Existing computer-vision tools collect player, ball, court, movement, and rally data with minimal manual entry.	Video, coordinates, trajectories, and event timestamps
Integration	Current-session outputs are linked with the player profile, match context, prior sessions, skill estimate, goals, and coach-authorized notes.	Unified player performance record
Reflection	Dashboards present annotated clips, trends, strengths, weaknesses, court-position summaries, and evidence supporting each insight.	Understandable performance report

Action	The system recommends a next skill, drill, goal adjustment, or positioning focus. The player chooses an action and applies it in the next session.	Training decision and updated goal
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Cooperative Coach Layer

- The player controls whether a coach may access performance information.
- The coach can review player-authorized reports, annotated clips, trends, and goals.
- The coach can add interpretations, drills, or approved feedback templates.
- System recommendations supplement rather than replace professional coaching judgment.
- The next session returns the player to the collection stage so the effect of the chosen action can be evaluated.

Appendix H

Supplementary Scope Summary and System Boundaries

Primary and Supporting Stakeholders

Stakeholder	Role in the System
Players	Primary users and direct beneficiaries of performance tracking, reflection, goal setting, recommendations, and progress monitoring.

Coaches	Authorized cooperative users who interpret evidence, provide drills, and review player progress.
Clubs or communities	Supporting stakeholders for skill-based game discovery, localized challenges, and community participation.
Court owners or administrators	Indirect stakeholders only. The system is not a court-booking, payment, or availability-management platform.

In-Scope Components

1. Core performance monitoring through recorded-video analysis, including player movement, validated ball tracking, court positioning, zone usage, rally information, and selected shot-related outputs.
2. A player dashboard showing strengths, weaknesses, match history, progress trends, and supporting visual evidence.
3. Personal informatics functions for goal setting, reflection, recommendations, and a recurring performance-feedback loop.
4. Light skill-level and play-style estimation to support understandable guidance and player matching.
5. Gameplay-context support for practice or training, casual play, tournaments, leveling games, singles, doubles, and wall or machine drills when technically feasible.
6. Find-a-game and localized microchallenge functions that support participation without becoming a court reservation system.
7. Player-authorized coach access to performance reports and recommendations.

8. RAG-supported coaching guidance generated from approved pickleball knowledge and converted into optional audio after processing or during natural pauses.

Out-of-Scope Components

- Court booking or reservation.
- Payment processing for court use.
- Court availability management by owners.
- Tournament or league scheduling and administration.
- Development of a new artificial-intelligence or machine-learning model from scratch.
- Medical diagnosis, injury prediction, medical treatment, or replacement of a qualified health professional.
- Automated officiating decisions presented as authoritative rulings.
- Wearable-device integration in the final MVP.

Appendix I

Consultation and Revision Log

The following log consolidates key directions from the project-development meetings. Informal wording from the original notes has been paraphrased into formal action items.

Date	Consolidated Direction and Resulting Revision
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June 11	Recenter the project on pickleball players rather than court owners; identify root causes and decision-making problems; use personal informatics; collect meaningful player and coach data; avoid becoming a booking platform; justify player groups and interview coaches.
June 26	Use existing technologies and models rather than building a model from scratch; define one clear solution type; follow the five personal-informatics stages; minimize manual collection; distinguish the personal, cooperative, and collaborative layers; solve information-quality problems within a manageable scope.
July 1	Confirm the specific APIs, repositories, and models to be used; determine how player and ball tracking will work; define pickleball-specific analytics; investigate RAG or MCP but avoid unnecessary architectural complexity; organize the system into player, in-game collection, post-game analytics, and community or coach functions.
July 11	Remove continuous real-time wording; use RAG-supported coaching and include audio in the methodology; remove wearable integration from the final scope; explain the computer-vision pipeline and pickleball-specific technology stack; state the exact data collected; simplify the solutions framework; use readable formatting and pickleball terminology; align scope, limitations, problems, and proposed solutions.

Resulting Finalization Principles

- The player remains the main stakeholder.
- The system is a personal informatics solution, not a booking or general club-management platform.
- Computer vision uses existing resources to collect objective gameplay information.
- Analytics must use explicit pickleball terms, metrics, and court zones.
- Recommendations must be traceable to detected evidence and approved knowledge.
- Audio coaching is supplementary and should not distract the player during active rallies.
- The scope, objectives, methodology, technology stack, and Gantt chart should describe the same set of final components.

Appendix J

Preliminary Related Systems Inventory

This inventory was assembled during early related-systems research. The systems should be discussed in the Review of Related Systems only after their current features and availability have been verified through authoritative sources.

System	Primary Relevance	Original Source	Note
Zepp Tennis	Racket-sport swing and performance tracking	App-store listing	
PlaySight SmartCourt	Video capture, replay, and sports analytics	playsight.com	

Arccos Caddie	Sensor-based golf tracking and recommendations	Google Play listing
Padelytics	Padel-focused analytics	padelytics.ai
HomeCourt	Camera-based basketball practice and analytics	homecourt.ai
OnForm	Video analysis and coaching collaboration	onform.com
Dartfish	Video analysis and performance feedback	dartfish.com
CoachNow	Coach-athlete communication and feedback	coachnow.com
DUPR	Pickleball player rating and match-result platform	dupr.com
Pickleheads	Pickleball court, game, and community discovery	pickleheads.com

Note. The main reference list should contain the complete, verified citations for all literature, technologies, repositories, and systems cited in the paper. A reference dump should not be placed inside the appendices unless the institution explicitly requires a separate supplementary bibliography.

Appendix A

Appendix A

Player Interview Guide

This interview guide was prepared to gather information about pickleball players' current performance-improvement practices, information needs, comfort with video-based analysis, and interest in community features. The interviewer should explain the study, request informed consent, and remind the participant that participation is voluntary before beginning.

A. Participant Background

1. How long have you been playing pickleball?
2. How often do you play pickleball? (Daily; 3–5 times per week; 1–2 times per week; occasionally)
3. What types of games do you usually play? (Practice or training; casual games; leveling games; tournaments; singles; doubles; other)
4. Are you part of a pickleball club, community, or regular playing group? If yes, please describe it briefly.
5. How did you learn to play pickleball? (Self-taught; coach; friends or family; online videos; clinics or workshops; other)
6. When you teach yourself, what resources do you use to improve your skills?

B. Current Performance-Improvement Practices

7. Do you currently track your pickleball performance or progress? Why or why not?
8. How do you evaluate whether you are improving as a pickleball player?
9. How confident are you in your ability to assess your own performance accurately? Why?

10. When you track your performance, what tools, devices, or methods do you use?
11. What information or metrics do you monitor?
12. Which metrics do you find most useful?
13. How satisfied are you with your current tracking method? Why?
14. What limitations or frustrations have you experienced?
15. When you do not track your performance, what prevents you from doing so?
16. What information would you want a performance-tracking system to provide?

C. Challenges and Information Needs

17. Which aspects of your game are the most difficult to improve?
18. What information do you wish you had after finishing a match?
19. Have you ever been unsure about what to improve next? Please describe the experience.
20. What do current sports or pickleball performance tools do well?
21. What do you think these tools are missing?

D. Validation of the Proposed System

22. Imagine a system that records a pickleball match and analyzes gameplay using computer vision. How useful would this system be for improving your performance? Why?
23. Which performance insights would be most valuable to you? (Shot classification; court positioning; player movement; court coverage; match statistics; strengths and weaknesses; progress over time; personalized recommendations; goal tracking; other)
24. How would you use these insights during future training sessions or matches?

25. Would reviewing your historical performance motivate you to continue improving?
Why or why not?

E. Computer Vision and Data Privacy

26. Would you be comfortable having your matches recorded for performance analysis?
Why or why not?
27. What concerns would you have regarding video-based performance analysis?
28. What privacy, access-control, or data-security measures would make you comfortable using the system?

F. Preliminary Wearable-Technology Questions

29. Do you currently use a smartwatch, fitness tracker, or sports wearable? If yes, what device do you use?
30. Would you be interested in combining wearable information with gameplay analysis?
Why or why not?
31. Which wearable information would you consider valuable? (Heart rate; calories burned; activity load; recovery status; fatigue; swing speed; swing angle; impact; injury-risk indicators; other)
32. What concerns would you have about using a wearable device or paddle-mounted sensor during play?

***Note.** These questions formed part of the preliminary requirements interview. Wearable integration was later removed from the final project scope, but the questions are retained here to document the requirements-development process.*

G. Matchmaking and Community Features

- 33. How do you usually find opponents or playing partners?
- 34. Have you experienced difficulty finding players with a similar skill level? Please describe.
- 35. Would a player-matching feature be useful to you? Why or why not?
- 36. Would you use a feature that recommends nearby games or events based on skill level and availability?
- 37. Would community challenges or achievement-based goals encourage you to play more consistently? Why or why not?

H. Closing Questions

- 38. If you could design the ideal pickleball performance-tracking system, which features would you include?
- 39. Is there anything else you would like to share that could improve the proposed system?

Appendix B

Appendix B

Player Interview Transcript: Participant P1

Note. The participant is identified by code to protect privacy. Responses were lightly edited for grammar and readability without changing their substantive meaning. “No response provided” indicates that the participant left an item unanswered.

A. Participant Background

How long have you been playing pickleball?

Response: Approximately three to four months.

How often do you play pickleball?

Response: One to two times per week.

What types of games do you usually play?

Response: Casual games.

Are you part of a pickleball club, community, or regular playing group?

Response: No.

How did you learn to play pickleball?

Response: Through friends or family.

If you are self-taught, what resources do you use?

Response: Not applicable.

B. Current Performance-Improvement Practices

How do you currently evaluate whether you are improving?

Response: I record myself while playing and compare my current games with previous ones. I also notice whether I make fewer mistakes, move better on the court, and become more consistent during rallies.

How confident are you in your ability to assess your own performance accurately?

Response: Somewhat confident. I can identify obvious mistakes, but I might miss technical details or bad habits because I do not have a coach or a performance-analysis tool.

Do you currently use a structured method or tool to track your performance?

Response: No. I do not really know how to track my performance or which tools and resources I should use.

What information would you want a performance-tracking system to provide?

Response: I would want it to show my progress over time, identify my strengths and weaknesses, point out the skills that need improvement, and recommend what I should work on next.

C. Challenges and Information Needs

Which aspect of your game is the most difficult to improve?

Response: Learning and consistently executing my backhand.

What information do you wish you had after finishing a match?

Response: I want to know what I did well, what mistakes I made, which areas need improvement, and what I should focus on in my next match.

Have you ever been unsure about what to improve next?

Response: Yes. Since I am still new to pickleball, I sometimes know that I did not play well but cannot tell whether the problem is my footwork, positioning, shot selection, or technique. Without feedback, it is difficult to know what to prioritize.

Are you familiar with existing pickleball or sports performance-tracking systems?

Response: No.

What do you think current performance-tracking tools do well?

Response: They can help players monitor performance and provide statistics that are useful for identifying strengths and weaknesses.

What do you think current tools are missing?

Response: They could provide more personalized feedback, easier-to-understand insights for beginners, and specific recommendations based on the player's performance.

D. Validation of the Proposed System

How useful would a system that records and analyzes your matches be for improving your performance?

Response: Very useful. It would provide objective feedback instead of relying only on my own observations. It would help me understand what I am doing correctly, what needs improvement, and how I can train more effectively.

Which types of performance insights would be most valuable?

Response: Player movement tracking; strengths and weaknesses; progress over time; personalized recommendations; and goal tracking.

How would you use these insights?

Response: I would focus my training on weak areas, monitor improvement over time, and apply the recommendations during practice and actual matches.

Would reviewing historical performance motivate you to continue improving?

Response: Yes. Seeing my progress would motivate me to keep practicing because I could clearly see how much I had improved and identify new goals.

E. Computer Vision and Data Privacy

Would you be comfortable having your matches recorded for performance analysis?

Response: Yes, provided that the recordings are secure, used only for performance analysis, and that I control who can access my data.

What concerns would you have regarding video-based performance analysis?

Response: No additional concern was stated.

What would make you comfortable using this type of system?

Response: No response provided.

F. Preliminary Wearable-Technology Questions

Do you currently use a smartwatch, fitness tracker, or sports wearable?

Response: No.

Would you be interested in combining wearable data with gameplay analysis?

Response: Yes. It could provide a more complete picture of performance by connecting physical activity with gameplay.

Which wearable information would you consider valuable?

Response: Calories burned, activity load, swing speed, swing angle, and swing impact.

How useful would workload- or exertion-based injury-risk warnings be?

Response: Very useful because they could indicate when I am overexerting myself or should rest before my condition becomes worse.

What concerns would you have about using wearable devices during pickleball?

Response: Comfort, battery life, accuracy, and whether the device might interfere with movement.

Would you use an attachable paddle sensor that measures swing speed, swing angle, and impact?

Response: Yes. It could help me understand my swing mechanics and identify ways to improve technique and consistency.

***Note.** Wearable integration was later removed from the final system scope. These responses remain part of the historical requirements evidence.*

G. Matchmaking and Community Features

How do you usually find opponents or playing partners?

Response: Through the website used by open-play organizers.

Have you experienced difficulty finding players with a similar skill level?

Response: Yes. There are several skill categories, such as beginner, semi-beginner, intermediate, and advanced. It is not always easy to find a close match, so games can become too easy or too difficult, affecting both learning and enjoyment.

Would a player-matching feature be useful?

Response: Yes. It would make it easier to find players with similar skill levels and create more balanced, enjoyable, and competitive games.

Would you use recommendations for nearby games or events based on skill level and availability?

Response: Yes.

Would community challenges or achievement-based goals encourage you to play more consistently?

Response: Yes. Goals, achievements, and challenges would make training more enjoyable, support regular play, and help me track progress.

H. Closing Questions

Which features would you include in an ideal pickleball performance-tracking system?

Response: No response provided.

Is there anything else you would like to share that could improve the proposed system?

Response: No response provided.

Appendix C

Appendix C

Coach Interview Guide

This semi-structured guide is intended for pickleball coaches or trainers. It examines current coaching practices, objective information needs, expectations for automated analytics, and the appropriate boundary between system recommendations and professional judgment.

A. Coach Background

1. How long have you been coaching pickleball?
2. How many years have you been playing pickleball?
3. What types or skill levels of players do you usually coach?
4. Do you coach individuals, groups, or both?
5. How often do you conduct coaching sessions?

B. Current Coaching Practices

6. How do you currently evaluate a player's performance?
7. Which aspects of performance do you observe during training sessions or matches?
8. How do you identify a player's strengths and weaknesses?
9. How do you monitor whether a player is improving over time?
10. Do you keep records or notes on player progress? If yes, what do you record?
11. What challenges do you encounter when evaluating player performance?
12. When do you wish you had more objective performance data? Please describe.

C. Performance Information Needs

13. Which performance metrics would help you provide better coaching?
14. How would you use those insights during coaching sessions?
15. How important is it to compare a player's current performance with previous matches?

D. Validation of the Proposed System

16. How useful would an automated video-analysis system be in supporting your coaching?
Why?
17. Which system features would provide the greatest value?
18. How could automated performance analytics improve your coaching process?
19. Which coaching decisions should remain based on human judgment rather than system-generated recommendations?
20. What types of recommendations would you expect the system to generate?

E. Computer Vision and Trust

21. Would you be comfortable using video-based match analysis as part of your coaching process? Why or why not?
22. What level of accuracy, validation, or supporting evidence would you require before trusting the system's analytics?

F. Player Development

23. Which factors do you consider when determining a player's skill level?
24. How do you identify a player's playing style, such as aggressive, defensive, or balanced?

25. How useful would historical performance trends be when planning future coaching sessions?
26. Would personal goal setting and progress monitoring between coaching sessions help your players? Why or why not?

G. Matchmaking and Community

27. How important is it for players to compete against others with similar skill levels?
28. Would player matching based on skill level and availability benefit your players? Why or why not?

H. Closing Questions

29. What concerns or recommendations would you have before adopting a system like this?
30. Is there anything else you would like to share that could improve the proposed system?

Appendix D

Appendix D

Proposed Pickleball Performance Metrics and Sample Reports

The following metrics were identified in the preliminary system-design notes. Final implementation should define the detection event, denominator, success condition, and minimum confidence threshold for every percentage-based metric.

Table D1

Proposed Core Pickleball Statistics

Stat	Meaning
PTS	Points scored
ACE	Service aces
WIN	Winning shots that directly end a rally
DRV%	Drive accuracy or success rate
DNK%	Dink accuracy or success rate
VOL%	Volley accuracy or success rate
UER	Unforced errors
KF	Kitchen or non-volley-zone faults
CC%	Court coverage efficiency
KLC%	Kitchen-line control

Example Player Stat Line

Juan Dela Cruz

11 PTS | 3 ACE | 8 WIN | 82% DRV | 76% DNK | 88% VOL | 4 UER | 1 KF

- Scored 11 points.
- Recorded three service aces.
- Hit eight winning shots.
- Achieved an 82% drive success rate.
- Achieved a 76% dink success rate.
- Achieved an 88% volley success rate.
- Committed four unforced errors.
- Committed one kitchen fault.

***Note.** The preliminary example does not contain values for court coverage efficiency (CC%) or kitchen-line control (KLC%). These fields should be populated only after their formulas and court-zone thresholds have been operationally defined.*

Example Doubles Match Report

Team A vs. Team B. Final score: 11–8.

Player	Stat	Value
Player A	PTS	6
Player A	ACE	2

Player A	WIN	5
Player A	DRV%	84%
Player A	DNK%	79%
Player A	VOL%	91%
Player A	UER	2
Player A	KF	0
Player B	PTS	5
Player B	ACE	1
Player B	WIN	3
Player B	DRV%	80%
Player B	DNK%	75%
Player B	VOL%	85%
Player B	UER	3
Player B	KF	1

Appendix E

Appendix E

RAG-Supported Audio Coaching Workflow

Note. The initial concept described continuous live coaching. The refined capstone scope uses RAG-supported coaching after video processing or during natural pauses, rather than constant audio during active rallies. This reduces distraction and makes the feature easier to test and defend as an MVP.

The workflow converts computer-vision outputs into short, contextual coaching cues. Retrieval-augmented generation (RAG) is used to retrieve approved pickleball knowledge, player history, and drill templates before the language model produces a recommendation.

Table E1

RAG-Supported Coaching Process

Stage	Process	Output
1. Start session	The player records or uploads a practice or match video and selects a session type.	Video and session metadata
2. Detect court and players	Computer vision identifies the court, player, and ball and tracks their positions.	Coordinates, tracking IDs, court zones
3. Extract performance metrics	Tracking data are converted into metrics such as kitchen-line distance, baseline time,	Structured performance metrics

	movement path, rally duration, and court coverage.	
4. Identify coaching events	Rules and thresholds identify moments such as staying too deep, failing to recover, or leaving a zone uncovered.	Detected gameplay event
5. Filter and prioritize	Confidence, repetition, severity, and cooldown rules prevent excessive or low-quality feedback.	Approved event
6. Build retrieval query	The event is translated into a query containing the player's skill level, session context, and detected behavior.	RAG query
7. Retrieve evidence	The system searches coach-approved tactics, shot guidance, drill templates, cue templates, and the player's history.	Relevant context
8. Generate recommendation	A language model produces one concise and actionable cue grounded in the retrieved material.	Text recommendation
9. Apply safety and quality checks	The system rejects medical advice, officiating decisions, unsupported claims, harsh language, or distracting instructions.	Approved cue

10. Convert to audio	Text-to-speech converts the approved recommendation into audio.	Audio coaching cue
11. Deliver feedback	The player receives the cue after processing, between drills, or during a natural pause.	Player-facing feedback
12. Log and reflect	The event, recommendation, timestamp, and later outcome are saved for review on the dashboard.	Session history and reflection data

Suggested Knowledge-Base Contents

Knowledge-Base Section	Example Contents
Pickleball tactics	Kitchen control, third-shot strategy, recovery position, doubles spacing
Shot-specific guidance	Dinks, drives, volleys, lobs, serves, and returns
Skill-level guidance	Beginner, intermediate, and advanced cue variants
Drill library	Dink drills, reset drills, return-and-advance drills, and footwork-recovery drills
Player history	Repeated weaknesses, recent progress, goals, and prior recommendations

Coach-approved templates	cue	Short, single-action, constructive feedback patterns
Safety rules		No medical diagnosis, no officiating decisions, and no unsafe or distracting commands

Illustrative Example

Detected event:

- The player remains near the baseline for 4.2 seconds after returning serve.
- The player is approximately 5.8 meters from the kitchen line.
- Session type: doubles training.
- Skill level: beginner to intermediate.
- Detection confidence: 0.86.

Retrieved coaching principle:

After returning serve in doubles, the returning team generally benefits from advancing toward the non-volley zone to establish stronger court control.

Retrieved drill:

Return-and-advance drill: return deep, move forward, and split-step before the opponent contacts the ball.

Generated cue:

“Move forward after your return. Take the kitchen earlier.”

Appendix F

Appendix F

Candidate Computer Vision Resources and Proposed Technology Stack

Note. The project should integrate existing models, repositories, and services rather than develop a new artificial-intelligence model from scratch. Every external resource remains subject to testing, licensing review, device requirements, and compatibility with the approved capstone scope.

A. Pickleball-Specific Candidate Resources

5urabhi/Pickle_ball_tracking

Tracks a pickleball in video, including bounce status and whether the ball is inside or outside court boundaries. Uses OpenCV, Ultralytics, NumPy, Pandas, Matplotlib, and FilterPy.

[Source](#)

kpp91302/Pickleball-Analytics

Computer-vision pipeline that detects players, ball position, and court layout and produces heatmaps, rally tempo, and kitchen-zone usage. [Source](#)

AndrewDettor/TrackNet-Pickleball

Deep-learning model for pickleball location tracking and candidate outputs such as ball velocity, shot type, and location. [Source](#)

Roboflow Pickleball Detection Dataset

Candidate annotated pickleball-detection dataset that may support evaluation or integration of an existing detector. [Source](#)

Pickleball Vision: CV-Driven Match Analytics

Pipeline described as using YOLOv8, a court-keypoint regressor, and homography refinement for player tracking, ball tracking, court geometry, a minimap, and speed or shot analytics. *Candidate public project/article; verify source before implementation*

bradautomates/claude-video

General video-processing utility that extracts frames and transcripts for model-assisted analysis. This is supplementary rather than a pickleball-specific tracker. [Source](#)

B. Proposed Integration Stack

Layer	Candidate Technology	Primary Output
Video input	Fixed court-side camera or smartphone recording	Match or training video
Object detection	Existing Ultralytics YOLOv8 model or approved pickleball-specific detector	Player, ball, and object detections
Tracking	ByteTrack, TrackNet-Pickleball, FilterPy-based tracking, or an existing repository	Frame-to-frame identities and ball trajectory
Court mapping	OpenCV perspective transformation and homography	Court-space coordinates and zones

Dataset management	Roboflow or an equivalent annotation and dataset-management platform	Dataset versions and evaluation samples
Metric extraction	Python and OpenCV rules using tracked coordinates and timestamps	Movement paths, zone occupancy, rally timing, and candidate shot features
Storage	Supabase/PostgreSQL and object storage	Player profiles, match records, video references, analytics, and recommendations
RAG retrieval	OpenAI vector stores, Supabase pgvector, or Pinecone	Retrieved coaching knowledge and player context
Recommendation generation	OpenAI API or another approved language-model API	Grounded coaching text
Audio output	Approved text-to-speech service or device TTS	Player-facing audio cue
User interface	Web application with player and coach views	Upload, reflection, trends, goals, and coach access

C. Minimum Computer-Vision Outputs

- Player location and movement trajectory.
- Ball location and trajectory when detector confidence is sufficient.
- Court boundaries, non-volley zone, baseline, and other defined court zones.
- Time spent near the baseline and non-volley zone.
- Court coverage and position-recovery patterns.
- Rally duration and event timestamps.
- Candidate shot classification only when technically validated using existing resources and an explicit operational definition.

Appendix G

Appendix G

Personal Informatics Framework Mapping

The proposed system is structured around the five stages of personal informatics: preparation, collection, integration, reflection, and action. The coach operates as a cooperative stakeholder who can interpret player-authorized information without replacing the player as the primary user.

Table G1

System Activities Across the Personal Informatics Stages

Stage	System Activity	Output
Preparation	The player selects a session type, reviews prior goals, and chooses a performance focus. A beginner may receive starter goals or explanations of pickleball terms.	Session objective, selected metrics, and goal
Collection	The player records or uploads gameplay. Existing computer-vision tools collect player, ball, court, movement, and rally data with minimal manual entry.	Video, coordinates, trajectories, and event timestamps
Integration	Current-session outputs are linked with the player profile, match context, prior sessions, skill estimate, goals, and coach-authorized notes.	Unified player performance record

Reflection	Dashboards present annotated clips, trends, strengths, weaknesses, court-position summaries, and evidence supporting each insight.	Understandable performance report
Action	The system recommends a next skill, drill, goal adjustment, or positioning focus. The player chooses an action and applies it in the next session.	Training decision and updated goal

Cooperative Coach Layer

- The player controls whether a coach may access performance information.
- The coach can review player-authorized reports, annotated clips, trends, and goals.
- The coach can add interpretations, drills, or approved feedback templates.
- System recommendations supplement rather than replace professional coaching judgment.
- The next session returns the player to the collection stage so the effect of the chosen action can be evaluated

Appendix H

Appendix H

Supplementary Scope Summary and System Boundaries

Primary and Supporting Stakeholders

Stakeholder	Role in the System
Players	Primary users and direct beneficiaries of performance tracking, reflection, goal setting, recommendations, and progress monitoring.
Coaches	Authorized cooperative users who interpret evidence, provide drills, and review player progress.
Clubs or communities	Supporting stakeholders for skill-based game discovery, localized challenges, and community participation.
Court owners or administrators	Indirect stakeholders only. The system is not a court-booking, payment, or availability-management platform.

In-Scope Components

1. Core performance monitoring through recorded-video analysis, including player movement, validated ball tracking, court positioning, zone usage, rally information, and selected shot-related outputs.
2. A player dashboard showing strengths, weaknesses, match history, progress trends, and supporting visual evidence.

3. Personal informatics functions for goal setting, reflection, recommendations, and a recurring performance-feedback loop.
4. Light skill-level and play-style estimation to support understandable guidance and player matching.
5. Gameplay-context support for practice or training, casual play, tournaments, leveling games, singles, doubles, and wall or machine drills when technically feasible.
6. Find-a-game and localized microchallenge functions that support participation without becoming a court reservation system.
7. Player-authorized coach access to performance reports and recommendations.
8. RAG-supported coaching guidance generated from approved pickleball knowledge and converted into optional audio after processing or during natural pauses.

Out-of-Scope Components

- Court booking or reservation.
- Payment processing for court use.
- Court availability management by owners.
- Tournament or league scheduling and administration.
- Development of a new artificial-intelligence or machine-learning model from scratch.
- Medical diagnosis, injury prediction, medical treatment, or replacement of a qualified health professional.
- Automated officiating decisions presented as authoritative rulings.
- Wearable-device integration in the final MVP.

Appendix I

Appendix I

Consultation and Revision Log

The following log consolidates key directions from the project-development meetings. Informal wording from the original notes has been paraphrased into formal action items.

Date	Consolidated Direction and Resulting Revision
June 11	Recenter the project on pickleball players rather than court owners; identify root causes and decision-making problems; use personal informatics; collect meaningful player and coach data; avoid becoming a booking platform; justify player groups and interview coaches.
June 26	Use existing technologies and models rather than building a model from scratch; define one clear solution type; follow the five personal-informatics stages; minimize manual collection; distinguish the personal, cooperative, and collaborative layers; solve information-quality problems within a manageable scope.
July 1	Confirm the specific APIs, repositories, and models to be used; determine how player and ball tracking will work; define pickleball-specific analytics; investigate RAG or MCP but avoid unnecessary architectural complexity; organize the system into player, in-game collection, post-game analytics, and community or coach functions.

July 11	Remove continuous real-time wording; use RAG-supported coaching and include audio in the methodology; remove wearable integration from the final scope; explain the computer-vision pipeline and pickleball-specific technology stack; state the exact data collected; simplify the solutions framework; use readable formatting and pickleball terminology; align scope, limitations, problems, and proposed solutions.
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Resulting Finalization Principles

- The player remains the main stakeholder.
- The system is a personal informatics solution, not a booking or general club-management platform.
- Computer vision uses existing resources to collect objective gameplay information.
- Analytics must use explicit pickleball terms, metrics, and court zones.
- Recommendations must be traceable to detected evidence and approved knowledge.
- Audio coaching is supplementary and should not distract the player during active rallies.
- The scope, objectives, methodology, technology stack, and Gantt chart should describe the same set of final components.

Appendix J

Appendix J

Preliminary Related Systems Inventory

This inventory was assembled during early related-systems research. The systems should be discussed in the Review of Related Systems only after their current features and availability have been verified through authoritative sources.

System	Primary Relevance	Original Source	Note
Zepp Tennis	Racket-sport swing and performance tracking	App-store listing	
PlaySight SmartCourt	Video capture, replay, and sports analytics	playsight.com	
Arccos Caddie	Sensor-based golf tracking and recommendations	Google Play listing	
Padelytics	Padel-focused analytics	padelytics.ai	
HomeCourt	Camera-based basketball practice and analytics	homecourt.ai	
OnForm	Video analysis and coaching collaboration	onform.com	
Dartfish	Video analysis and performance feedback	dartfish.com	
CoachNow	Coach-athlete communication and feedback	coachnow.com	
DUPR	Pickleball player rating and match-result platform	dupr.com	

Pickleheads	Pickleball court, game, and community discovery	pickleheads.com
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***Note.** The main reference list should contain the complete, verified citations for all literature, technologies, repositories, and systems cited in the paper. A reference dump should not be placed inside the appendices unless the institution explicitly requires a separate supplementary bibliography.*